The aerial bombing of Cambodia and the recovery of communities*

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Abstract

Communities exposed to the same conflict often follow different recovery paths. One reason may be that conflict leaves two distinct legacies. Unexploded ordnance (UXO) creates a lasting hazard, while ordnance that detonates causes destruction but leaves room to rebuild. We study the US bombing of Cambodia (1965–1973), using soil type to predict the likelihood of detonation. With geo-coded data on every recorded strike and a spatial matching approach, we test whether bombed communities still lag behind similar unbombed neighbors on the same soil decades later. On soft soil, where bombs were less likely to detonate, bombed communities face more UXO incidents, worse health, lower development, reduced income, and constrained agricultural productivity, consistent with a *risk* channel. On hard soil, where ordnance was more likely to detonate, bombed communities instead show stronger health, development, and household welfare, consistent with a *reset* channel. Destruction alone need not prevent long-run recovery, whereas lingering hazards can.

JEL-Codes: D74, N45, I15, O12, O13

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1 Introduction

Conflict devastates communities, and many experience lasting harm, trapped in post-conflict poverty rather than returning to growth (Leon 2012, Blattman & Miguel 2010, Rohner & Thoenig 2021), a pattern further exacerbated by risks due to unexploded ordnance (UXO) (Guo 2020, Nguyen et al. 2024, Riaño & Valencia Caicedo 2024). Yet recovery varies sharply. Countries such as Vietnam, Japan, and Germany recovered from widespread destruction (Fenske & Voth 2024), while within the same conflict some communities recover and their neighbors do not. One reason may be that conflict leaves two distinct legacies. Ordnance that fails to detonate creates a lasting hazard that restricts land use. Meanwhile, ordnance that detonates causes immediate destruction but leaves room for rebuilding and a possible reset of productive capacity. We ask whether bombed communities still lag behind their unbombed neighbors decades later, and whether that depends on which legacy the bombing left.

Leftover hazards constrain land use, investment, and trade long after a conflict. Farmers may need to work cautiously or leave land uncultivated (Lin 2024), and hazards also restrict transport and market access (Chiovelli et al. 2025). In a standard growth model, we describe this as a tax on productivity that prevents re-convergence to the path a community would otherwise have followed. Where ordnance detonated, this tax is absent and theory predicts convergence. Missed in this prediction, however, are coordination failures and institutional inertia that may keep communities tied to low-productivity capital. Destruction could relax these conditions and allow rebuilding with more productive capital.[1] We label these the risk and reset channels and combine them in a growth model that guides our design and interpretation. It predicts that bombed communities fall behind their unbombed neighbors where a hazard persists but can recover or even pull ahead where it does not (Section 3).

Militaries select targets for strategic value, so bombed and unbombed communities differ in ways that shape long-run development. Destruction and unexploded ordnance

[1] Hornbeck & Keniston (2017) document such gains after the Boston Fire, although their wealthy urban setting differs from rural Cambodia.

are also hard to separate, because heavier bombing typically brings both. The empirical challenge is to isolate bombing effects from pre-existing differences, then to distinguish their consequences where ordnance was more or less likely to detonate.

We address this challenge in Cambodia. In the late 1960s and early 1970s, the US conducted airstrikes across 115,273 recorded sites, dropping approximately half a million tons of munitions over Cambodia. We use geo-coded data on the universe of these strikes from the Yale University Cambodian Genocide Program. Borrowing a grid-based method from epidemiology, we group contiguous bombed cells into zones, preserving the spatial distribution of strikes. The interface between bombed and unbombed zones defines our *bombing boundaries*. To distinguish locations with high versus low UXO risk, we draw on Lin's (2022) extensive ethnographic fieldwork showing that bombs were less likely to detonate on soft soil, which absorbs impact energy rather than transferring it to the fuse mechanism. We use a pre-bombing soil map to classify locations as soft or hard soil (Crocker 1962).

We geographically match these bombing and soil measures to a range of outcomes. Four waves of the Cambodian Demographic and Health Survey (DHS, 2000–2014) record five physical health measures for women in geo-located DHS villages — height-for-age, weight-for-height, BMI, corpulence index, and anemia — which we combine into an index following Anderson (2008). The 2008 Population Census locates the universe of villages, allowing us to link UXO incidents to each one and to build an index of economic development from access to public services, physical infrastructure, and human capital. The Cambodia Socio-Economic Survey (CSES) and Agricultural Census provide further household and field-level evidence on income, agricultural productivity and risk-taking behavior.[2]

To identify long-run effects of the airstrikes, we develop a spatial matching approach that compares bombed communities with similar unbombed neighbors on the same soil, across bombing boundaries. These within-soil comparisons form our primary evidence. Identification requires that, in the absence of bombing, outcomes vary smoothly over

[2] See Section 4 for further details.

space in a way our geographic controls capture, and that people did not sort across the boundaries for reasons other than the bombing itself. Compared with their unbombed neighbors on the same soil, bombed communities are balanced on pre-bombing characteristics, including settlement mapped before the war (Section 5.3). Soil type does not predict where bombs fell, how many, or whether they included cluster munitions or B-52 strikes, supporting the use of the soil split to isolate detonation risk rather than targeting differences. Interpreting the across-soil contrast requires stronger assumptions about the unbombed counterfactual, so we treat it as secondary evidence.

We first test whether UXO incidents follow the predicted detonation pattern. In soft-soil areas, bombing raises the probability of a UXO incident by 3.9 percentage points against a mean of 7.6 percent, and increases the number of incidents by more than half relative to its mean. In hard-soil areas we find no comparable increase. This first stage is consistent with softer ground reducing detonation and leaving more live ordnance decades later.

By holding communities back, ordnance left in the ground should weigh on people's health, a dimension of welfare that is also associated with economic activity (Sen 1999, Deaton 2008, Fleurbaey 2009). Decades after the bombing, women in bombed soft-soil communities have significantly worse health than their unbombed neighbors, with the index about 7% of a standard deviation lower. On hard soil, where bombs were more likely to detonate, the effect reverses and women in bombed communities are healthier by about 6% of a standard deviation. Both effects survive a wide range of sensitivity and placebo tests.

Community economic development and household conditions again follow the same pattern. In soft-soil areas, bombed villages are significantly less developed than their unbombed neighbors, by about 12% of a standard deviation on our index. In hard-soil areas, development rises by about 6% of a standard deviation. Evidence from the CSES household survey reinforces these village-level results. In bombed soft-soil communities, households report lower income, less valuable durable goods and lower food security, while on hard soil they report higher income and food security. Together, these results

show that, decades later, bombed communities lag behind their unbombed neighbors where ordnance was less likely to detonate and fare better where detonation was more likely.

To see how these differences arise, we look at agriculture down to individual fields. Ethnographic research documents that farmers in UXO-affected areas use hand tools rather than plows, move slowly, and avoid clearing new land to reduce the chance of detonation (Lin 2024). Our CSES field data show that households in bombed soft-soil communities cultivate less land, earn lower crop revenues and suffer higher post-harvest losses than households in unbombed communities on the same soil. They also borrow less for agriculture and, in the 2013 Agricultural Census, hold smaller herds and use forests less, without moving into non-farm work. On hard soil, bombed households are more likely to produce for sale and allocate more labor to non-farm work than their unbombed neighbors. Activity thus contracts on bombed soft soil and reorganizes on bombed hard soil.

The contraction in bombed soft-soil communities fits the persistent productivity tax in our model. Higher rates of UXO incidents show that the hazard behind it is still present, and the health, development, household and field results measure its cost to welfare and production. In the model, the tax reduces current output and holds back capital accumulation, through lower savings and investment, potentially outweighing gains from rebuilding. An illustrative calibration under the model's assumptions puts the tax as high as 30% (Section 6.7).

In turn, long-run gains on bombed hard soil are consistent with the reset side of our model. Better health and development carry through to households, who are more likely to produce for sale and work off the farm, consistent with a reorganization of economic activity. Heavier bombing should also mean more destruction and more room for a reset. We find the gains are indeed largest where bombing was most intense (Section 6.4). A general advantage of hard soil cannot explain these patterns, because we compare bombed and unbombed communities on the same soil. Reset nevertheless remains a model-based interpretation, as we have no separate first stage identifying

it. We therefore also consider whether targeted reconstruction, transfers or population movement could account for the gains, and find little evidence of a lasting concentration of NGO projects or people in bombed hard-soil villages, though we cannot rule out every alternative. We conclude that destruction without persistent hazard does not, by itself, prevent long-run recovery, and can even leave communities modestly ahead of their unbombed neighbors.

Our findings contribute to several literatures. First, we distinguish the long-run effects of bombing with and without persistent hazards. Studies of post-conflict recovery find mixed evidence, with some economies returning to balanced growth (Davis & Weinstein 2002, Miguel & Roland 2011, Fenske & Voth 2024) and others suffering persistent losses (Abadie & Gardeazabal 2003, Yamada & Yamada 2021, Kešeljević et al. 2026), and a growing literature documents the long-run costs of UXO (Guo 2020, Nguyen et al. 2024, Riaño & Valencia Caicedo 2024). Because destruction and contamination usually arrive together, it is hard to tell which legacy holds communities back. Soil-based variation in detonation risk lets us study these effects by comparing bombed communities with unbombed neighbors on the same soil. We find that destruction alone need not prevent long-run recovery, whereas lingering hazards can. We then trace the costs of these hazards from UXO incidents, through community infrastructure and household welfare, down to individual fields.

Second, we add to the literature on the long-term health consequences of conflict. Most studies focus on direct exposure *in utero* or in early childhood and on intergenerational transmission (Akbulut-Yuksel 2014, Akresh et al. 2012, 2023, Camacho 2008, Duque 2017, Islam et al. 2017, Mansour & Rees 2012, Moyano 2017). Few examine how conflict shapes adult health decades after the fighting ends (Palmer et al. 2019, Yamashita & Trinh 2022).[3] We show that health can diverge across communities exposed to the same conflict, depending on whether the war left a lasting hazard.

Third, we speak to the literature on how historical shocks shape long-run develop-

[3] Palmer et al. (2019) find more disability in locations exposed to intense bombing in Vietnam; Yamashita & Trinh (2022) find that Agent Orange exposure in southern Vietnam is still associated with greater disability 30 years later.

ment (Dell & Olken 2020, Dell et al. 2018, Dell 2010). Conflict destroys infrastructure, food systems, and human capital (Levy 2002, Frost et al. 2017, Ghobarah et al. 2003, Leon 2012), but post-war investment can gradually offset these losses (Miguel & Roland 2011, Strauss & Thomas 2008). Beyond physical destruction, conflict can leave durable legacies, including persistent mistrust (Nunn & Wantchekon 2011), intergenerational prejudice (Voigtländer & Voth 2012), and shifts in cooperation and civic participation (Bauer et al. 2016). We complement this literature by adding a physical channel, as ordnance left in the ground constrains economic activity for as long as it remains and, in our model, its costs compound through capital accumulation.

Fourth, our household and field evidence shows that where UXO persists, farmers borrow less, cultivate less land, and earn less from their crops. These findings connect to a broader literature showing that risk is a binding constraint on agricultural productivity in developing countries (Rosenzweig & Binswanger 1993, Dercon & Christiaensen 2011, Karlan et al. 2014, Goldstein & Udry 2008). Our evidence extends this logic to persistent physical hazard from unexploded ordnance. Persistent hazards thus belong among the risks that hold back agricultural productivity, with costs that last for decades.

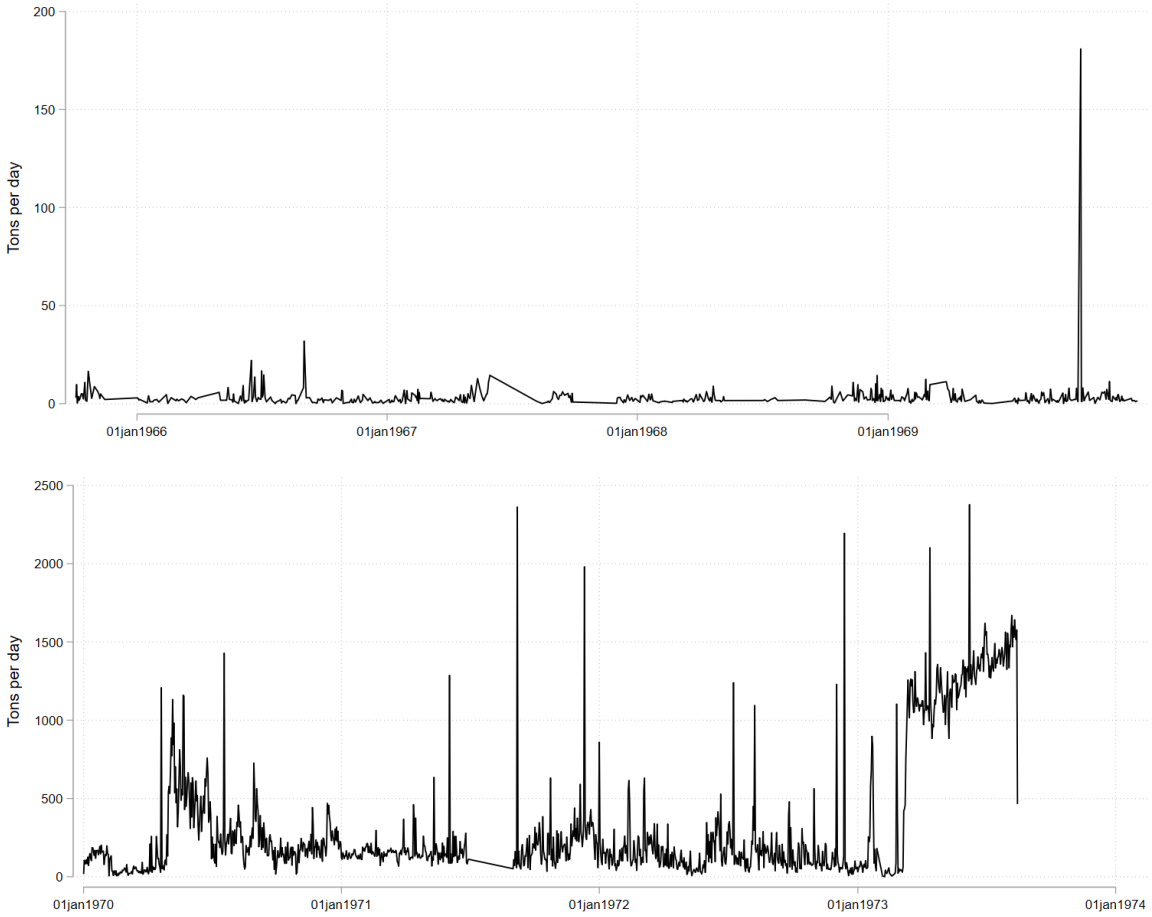
We next present Cambodia's historical background (Section 2) and develop the conceptual framework (Section 3). We then describe the data (Section 4), lay out the empirical strategy and validity checks (Section 5), and report the results (Section 6). Section 7 concludes.

2 Historical background

Over the last century, Cambodia endured colonization, civil wars, and genocide (Rany et al. 2012, Chandler 2018). After 90 years as a French protectorate (1863–1953), the country gained independence on 9 November 1953. A coup d'état on 18 March 1970 deposed Prince Sihanouk and installed the Lon Nol Government, setting off a civil war that lasted until 1975. In April 1975, the Khmer Rouge under Pol Pot seized power, beginning a period of genocide. The regime fell in 1979 when Vietnamese forces installed

a new government, and civil war continued until the 1991 Paris Peace Agreements opened a UN-supervised transition and reconstruction, with continued fighting largely confined to areas along the Thai border until the Khmer Rouge insurgency collapsed in the late 1990s (Rany et al. 2012, Chandler 2018, Bühler & Madestam 2024).

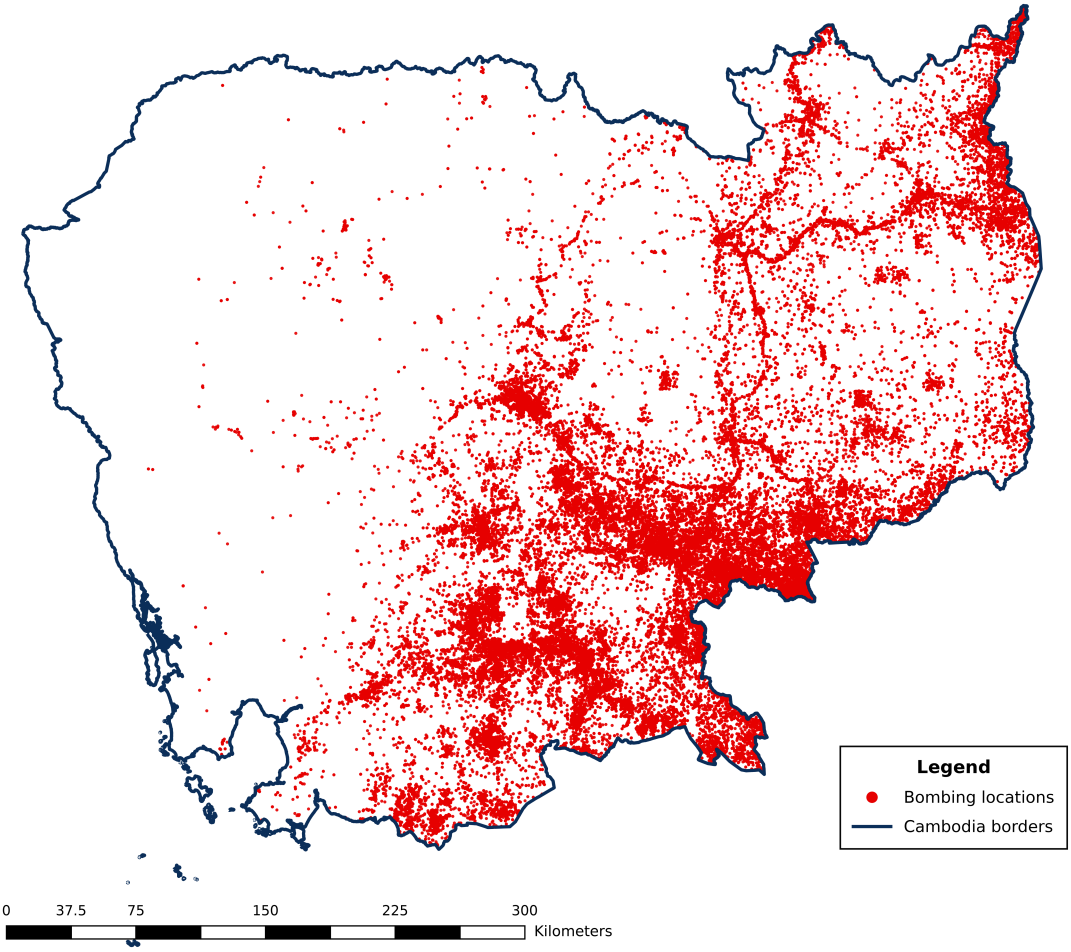
Figure 1: US ordnance dropped on Cambodia



Notes: Calculations by the authors from the Yale University Cambodia Genocide Project data on the universe of US airstrikes in Cambodia with detailed information on payload sizes for each strike.

The Vietnam War spilled into Cambodia through a US bombing campaign that began in 1965 and was heaviest from 1969 until it ended in August 1973 (Owen & Kiernan 2006). Under the Johnson administration, the US bombed Cambodia to disrupt supply lines and destroy Communist bases. The first intense phase, Nixon's secret Menu campaign, targeted border areas from March 1969 and ended in May 1970 after the coup. From 1970 onward, the campaign expanded as the US sought both to eradicate Vietnamese Communist forces and to support Lon Nol's regime in the civil war. Congress

Figure 2: Bombing sites targeted in Cambodia



Notes: Red dots give the locations of recorded bombing sites, nearly all between October 1965 and August 1973. Data provided by Yale University (Cambodian Genocide Program).

halted funding in 1973 upon discovering Nixon's deception about the military campaign (Owen & Kiernan 2006).

Data from Yale University (Cambodian Genocide Program) record 115,273 bombing sites in Cambodia (Owen & Kiernan 2006, Cambodian Genocide Program 2005*b*). We estimate that these strikes delivered roughly 400,000 to 500,000 tons of munitions, consistent with the revised estimate of approximately 500,000 tons discussed by Owen

& Kiernan (2015).[4] The bombing intensified sharply from 1970, as daily tonnage rose from around three tons to several hundred, peaking above two thousand (Figure 1). Although strikes concentrated in eastern Cambodia near the Vietnamese border, the campaign reached much of the interior (Figure 2). Early estimates placed civilian casualties between 50,000 and 150,000, and later evidence on the scale of the campaign suggests the true toll may have been substantially higher (Owen & Kiernan 2006).

Some bombs failed to explode, leaving hazards across the land that mines and other ordnance from the conflicts surrounding the Khmer Rouge takeover and Vietnam's 1979 invasion later compounded. Cambodia became one of the world's most heavily contaminated countries, with more than 64,700 mine and UXO casualties recorded since 1979, including over 19,700 deaths (Moyes et al. 2002, Martin et al. 2019).

3 Conceptual framework: destruction, risk, and recovery

We use a simple growth framework to explain how bombing can produce different long-run outcomes across communities, depending on whether destruction leaves a persistent hazard. Cambodia experienced displacement under the Khmer Rouge, Vietnamese occupation and civil war into the 1990s, so we do not expect the communities in our 2000–2014 data to have reached a steady state. The steady states below therefore characterize long-run tendencies rather than equilibria observed in the data, and help us interpret the treatment effects we estimate.

Vintage capital. Community infrastructure — roads, irrigation, schools and agricultural systems — embodies the technology available when it was built (Solow 1960). Replacing working infrastructure means forgoing its output, while coordination problems, institutional inertia and credit constraints can prevent collective replacement even where installed capital is scarce (Hornbeck & Keniston 2017, Henderson et al. 2021,

[4]The strike file combines two source databases with different conventions for recording loads. The earlier records report weight per weapon, whereas later records report strike totals. We reconstruct tonnage from recorded weapon counts and weights for each ordnance class. The range reflects alternative weight calibrations, using the earlier per-weapon records or nominal weights implied by weapon descriptions.

Banerjee & Duflo 2014). For shared infrastructure, no household may capture enough of the gain to bear the cost alone, and low rural incomes make the switch harder to finance. To capture the distinction between capital and its vintage, we start with a Solow growth model (Solow 1960) and $y = Ak^{\alpha}$, where k is capital per capita, A its productivity and $\alpha \in (0, 1)$ the capital share. Let A_o denote old-vintage technology and $A_n > A_o$ the frontier. We assume these constraints keep unbombed communities at the old technology, with steady-state capital k_o^* derived in Appendix A.

Destruction, risk and reset. Now suppose bombing destroys the existing stock. We take frontier technology A_n as the benchmark for hazard-free rebuilding, but UXO can constrain the use of land and capital and obstruct upgrading. We summarize both sources of the productivity shortfall as a tax $\tau \in [0, 1)$, giving $y = (1 - \tau)A_n k^{\alpha}$. Lower output reduces saving and capital accumulation, magnifying the initial productivity shortfall. With a constant savings rate s and depreciation rate δ, the resulting steady state is $k_\tau^* = (s(1 - \tau)A_n/\delta)^{1/(1-\alpha)}$. Appendix A gives the derivation and its use in a later illustrative calibration.

Without persistent hazards ($\tau = 0$), destruction removes the old stock and, by assumption, the constraints that prevented its replacement. With no working stock to preserve, rebuilding then leads to $k^*_{\text{reset}} = k_n^* > k_o^*$, a *reset* to higher productive capacity rather than convergence back to the same path.[5] Where hazards remain ($\tau > 0$), however, $k^*_{\text{risk}} = k_\tau^*$ falls below k_o^* if $(1 - \tau)A_n < A_o$: the productivity drag exceeds the potential gain from rebuilding. This is the *risk* case in which bombed communities fall behind their unbombed neighbors.

Historical and ethnographic accounts from Cambodia help motivate these mechanisms. Rural Cambodia held little installed capital, mostly dwellings and simple structures, but productive activity also depended on access to land, water and paths. Chandler (2018) documents a long-standing preference against change at the village level that

[5] Hornbeck & Keniston (2017) document gains from rebuilding after the Boston Fire. Their downtown land-assembly mechanism differs from rural Cambodia. The parallel here is narrower: destruction can allow rebuilding with more productive capital.

could reinforce constraints on agrarian practice and infrastructure. His account of post-conflict recovery includes renewed cultivation and new local markets. Such recovery fits the model's reset interpretation if rebuilding raised productive capacity beyond what the old stock allowed, whether financed locally or from outside. Where UXO persists, by contrast, farmers cultivate less land, use hand tools rather than plows, move slowly and avoid deep tilling to reduce the risk of detonation (Lin 2022). Hazards also obstruct irrigation and infrastructure development (Hamlin & Jaupi 2018, Martin et al. 2019).

Empirical comparisons. Writing $k^*_{\text{no-shock}} = k^*_o$ for unbombed communities, we summarize the two cases as

$$k^*_{\text{risk}} < k^*_{\text{no-shock}} < k^*_{\text{reset}}. \tag{1}$$

These cases frame the empirical questions our design addresses. Distinguishing between them requires knowing where detonation was likely to fail.

To capture this likelihood, we draw on Lin's (2022) extensive ethnographic fieldwork, which documents that bombs were more likely to fail on soft ground, leaving persistent UXO. Based on this, we use pre-bombing soil conditions to distinguish locations where bombing was more or less likely to leave a lasting hazard.

Table 1: The research design as a 2×2

	Bombed	Unbombed
Hard soil	k^*_{reset}	$k^*_{\text{no-shock}}$
Soft soil	k^*_{risk}	$k^*_{\text{no-shock}}$

Notes: Long-run steady-state capital k^* by pre-bombing soil type and bombing exposure. $k^*_{\text{no-shock}}$ is an unbombed community, k^*_{risk} a bombed community left with a persistent hazard, and k^*_{reset} a bombed community left without one. Section 3 defines each.

Table 1 maps three comparisons. Within soft soil, comparing bombed with unbombed communities gives the risk contrast, $k^*_{\text{risk}} - k^*_{\text{no-shock}}$. Within hard soil this comparison gives the reset contrast, $k^*_{\text{reset}} - k^*_{\text{no-shock}}$. These two contrasts are our primary comparisons. Each holds soil type fixed, so a general advantage of one soil cannot explain any bombing effect. The unbombed benchmark may differ across soils without

invalidating either within-soil comparison. The across-soil divergence, $k^*_{\text{risk}} - k^*_{\text{reset}}$, we treat as secondary. Interpreting the difference in bombing effects as this divergence requires a common unbombed reference across soils, despite possible differences in agricultural potential. We return to this stronger assumption in Section 5.3.

The two mechanisms also rest on different evidence. Risk has a measurable first stage in UXO incidents, while reset has no first stage of its own. It therefore remains a model-based interpretation, which we weigh against alternative explanations in the results. We return to the model in Section 6.7 to illustrate the size of the productivity tax.

4 Data

We assemble data from multiple sources, collected decades after the bombing. Health is measured from individual records in the DHS, UXO incidents and economic development are measured for census villages, and household conditions, field productivity and risk-taking come from the CSES and the 2013 Agricultural Census. Village coordinates from the 2008 Cambodia Population Census provide the common geographic reference for all but the DHS. Detailed variable definitions, construction methods and full data-source references are in Appendix B.

4.1 Bombing and the identification of bombing areas

We use geo-coded data on 115,273 US bombing sites in Cambodia (October 1965 – August 1973) from the Yale University Cambodian Genocide Program.[6] The data record bombing dates, precise locations, aircraft, bombing loads, and ordnance types. To identify bombing areas, we apply a grid-based clustering technique that captures the spatial concentration of strikes, a method widely used in geography, public health, and ecology (Aldstadt 2010, Grubesic et al. 2014).[7]

[6] Apart from 11 records dated May 1975, the data end on 15 August 1973, when the US bombing of Cambodia stopped. The total of 115,273 includes these 11 records.

[7] See Appendix B Section B.1 for further details.

We divide the country into small geographic grid cells. Cell size matters, as setting it too small produces noise, while too large over-smooths (Ankerst et al. 1999, Cheng et al. 2018). Following Boots & Getls (1988), we set cell size as $I = \sqrt{2 \times A/n}$, where A is the area of the rectangular analysis extent covering Cambodia (337,561 km^2), before clipping to the study boundary, and n is the number of strikes (115,273). This yields cells of 5.856 km^2 (2.42 km per side), dividing the country into 31,799 grid cells after excluding cells outside national boundaries or over permanent water bodies.[8] Cells receiving at least one air-delivered bomb are classified as bombing areas.[9] Because strikes cluster spatially, bombed and unbombed cells each form contiguous zones. A bombing boundary is a cell edge where a bombed zone meets an adjacent unbombed zone (Figure B.1); edges internal to a single zone are not boundaries.

Appendix Figure B.2 shows the resulting bombing areas, which preserve the spatial distribution of strike locations. Bombing areas concentrate in eastern and southern Cambodia along the Vietnamese border but extend into the interior. Our treatment variable equals 1 for locations within bombing areas and 0 otherwise.[10]

4.2 Outcome variables

Unexploded ordnance. To measure UXO prevalence, we use data on Mine/ERW casualties (2005–2013), which record the location of landmine and explosive remnants of war incidents. We restrict to ERW incidents plausibly associated with aerial bombing and exclude landmines and other ground-based weapons. For each census village, we count incidents within a 3-km radius and construct a binary indicator equal to 1 if any incidents occurred. See Appendix B.2.3 for more details.

[8] This is similar to the cell size selected by Kohama et al. (2020), who examine how the economic characteristics of conflict zones influence the choice of military strategies.

[9] A small number of records in the strike data are not bombs: parachute flares, gunship cannon and machine-gun rounds, air-dropped sensors and leaflets. These are logged by round count rather than by weight and carry no ordnance load, so cells whose only records are of this kind are not classified as bombed. Ninety-five cells, 0.3 percent of the grid, fall into this category, and every record of an air-delivered bomb sits in a cell classified as bombed.

[10] In robustness checks (Appendix E.3), we treat locations as bombed only if their bombing load exceeds the 2nd, 5th or 10th percentile of positive loads. Lightly bombed areas then join unbombed ones in the comparison group, which should, if anything, attenuate the estimates.

Health. Our primary measure of individual well-being is a *Health Index* constructed from four waves of the Cambodia Demographic and Health Surveys (2000, 2005, 2010, and 2014). Although the DHS covers a subset of villages rather than the full universe, it provides the richest source of individual-level health measures, surveying a large sample of women across multiple waves. Following Anderson (2008), we combine five standardized health outcomes into a weighted summary index: height-for-age Z-scores, weight-for-height Z-scores, a healthy body mass index, a healthy corpulence index, and an indicator for no moderate or severe anemia. To identify their locations, we use DHS-provided geolocations for sampled groups of 25–30 households, which we refer to as *DHS villages* to distinguish them from *census villages* in the 2008 Population Census.[11] To preserve confidentiality the DHS displaces village coordinates at random, by up to 2 km in urban areas and up to 5 km in rural ones. Random displacement can nonetheless produce nonclassical errors in treatment and soil assignment. Displacement is confined to this outcome, as everything else we use is tied to exact 2008 Census village coordinates. Section 6.2 discusses the implications and robustness checks, and Appendix B.2.1 gives further detail.

Economic development. To capture village-level infrastructure and living standards, we construct an *Economic Development Index* (or *Economic Index*) following Anderson (2008). We merge the universe of village locations from the 2008 Population Census with data on distance to schools (2012), distance to hospitals (2010), and coverage by major radio networks (2012). We add census indicators on the share of households with improved water, electricity and sanitation, the share living in owner-occupied dwellings, and rates of adult literacy, school attendance and employment. Together, these form a composite index of tangible village-level development.[12]

Additional outcomes. We use the Cambodia Socio-Economic Surveys (CSES) of 2009 and 2014 to measure household income, expenditure, assets, food security, field pro-

[11] See Appendix B Section B.2.1 for further details on variable construction.
[12] See Appendix B Section B.2.2 for further details.

ductivity and risk-taking. We link sampled villages to 2008 Census coordinates using official administrative village identifiers. The 2013 Cambodia National Agricultural Census, a full enumeration of agricultural households linked through the same identifiers, adds measures of crop choice, livestock, land use and labor allocation. Finally, geocoded Khmer Rouge burial sites from Yale's Cambodian Genocide Program allow us to test whether bombing exposure predicts subsequent violence.

4.3 Pre-bombing data and controls

Soil type. We use the distribution of soil types in Cambodia from Crocker (1962) to classify pre-bombing soil conditions.[13] Following the classification in White et al. (1997) and Kohama et al. (2020), we create a binary variable equal to 1 if a village is located on a soil type classified as fertile and 0 otherwise.[14] We label this indicator *Soft Soil* and refer to its two categories as soft and hard soil throughout the empirical analysis. These labels describe the fertility-based proxy, not a direct classification of mechanical softness. Because fertility and mechanical softness are related but distinct properties, Appendix B.3 sets out each Crocker type, classifies it by the mechanical behavior that governs whether a fuze fails, and reports how closely that classification tracks the fertility binary we use.

Geographic controls and historical settlement. We digitize transportation links, agricultural activity and population density from the 1970 Indochina Atlas. These maps precede the bulk of the bombing, which escalated sharply from 1970 onward (Figure 1). We also use the FAO–IIASA Global Agro-Ecological Zones classification to capture climate and agricultural potential. For finer measures of historical settlement, we use the digitized building data of Bühler & Madestam (2024), based on US Army topographic maps hosted by Yale's Cambodian Genocide Program. These sheets contain information dating from 1966 to the early 1970s. Building counts and settlement

[13] The data were provided to Open Development Cambodia in ESRI Shapefile format by Save Cambodia's Wildlife's 2013 Atlas Working Group (Open Development Cambodia n.d.*b*).

[14] Among sixteen soil types in Crocker (1962), six are classified as fertile: Latosols, Alluvial soils, Brown alluvial soils, Lacustrine alluvial soils, Regurs, and Brown hydromorphics (Kohama et al. 2020).

layout provide additional balance tests and let us ask whether bombing effects depend on earlier build-up. Appendix B.4 details these measures, and Appendix B.5 reports descriptive statistics.

5 Empirical strategy and validity

Bombing locations were not randomly chosen, so comparisons of bombed and unbombed areas may confound bombing effects with the factors that drove targeting. Our spatial matching approach restricts comparisons to locations within a narrow band around bombing boundaries, where local geographic and pre-bombing economic conditions are similar. We estimate effects separately on soft and hard soil, using pre-bombing soil type to distinguish areas with higher and lower UXO risk. This section sets out the identifying assumptions and assesses their plausibility through validity checks.

5.1 Specification

We implement these within-soil comparisons drawing on the spatial designs of Bühler (2023) and Dell & Olken (2020), adapted to many borders. For each soil type $s \in \{\text{soft}, \text{hard}\}$, we estimate:

$$Y_{icgpt} = \alpha^s + \beta_1^s Bombing_c + f^s(Geo_c) + \boldsymbol{\lambda}^{s\prime}\mathbf{X}_c + \theta_g^s + \delta_p^s + \mu_t^s + \epsilon_{icgpt}. \tag{2}$$

The unit i is an individual, household or field in village c, depending on the outcome, and for village-level outcomes, there is one observation per c. The indices g, p and t denote geographic grid, province and survey year. For the census-village outcomes, we have Y_{cgp}. A location is bombed ($Bombing_c = 1$) if it falls within an identified bombing area (Appendix Figure B.2).

All coefficients vary freely by soil type (s). In the 2×2 of Table 1, β_1^{soft} maps to the risk contrast, $k^*_{\text{risk}} - k^*_{\text{no-shock}}$, and β_1^{hard} to the reset contrast, $k^*_{\text{reset}} - k^*_{\text{no-shock}}$. These are model interpretations of the within-soil long-term treatment effects, not estimates of steady-state capital.

To match similar locations, we restrict the sample to locations within 2km of a bombing boundary, where bombed and unbombed zones meet (Figure B.1). Edges internal to either zone do not form boundaries, so locations deep inside bombed or unbombed areas are excluded. We use a triangular kernel to give more weight to observations closer to the boundary, and borrow the bandwidth and kernel from the regression discontinuity literature to localize the comparison.[15] Unless stated otherwise, all analyses use this bandwidth.

Geographic controls and fixed effects. The function $f^s(Geo_c)$ captures smooth spatial variation through a local linear polynomial in demeaned x- and y-coordinates. To identify the effects, our design requires that, within each soil type, outcomes in the absence of bombing vary smoothly over space in a way our geographic controls capture. Because boundaries run through different parts of the country, we also include 50×50km grid fixed effects (θ_g^s) and province fixed effects (δ_p^s), keeping comparisons within similar geographic and economic environments (Dell 2010, Dell et al. 2018, Dell & Olken 2020, Lehner 2024). Alongside the results we also assess sensitivity to tighter local comparisons and alternative spatial polynomials.

Additional controls. The vector $\mathbf{X}_c$ includes elevation, tropics/lowland classification, the 1970 Atlas measures and distances to the capital and Vietnamese border (Section 4.3). For DHS and CSES outcomes, μ_t^s absorbs survey-year effects.

Inference. We cluster standard errors at the village for DHS, CSES and Agricultural Census outcomes. For census-village outcomes, we cluster at the commune, allowing for correlation among neighboring villages. Because bombing is assigned at a coarser level than the village, we also check commune clustering for household, field and Agricultural Census outcomes. As a further check on spatial correlation, we report Conley (1999) standard errors for the principal outcomes and CSES results.

We also report a pooled specification with a soft-soil indicator and its interaction with bombing, with common control coefficients and fixed effects across soils. Denoting its bombing coefficient by β_1 and its interaction by β_2, the soft-soil effect is $\beta_1 + \beta_2$,

[15] We use the RD bandwidth estimator of Calonico et al. (2020), which returns approximately 2km for both the *Health Index* and the *Economic Development Index*.

compared with β_1 on hard soil. Their difference, β_2, maps to the across-soil contrast in Table 1, $k^*_{\text{risk}} - k^*_{\text{reset}}$, under the additional assumption of a common unbombed counterfactual across soils. We therefore treat this interaction as secondary and examine that assumption in Section 5.3.

5.2 Leveraging our boundaries

Unlike in standard spatial RD designs, bombing exposure has no pre-existing administrative boundary (Dell 2010, Dell et al. 2018). Our spatial matching approach compares nearby locations across boundaries between contiguous bombed and unbombed zones. These zones reflect the spatial clustering of strikes, with boundaries defined where bombed and unbombed grid cells meet (see Figure B.1 for an example). Our constructed boundaries thus follow the spatial distribution of bombing, allowing local comparisons throughout Cambodia rather than only near institutional borders.

Why should these constructed boundaries provide useful local comparisons? Our boundaries are drawn from the clustering of the treatment itself, not from an administrative or geographic line that exists independently of bombing. Inference across such a boundary is therefore non-standard. In practice, any set of bombing boundaries approximates a latent truth, so ours are inevitably measured with error. Two features work in our favor. First, our grid-based method (Section 4) draws borders that are both consistent with the spatial dispersion of strikes (Figure B.1) and largely orthogonal to local features, as the balance tests in Section 5.3 show. Second, to the extent that boundary error is random, it attenuates our estimates toward zero, working against us finding effects.

For these boundaries to support local comparisons, they must separate locations with different bombing exposure. We check this by counting airstrikes within several radii of each census village, without using the grid to define these counts. Exposure is higher on the bombed side of the boundary in both soil types at every radius examined (Appendix Table C.1). The boundaries therefore capture differences in exposure beyond the grid-based treatment classification. Later, we show that shifting the boundaries does not

reproduce our health results, further supporting the relevance of their actual location.

The within-soil comparisons rest on local comparability in the absence of bombing and on people not sorting precisely around the boundaries independently of bombing itself. Before testing these conditions, we check that soil type does not predict the bombing a location received.

5.3 Validity

Independence of Bombing and 1962 Soil Types. For the soil split to capture detonation risk rather than differences in bombing exposure, soil type should not systematically predict bombing. Table 2 supports this condition. Soil type is not significantly associated with the treatment indicator (column 1) or with a range of bombing measures known to produce unexploded ordnance (Lin 2022), including intensity based on the strikes recorded in a location's own grid cell (column 2) and the munitions those strikes carried (column 3), as well as cluster munitions (column 4) and bombs delivered by B-52 aircraft (column 5). We conduct the analysis on the DHS sample and the universe of census villages, the two core samples we use. In both samples, the evidence supports our strategy of using 1962 soil types to separate bombing effects.

Smooth changes in pre-bombing characteristics at boundaries. To identify the effects of bombing, our design requires factors that could impact outcomes to vary smoothly over space. We test seven pre-bombing characteristics within each soil type in both the DHS and census samples using our boundary regressions (Dell 2010) (Appendix Table C.2 and Table C.3). Of these 28 balance tests, 25 show no significant imbalance. The three exceptions are among census villages: tropics/lowland and 1970 population density on soft soil, and distance to the capital on hard soil. The first two weaken when we narrow the estimation band to 1km, and the third is no longer significant (Appendix Table C.4), supporting greater local comparability nearer the boundary.

Balance is weaker across soil types. Six of fourteen interactions show imbalance: elevation, tropics/lowland, population density and distance to the capital among census

Table 2: Independence of Bombing and 1962 Soil Types

	Dependent variable is:				
	(1) Bombing	(2) Ln(Strikes + 1)	(3) Ln(Munitions + 1)	(4) Cluster Bombs	(5) B52
Panel A: Population Census Villages					
Soft Soil	-0.010	0.010	-0.006	-0.009	-0.004
	(0.022)	(0.048)	(0.103)	(0.014)	(0.008)
Mean	0.559	0.977	2.462	0.139	0.043
Relative magnitude	0.017	0.010	0.006	0.062	0.082
Observations	7013	7013	7013	7013	7013
Clusters	1104	1104	1104	1104	1104
Panel B: DHS Villages					
Soft Soil	-0.006	0.107	0.292	-0.002	0.013
	(0.049)	(0.095)	(0.219)	(0.028)	(0.016)
Mean	0.502	0.902	2.266	0.120	0.056
Relative magnitude	0.013	0.107	0.292	0.019	0.232
Observations	1043	1043	1043	1043	1043
Clusters	602	602	602	602	602

Note: The sample is restricted to villages within 2km of a treated-control interface (a bombing boundary). All regressions use a local linear polynomial in spatial coordinates with a triangular kernel weight. We control for 50x50km grid fixed effects and province fixed effects in all regressions. *Soft Soil* is a dummy variable equal to one on soil types the 1962 classification, which predates the bombing, rates as fertile. *Bombing* is our treatment variable indicating locations inside our defined bombing areas. *Ln(Strikes + 1)* and *Ln(Munitions + 1)* are the two intensity measures used in Table 6, the recorded airstrikes in a location's own bombing grid cell and the number of munitions those strikes carried; both are heavily right-skewed and enter in logs. *Cluster Bombs* is a dummy variable indicating whether the bombing grid contains cluster bombs. *B52* is a dummy variable equal to 1 if the grid was bombed by B-52 aircraft and 0 otherwise. *Relative magnitude* is the coefficient divided by the outcome mean, or the coefficient itself for the two log outcomes, where it is already a proportional difference. Standard errors reported in parentheses are clustered at the commune level. ***(**)(*) indicates significance at the 1%(5%)(10%) level.

villages, and elevation and economic activity among DHS villages. Tropics/lowland differs by 14% of its mean in the census sample, and economic activity by 22% in the DHS. Elevation differs by two and twelve meters respectively; the remaining significant differences are below 8% of their means. These residual differences are another reason to treat the across-soil contrast as secondary. Each of these variables enters our controls, and we later confirm that the results do not depend on them, within or across soil types (Section 6.2).

Historical building counts and settlement layout provide five further balance measures (Appendix Table C.5). Although some imbalances remain across soil types, bombed and unbombed villages are balanced on these measures within soil type, with only one of ten estimates marginally significant, at about 5% of its mean. This evidence on earlier

settlement therefore reinforces the local comparisons on which our design rests.

No selective sorting across bombing boundaries. Our comparisons require local comparability in the absence of bombing, not an absence of migration caused by bombing. The boundaries are drawn retrospectively from strike records and have no physical or administrative status that would itself attract settlement. Movement in response to bombing can nevertheless form part of its long-run effect on communities.[16] The available evidence provides little indication of differential population concentration. The 1970 population density is balanced across boundaries in the DHS sample on both soil types (column 4 in Table C.2), and differs by under three percent of its mean among census villages (Table C.3). Within each soil type, migration status also shows no consistent difference across the bombing boundaries, and bombing exposure does not significantly predict village population in either 1998 or 2008 (Appendix Tables F.3 and F.4). Sorting across the boundaries is therefore unlikely to explain the differences we estimate within soil type.

No spillovers across boundaries (SUTVA). Our estimates assume that treatment does not spill over to control locations (Keele & Titiunik 2015). Two spillovers could narrow our within-soil comparisons. Households leaving bombed soft-soil areas may arrive in nearby unbombed areas with few resources, depressing outcomes among controls. Nearby bombing may also affect locations classified as unbombed. To the extent that controls experience the same hazards or opportunities for rebuilding as treated locations on the same soil, the differences we estimate would shrink. The migration evidence above points against the first.

For the second, we count strikes within 3km of locations in unbombed grids and test whether outcomes in unbombed locations vary with this exposure (Appendix Table C.6). We find no evidence of a significant effect either individually or jointly. The point estimates nevertheless follow a pattern that supports the attenuation logic. On soft soil, all are negative, while on hard soil the coefficients rise with exposure and are

[16] Dell (2010) similarly considers treatment-induced migration as a potential channel of persistence.

positive at medium and high exposure in both samples. Worse outcomes among soft-soil controls would narrow the negative bombing effect, while better outcomes among hard-soil controls would narrow the positive effect. Both movements bring controls closer to treated locations on the same soil, consistent with spillovers attenuating our estimates.

6 The Long-Run Effects of Bombing on Communities

The conceptual framework (Section 3) predicts that persistent *risk* from unexploded ordnance leaves bombed communities behind their unbombed neighbors in the long run, while a *reset* where destruction left no lasting hazard can let them recover or even pull ahead. To test this, we estimate the effects of bombing separately across soft-soil areas (high UXO risk) and hard-soil areas (low UXO risk). We begin with a first-stage analysis of UXO incidents, then turn to health as our measure of well-being, and finally assess community-level economic development, household income, and field productivity.

6.1 First-stage: UXO Incidents in Communities

If soft ground increased detonation failure, bombing should leave a stronger legacy of UXO incidents on soft soil. Table 3 examines the probability of any incident and the number of incidents within 3 km of census villages. Within soft soil, bombing raises incident probability by 3.9 percentage points relative to unbombed villages on the same soil and also increases incident counts. Within hard soil, both estimates are slightly negative, including a 1.8-point decline in incident probability. The central pattern is therefore that bombing raises subsequent UXO risk on soft soil but not on hard soil.[17] To check that landmines from later conflicts are not driving our results, we also report estimates excluding provinces along the K-5 mine belt, an area of extensive landmine contamination in northwestern Cambodia. The first-stage pattern is unchanged.

[17] Ordnance from later conflicts also contributes to ERW incidents, including in unbombed locations. UXO from US aerial bombing predominates in the east and northeast (Roberts 2011, Martin et al. 2019).

We use landmine incidents as a placebo to check whether the first stage reflects residual UXO rather than where people live, farm or report accidents. Buried ordnance enters the casualty record only when someone encounters it, so these activities could generate differences in recorded incidents even without differences in contamination. The database separately identifies landmine incidents, which we exclude from our UXO measure. Unlike air-dropped bombs, landmines were deliberately placed by ground forces, so their presence does not depend on detonation failure during bombing. Their recorded incidents nevertheless depend on where people live and work, whether accidents are reported and where clearance has occurred. Yet bombing does not significantly predict landmine incidents within either soil type, and the pooled interaction is essentially zero (Panel C). This contrast supports interpreting the UXO first stage as a legacy of aerial bombing rather than a general pattern in encounters or reporting.

We next ask whether the soil pattern reflects greater contamination or simply more encounters with ordnance on productive farmland. Our soil indicator comes from a fertility classification, while the detonation mechanism depends on how soft ground absorbs a bomb's impact (Appendix B.3). Rainfall softens the ground, so more bombs should fail during the wet season, especially on soft soil. We divide bombed villages at the median share of strikes falling in the wet season. On soft soil, bombing raises incident probability by an estimated 5.5 percentage points where more bombs fell during the wet season, compared with 2.0 points where fewer fell. The hard-soil estimates are similar across the two groups (Appendix Table D.1). The larger soft-soil first stage where more bombing occurred in wet conditions supports the detonation explanation rather than differences in agricultural potential alone.

We also test whether the UXO result is an artifact of counting incidents within 3 km of villages by re-estimating it at radii from 1 to 5 km (Appendix D.2). The soft-soil estimate changes little between roughly 2.75 and 3.75 km but weakens at either end of the range, for reasons that lie in the data. At 1 km, 98.8 percent of villages have no recorded casualty within the buffer, leaving little variation with which to compare villages. Wide buffers instead blur the exposure contrast, with all sampled villages

falling within 2 km of a bombing boundary, so at 5 km their buffers cross it by a wide margin and increasingly count the same incidents on both sides. The broad plateau around 3 km shows that our result does not hinge on a precisely chosen radius, and the hard-soil estimate stays small and negative at every radius. Weighting incidents by the number of victims produces a similar profile (Appendix Figure D.2).

The first stage also holds under alternative spatial inference and tighter local comparisons. The pooled UXO-likelihood interaction remains significant under Conley spatial standard errors (Appendix Table C.12). We further assign villages to their nearest boundary segment and use segment fixed effects to compare those sharing the same local stretch of boundary. This leads to a demanding fixed-effects specification, so we apply it only to the pooled model. The interaction increases from 0.050 to 0.075 and remains significant at the 1 percent level, with or without the K-5 provinces (Appendix Table C.8).

This evidence provides the first-stage foundation for the *risk* interpretation. *Reset* has no equivalent first stage, as we do not observe destruction directly. We next examine whether long-run health and economic outcomes follow the pattern these differences in persistent hazard would imply.

6.2 Health

We begin with health because community capital shapes residents' well-being, and health is fundamental to it (Sen 1999, Deaton 2008, Fleurbaey 2009). Across four survey waves, the DHS provides detailed health measures for multiple women in each sampled village, allowing us to examine how these community differences register in individual outcomes.

Bombed communities have worse health on soft soil and better health on hard soil than their unbombed neighbors. In Table 4, the *Health Index* is lower by 0.032 points on soft soil ($p < 0.05$), about 7 percent of a standard deviation, and higher by 0.027 points on hard soil ($p < 0.01$), about 6 percent.[18] The pooled model confirms a significant

[18]The index standard deviation is 0.43 (Appendix Table B.2).

Table 3: First-Stage: UXO incidents, with a landmine placebo

	All areas		Soft soil		Hard soil	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: UXO likelihood						
Bombing	-0.016	-0.016	0.039**	0.034*	-0.018*	-0.017*
	(0.010)	(0.010)	(0.019)	(0.019)	(0.010)	(0.010)
Bombing × Soft Soil	0.050**	0.047**				
	(0.021)	(0.022)				
Mean	0.065	0.059	0.076	0.062	0.059	0.057
Observations	7012	6613	2521	2396	4489	4215
Clusters	1104	1013	559	522	827	752
Exclude K5 Belt	No	Yes	No	Yes	No	Yes
Panel B. Number of UXO Incidents						
Bombing	-0.018	-0.021*	0.048**	0.033*	-0.023*	-0.021*
	(0.012)	(0.011)	(0.022)	(0.019)	(0.012)	(0.012)
Bombing × Soft Soil	0.053**	0.048**				
	(0.025)	(0.023)				
Mean	0.076	0.063	0.090	0.065	0.068	0.063
Observations	7012	6613	2521	2396	4489	4215
Clusters	1104	1013	559	522	827	752
Exclude K5 Belt	No	Yes	No	Yes	No	Yes
Panel C. Landmine likelihood (placebo)						
Bombing	0.005	0.004	0.001	0.001	0.005	0.004
	(0.004)	(0.004)	(0.002)	(0.002)	(0.004)	(0.004)
Bombing × Soft Soil	-0.001	-0.000				
	(0.005)	(0.005)				
Mean	0.013	0.011	0.008	0.007	0.015	0.014
Observations	7012	6613	2521	2396	4489	4215
Clusters	1104	1013	559	522	827	752
Exclude K5 Belt	No	Yes	No	Yes	No	Yes

Note: Observations are 2008 Census villages within 2km of a bombing boundary. Regressions use linear spatial controls, triangular kernel weights, 50x50km grid and province fixed effects, and the baseline pre-bombing controls. Columns (2), (4) and (6) exclude K-5 provinces. Panels A and B report an indicator for any UXO incident and the number of incidents within 3km, respectively, restricted to ordnance types plausibly associated with aerial bombing. Panel C reports an indicator for any landmine incident within 3km from the same casualty database. Standard errors reported in parentheses are clustered at the commune level. ***(**)(*) indicates significance at the 1%(5%)(10%) level.

difference between these bombing effects, with an interaction of −0.052 ($p < 0.01$, column 1). Excluding provinces along the K-5 mine belt leaves the estimates close to their baseline values (columns 2, 4 and 6). The individual health measures broadly

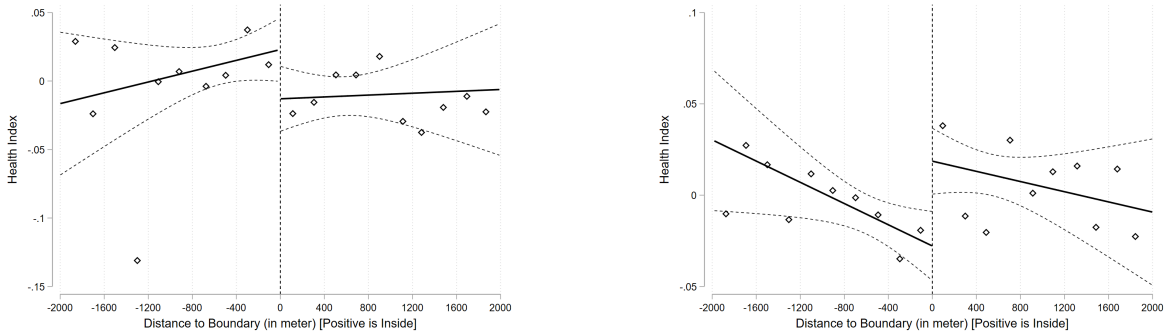
support the same pattern (Appendix Table E.1), and Figure 3 shows the opposing movements in health at bombing boundaries on the two soils.

Table 4: The long-term effects of local area exposure to bombing on health

	All areas		Soft soil		Hard soil	
	(1)	(2)	(3)	(4)	(5)	(6)
Bombing	0.024**	0.027***	-0.032**	-0.030*	0.027***	0.029***
	(0.010)	(0.010)	(0.014)	(0.015)	(0.010)	(0.011)
Bombing × Soft Soil	-0.052***	-0.056***				
	(0.017)	(0.018)				
Mean	-0.013	-0.013	-0.010	-0.015	-0.015	-0.011
Observations	17268	15647	6242	5801	11026	9846
Clusters	1051	954	381	355	670	599
Exclude K5 Belt	No	Yes	No	Yes	No	Yes

Note: The unit of analysis is DHS survey respondents. The sample is restricted to those living within 2km of a treated-control interface (a bombing boundary). All regressions use a local linear polynomial in spatial coordinates with a triangular kernel weight and control for 50x50km grid fixed effects, province fixed effects, distance to the capital, distance to Vietnam borders and other pre-bombing characteristics. Regressions (2) (4) (6) exclude provinces along the K-5 mine belt. Standard errors reported in parentheses are clustered at the DHS village level. *Health Index* is the standardized weighted summary index variable created from multiple health outcomes, including Height-for-age Z-scores, Weight-for-height Z-scores, a healthy Body Mass Index, a healthy Corpulence Index, and an indicator for no moderate or severe anemia (See Table E.1 for results on individual health outcomes). ***(**)(*) indicates significance at the 1%(5%)(10%) level.

Figure 3: Local area exposure to bombing on health: Graphical Depiction



(a) Soft soil (b) Hard soil

Note: The points represent binned residuals derived from a main regression of the outcome variable on a linear polynomial in spatial coordinates and other control variables. Solid lines depict local linear regressions, separately estimated on each side of the bombing boundary, while dashed lines represent 95% confidence intervals. Negative values of distance indicate locations outside the bombing areas.

This health pattern fits the two mechanisms developed in Section 3. On soft soil,

the first stage links bombing to persistent UXO, and the health decline is consistent with the *risk* channel reaching people's well-being through constrained livelihoods. On hard soil, the gain is measured against unbombed hard-soil neighbors, so the finding is not merely the absence of a soft-soil penalty. Our preferred interpretation follows the model's *reset* channel, in which destruction makes room for newer, more productive capital and thereby improves the conditions in which people live.

Children's survival offers a further way to assess whether the health pattern reflects lasting differences in community conditions. Because the health index is observed only for surviving, resident women, it can reflect who is surveyed as well as the conditions they face. Their birth histories also record children who died, extending the evidence beyond survivors' current health. On hard soil, women in bombed communities are 2.2 percentage points less likely to have experienced a child's death ($p < 0.01$). On soft soil, the estimate points toward higher child mortality, consistent with the risk channel, though it is not significant. The pooled difference between these bombing effects is 4.0 percentage points ($p < 0.01$, Appendix Table E.2). Together, these results strengthen our interpretation that bombing shaped the long-run conditions in which families live and raise children.

Two alternatives bear on this interpretation. First, because our soil measure is a fertility classification, bombing might affect productive farmland differently for reasons other than detonation failure. Fertility need not compete with the risk explanation, however. Where land is valuable for farming, leaving it idle or working it cautiously to avoid UXO carries a greater cost. Fertile soft ground can therefore both retain more unexploded ordnance and make that hazard more costly. The seasonal comparison from the first stage fits this account. On soft soil, places with more wet-season bombing show a larger increase in UXO incidents and worse health relative to unbombed villages. The health estimate is negative where less bombing fell during the wet season too (Appendix D.1). Second, health differences may reflect direct wartime exposure rather than recovery conditions. If direct exposure were the explanation, the pattern should fade among women born long after the bombing. It does not. The pattern holds

among women born from 1990, as among those born before 1980. It is absent for the 1980s cohort, whose early years coincided with continued conflict and deprivation that may have obscured local differences (Appendix E.4). Its presence among the youngest women supports an effect through the conditions in which communities recovered, rather than direct exposure to bombing alone.

Checks on the bombing estimates. We first ask whether uncertainty in DHS locations could generate this health pattern. Rural displacement of up to 5 km exceeds our 2 km estimation band, so the error is not classical (Section 4). We therefore examine urban DHS villages, where displacement is limited to 2 km. The hard-soil effect remains positive and significant at 0.059 ($p < 0.05$), against 0.027 in the full sample (Appendix Table E.3). For soft soil this sample is less informative, as urban households depend less on working contaminated land, and the estimate is indeed small and insignificant. We also vary the estimation band. Misassignment is likeliest for DHS villages recorded close to a boundary, so the share of the sample exposed to it is largest at 1 km and falls as the band widens toward the 5 km bound. Both within-soil estimates keep their sign and significance from 1 to 5 km (Appendix Figure E.1), and Appendix E.2 provides the details.

A separate concern is that bombing exposure might stand in for subsequent Khmer Rouge violence. We assess this using nearby mass graves. In the census-village sample, bombing does not significantly predict mass graves within 3 km (Appendix Table C.10). More importantly for the health results, controlling for the logged number of nearby mass graves leaves both within-soil effects and the pooled interaction unchanged (Appendix Table C.11). The health pattern therefore does not appear to be explained by local differences in subsequent Khmer Rouge violence.

Section 5.3 reports a handful of imbalances, mostly across soil types. Dropping first the 1970 measures and then the other location covariates leaves both within-soil effects and the pooled interaction essentially unchanged (Appendix Table C.7). The spatial polynomial, kernel weights and fixed effects remain throughout. The covariates

therefore do not drive our results, within or across soil type.

We also ask whether the estimates survive small perturbations to the constructed boundary. Because it follows grid edges, the boundary traces a staircase, and progressively rounding those corners changes each location's distance to it while leaving its bombing status fixed. Both within-soil effects and their difference remain stable (Appendix Table C.9), so the findings do not hinge on the precise path the grid draws.

Alternative specifications leave the within-soil pattern in place, including different polynomial orders, grid fixed effects, bombing-intensity thresholds, and distance to the boundary as the geographic control (Appendix E.3). As a placebo, shifting the boundaries by 5 km and reassigning treatment does not reproduce the health effects within either soil type or in the pooled model (Appendix Table E.4). This supports locating the health pattern at the bombing boundaries rather than elsewhere. The graphical pattern is also preserved at 1 km and 3 km plotting windows (Appendix Figure E.2).

The across-soil comparison also holds when we allow for spatial correlation and make the comparisons more local. The pooled interaction remains significant with Conley standard errors (Appendix Table C.12). As in the first stage, we retain the full sample for border-segment fixed effects, comparing villages along the same local stretch of boundary. The interaction is −0.049, or −0.053 excluding the K-5 provinces, against −0.052 at baseline (Appendix Table C.8). Thus, tighter local comparisons leave the difference between the bombing effects essentially unchanged.

Taken together, lower health on bombed soft soil and higher health on bombed hard soil match the framework's ordering, $k^*_{\text{risk}} < k^*_{\text{no-shock}} < k^*_{\text{reset}}$, though we interpret these as long-run treatment effects rather than measured steady states. We next turn to economic development to examine how persistent hazard and rebuilding shape communities.

6.3 Communities' economic development

Community infrastructure should follow a similar pattern to health. Through the risk channel, we expect bombing on soft soil to constrain development. On hard soil, the model predicts fewer constraints on investment among bombed villages than among similar unbombed ones, because destruction makes room to rebuild.

We test these implications in Table 5 using our *Economic Development Index*, measured across the universe of census villages. On soft soil, economic development is lower in bombed villages by roughly 12% of a standard deviation ($p < 0.05$, column 3), with little change when we exclude provinces along the K-5 mine belt (column 4). On hard soil, bombed villages show stronger development by about 6% of a standard deviation ($p < 0.10$, column 5), with a slightly larger gain when we exclude the K-5 provinces ($p < 0.05$, column 6). The pooled interaction confirms a significant across-soil difference between these bombing effects, equivalent to about 14% of a standard deviation ($p < 0.05$, column 1).[19] Figure 4 shows this contrast graphically, with development falling at bombing boundaries on soft soil and rising on hard soil. Estimates on each component of the index broadly support the same pattern across the underlying measures of infrastructure and economic opportunity, although most are individually imprecise (Appendix Table F.1).

The economic results extend the health pattern to communities' infrastructure, services and economic opportunities. The soft-soil decline is consistent with the *risk* channel constraining development. On hard soil, the gain is relative to unbombed hard-soil neighbors, so the absence of a persistent hazard alone cannot explain it. Here, bombing has not blocked long-run recovery, and the modest advantage of bombed villages is consistent with the model's *reset* interpretation, in which rebuilding replaces older capital with more productive capital.

As with health, seasonal variation helps assess whether the soft-soil penalty reflects the ground's mechanical properties rather than fertility alone. The estimated decline

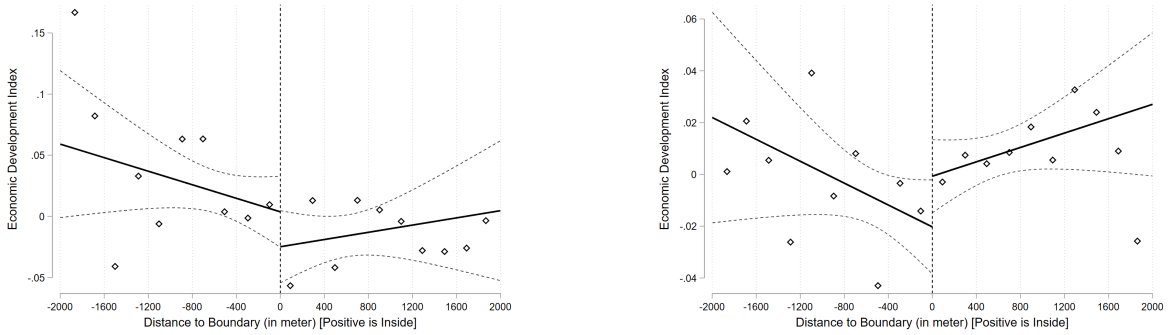
[19] The index standard deviation is 0.32 (Appendix Table B.3).

Table 5: Economic Development Index *(Unit: Census Villages)*

	All areas		Soft soil		Hard soil	
	(1)	(2)	(3)	(4)	(5)	(6)
Bombing	0.014	0.018*	-0.040**	-0.043**	0.018*	0.022**
	(0.010)	(0.010)	(0.017)	(0.017)	(0.009)	(0.010)
Bombing × Soft Soil	-0.046**	-0.052**				
	(0.021)	(0.022)				
Mean	0.023	0.029	0.027	0.032	0.020	0.028
Observations	7012	6613	2521	2396	4489	4215
Clusters	1104	1013	559	522	827	752
Exclude K5 Belt	No	Yes	No	Yes	No	Yes

Note: The unit of analysis is 2008 Population Census villages. The sample is restricted to villages within 2km of a treated-control interface (a bombing boundary). All regressions use a local linear polynomial in spatial coordinates with a triangular kernel weight and control for 50x50km grid fixed effects, province fixed effects, distance to the capital, distance to Vietnam borders and other pre-bombing characteristics. Regressions (2) (4) (6) exclude provinces along the K-5 mine belt. Standard errors reported in parentheses are clustered at the commune level. *Economic Development Index* is the standardized weighted summary index variable created from multiple outcomes including distance to schools, hospitals, locations within big radio network and other indicators of development such as the share of households with access to water, electricity, toilets, own dwellings, literacy, school attendance and employment rate (See Table F.1 for results on individual economic outcomes). ***(**)(*) indicates significance at the 1%(5%)(10%) level.

Figure 4: Economic Development Index (Census Villages): Graphical Depiction



(a) Soft soil (b) Hard soil

Note: The points represent binned residuals derived from a main regression of the outcome variable on a linear polynomial in spatial coordinates and other control variables. Solid lines depict local linear regressions, separately estimated on each side of the bombing boundary, while dashed lines represent 95% confidence intervals. Negative values of distance indicate locations outside the bombing areas.

is nearly twice as large in locations with a higher wet-season bombing share, −0.051 against −0.027, while the hard-soil gains are similar across the two groups (Appendix Table D.1). The larger penalty where bombs were more likely to meet wetter, softer

ground fits the risk interpretation, with a more pronounced seasonal pattern than for health.

The hard-soil gains could alternatively reflect where people and post-war investment settled. In a Malthusian setting, productivity gains can be absorbed by population growth rather than sustained increases in income per capita (Ashraf & Galor 2011). If households moved away from contaminated soft soil and public goods followed them, population reallocation and agglomeration could generate a divergence between soils. Movement toward safer soil bears more directly on the across-soil comparison, which we treat as secondary. Within hard soil, the question is why bombed villages experienced better health and economic outcomes than their similarly safe unbombed neighbors. For reallocation to explain this gain, it would need to favor bombed hard-soil villages specifically, not hard-soil areas generally. Balance on historical settlement around present-day villages supports their initial comparability (Section 5.3 and Appendix Table C.5). We next examine whether later population, migration and NGO projects show such a concentration in bombed hard-soil villages.

The CSES data provide little evidence of NGO concentration in bombed hard-soil villages. Neither the probability of hosting a project nor the number of projects differs significantly across their bombing boundaries (Appendix Table F.2). While we cannot directly test government reconstruction spending, reconstruction financed by the state or donors could itself fit the reset interpretation if it replaced older capital and thereby raised activity. The model concerns what rebuilding achieves, not who pays for it. The development index measures access to services, education and employment rather than income or aid receipts. Although it does not identify the source of the gains, its improvement reflects broader economic recovery.

Likewise, migration and population checks provide little evidence of sustained concentration in bombed hard-soil villages. Migration status does not differ significantly in either the DHS or CSES (Appendix Table F.3), and bombing does not significantly predict population on either soil in 1998 or 2008 (Appendix Table F.4). These data cannot establish whether displacement or resettlement before 1998 left lasting pub-

lic investment after population differences diminished, nor do village totals capture changes in population composition. They nevertheless provide no evidence that the hard-soil gains were driven by a lasting concentration of people in bombed villages.

The 1998 Census also lets us examine whether the development pattern was already present a decade before our main observations. We construct indices restricted to six common dimensions in each year (Appendix Table F.5). The soft-soil penalty is present in both censuses, providing further support for the risk channel. Bombed hard-soil villages lag behind in 1998, but the 2008 estimate is positive and close in magnitude to our main estimate, although less precisely estimated on the restricted index. This pattern is consistent with recovery over time and the model's reset interpretation.

Neither subsequent local violence nor the choice of controls appears to explain the development pattern. Controlling for the logged number of Khmer Rouge mass graves within 3 km leaves both within-soil effects and the pooled interaction close to the baseline (Appendix Table C.11). Removing the pre-bombing covariates in stages similarly leaves the within-soil estimates stable and the interaction significant, confirming that the controls do not drive the results (Appendix Table C.7).

Our findings are additionally insensitive to small perturbations of the grid-based boundary. Rounding its staircase corners while holding treatment fixed leaves both within-soil effects and their difference stable (Appendix Table C.9). The graphical pattern is not sensitive to the plotting window either. Using villages within 1 km or 3 km of the boundary still shows development falling on the bombed side on soft soil and rising on hard soil (Appendix Figure F.1).

For the pooled comparison, we check whether the result depends on how we account for spatial correlation or on how locally we compare villages. The interaction remains significant with Conley standard errors (Appendix Table C.12). Adding border-segment fixed effects restricts comparisons to villages along the same local stretch of boundary, yielding estimates of -0.027, or -0.033 excluding the K-5 provinces (Appendix Table C.8). These lose significance under this demanding fixed-effects specification but remain qualitatively consistent with our baseline results.

Across health and economic development, bombed communities fare worse than their unbombed neighbors on soft soil and better on hard soil, consistent with persistent hazard constraining development and rebuilding permitting gains where that hazard is largely absent. We next ask whether these effects scale with the intensity of strikes.

6.4 Does the effect scale with the dose?

Treatment has so far captured whether bombs fell. Intensity lets us ask whether more bombing strengthens the *risk* channel on soft soil and the *reset* channel on hard soil, where more impacts increase the scope for destruction and rebuilding. We replace the binary treatment with low, medium and high intensity, defined by terciles among bombed locations, with unbombed locations as the reference group. We measure intensity using strikes and munitions recorded in each location's bombing grid cell. We estimate these effects separately within each soil type.

With strike counts, Panel A of Table 6 shows a dose-response pattern in both health and economic development. On soft soil, the estimates are small and insignificant at low intensity, with substantial negative effects appearing at medium and high intensity. On hard soil, both outcomes are positive at every dose and the gains are largest where bombing was heaviest, with economic gains rising steadily with strike intensity.

Counting munitions in Panel B gives a clearer economic gradient but a noisier health pattern. The soft-soil development penalty deepens and the hard-soil gain increases with munition intensity, reaching −0.068 and 0.033, respectively. Health retains the same signs, but the soft-soil penalty is significant at low and medium intensity rather than at the highest. Location error offers a possible explanation for this contrast. Census villages have exact coordinates, whereas DHS displacement can assign a DHS village to the wrong bombing cell. Cells with similar strike counts can have very different munition loads, so assigning a DHS village to the wrong cell may distort munition intensity more than strike intensity. Panel B therefore reinforces the economic dose response, while the noisier health estimates remain consistent with the direction of the baseline effects.

The dose-response evidence challenges an alternative interpretation that soft-soil communities lag simply because fewer bombs detonated, leaving less old capital destroyed and less to rebuild. If this were the whole explanation, heavier bombing should reduce the disadvantage by creating more scope for rebuilding. Instead, the penalty is small where strikes were few and appears at medium and high intensity, consistent with more bombing leaving more persistent UXO risk. On hard soil, the largest gains where bombing was heaviest fit the reset interpretation.

Pre-war settlement provides a complementary way to ask whether the gains reflect opportunities to rebuild. We match each present-day census village to the mapped pre-war settlements within 2 km, then divide villages at the median building count of the largest nearby settlement. On hard soil, the estimated bombing effect on economic development is closer to zero and insignificant where settlements were smaller and positive and larger where settlements were bigger. The soft-soil penalty shows no such clear gradient (Appendix Table F.6). Counting all mapped pre-war buildings within 3 km of each census village gives a similar pattern. The differences in bombing effects between settlement groups are not statistically significant, but their direction supports rebuilding where more stood to be destroyed.

Together, these exercises connect the hard-soil gains to both how much bombing occurred and how much there was to rebuild. While neither directly measures destruction, both strengthen the model-based reset interpretation. On soft soil, the larger penalties at higher bombing intensity provide further support for persistent risk constraining development.

6.5 Household welfare and agricultural productivity

We next ask whether this pattern extends to household-level economic conditions and agricultural productivity. If the *risk* and *reset* channels shape community-level development, they should also reach individual households. In soft-soil areas, we therefore expect lower household income, fewer assets, and greater food vulnerability. Fieldwork in Cambodia reinforces this prediction, documenting that where UXO risk is high, farm-

Table 6: Bombing Intensity: Health Index and Economic Development Index

	Health index (DHS women)		Economic index (Census villages)	
	Soft soil (1)	Hard soil (2)	Soft soil (3)	Hard soil (4)
Panel A: Strikes in the village's own bombing cell				
Low bombing intensity	-0.007	0.025**	-0.015	0.006
	(0.019)	(0.012)	(0.020)	(0.012)
Medium bombing intensity	-0.051**	0.020	-0.063***	0.023*
	(0.024)	(0.014)	(0.024)	(0.012)
High bombing intensity	-0.050**	0.039**	-0.048*	0.032**
	(0.022)	(0.016)	(0.029)	(0.013)
Panel B: Munitions dropped in the village's own bombing cell				
Low bombing intensity	-0.033*	0.025*	-0.019	0.007
	(0.020)	(0.013)	(0.022)	(0.012)
Medium bombing intensity	-0.042*	0.025*	-0.038	0.016
	(0.022)	(0.014)	(0.024)	(0.013)
High bombing intensity	-0.015	0.030**	-0.068***	0.033***
	(0.019)	(0.014)	(0.025)	(0.012)
Mean	-0.00982	-0.0149	0.0279	0.0205
Observations	6242	11026	2521	4489
Clusters	381	670	559	827

Note: Bombing intensity is the number of recorded airstrikes in a unit's own 2.42km bombing grid cell in Panel A, and the number of munitions those strikes carried in Panel B. Bombed units are split into terciles of that count, with unbombed units as the omitted category, so each coefficient compares a bombed group against unbombed units on the same soil. Terciles are computed among bombed units within the estimation band. Health is measured on DHS women, with survey-year fixed effects and standard errors clustered at the DHS village; the economic index is measured on 2008 Population Census villages, with standard errors clustered at the commune. Samples are restricted to units within 2km of a treated–control interface (a bombing boundary), and all regressions use a local linear polynomial in spatial coordinates with a triangular kernel weight, 50x50km grid fixed effects, province fixed effects, distance to the capital, distance to the Vietnamese border and other pre-bombing characteristics. ***(**)(*) indicates significance at the 1%(5%)(10%) level.

ers reduce the area they cultivate, avoid investing in their land, and work with constant caution (Lin 2022, 2024). In hard-soil areas, our model suggests that vintage upgrading can operate without persistent hazard, so households should share in the broader recovery. We test these predictions using the Cambodia Socio-Economic Survey (CSES), which provides household-level data on income, expenditure, assets, food security, agricultural productivity, and borrowing behavior.

Household incomes and food security follow the same pattern as community devel-

Table 7: Post-bombing Development at household level (CSES)

	Dependent variable is: *Unit of Analysis: Households*			
	Ln(Income) (1)	Ln(Expenditure) (2)	Ln(Durable) (3)	Food Secure (4)
Panel A: All observations				
Bombing	0.297***	0.020	0.126*	0.041*
	(0.107)	(0.034)	(0.071)	(0.022)
	[0.038]	[0.580]	[0.249]	[0.249]
Bombing × Soft Soil	-0.541***	-0.054	-0.337***	-0.106***
	(0.160)	(0.050)	(0.116)	(0.034)
	[0.011]	[0.461]	[0.031]	[0.024]
Mean	14.048	16.412	14.104	0.862
Observations	11135	11135	11135	11135
Clusters	746	746	746	746
Panel B: Soft soil				
Bombing	-0.264**	-0.049	-0.255***	-0.076***
	(0.124)	(0.039)	(0.096)	(0.027)
	[0.084]	[0.233]	[0.033]	[0.032]
Mean	13.939	16.547	14.398	0.875
Observations	4674	4674	4674	4674
Clusters	323	323	323	323
Panel C: Hard soil				
Bombing	0.275**	0.020	0.114	0.042*
	(0.109)	(0.034)	(0.070)	(0.022)
	[0.080]	[0.590]	[0.250]	[0.217]
Mean	14.126	16.315	13.892	0.852
Observations	6461	6461	6461	6461
Clusters	423	423	423	423

Note: All regressions include households living within 2km of a treated-control interface (a bombing boundary). *Income* is households' income from agricultural and non-agricultural activities. *Expenditure* indicates households' total expenditure, including food, non-food, and housing expenses. *Durable* represents the total value of durable goods (TV, bicycle, radio, phone, etc.). All of these variables are transformed ln(Cambodian riel+10000). *Food Secure* is a dummy variable equal to 1 if households report that they do not go hungry. All regressions use a local linear polynomial in spatial coordinates with a triangular kernel weight and control for 50x50km grid fixed effects, province fixed effects, distance to the capital, distance to Vietnam borders, and other pre-bombing characteristics. Standard errors reported in parentheses are clustered at the village level. Romano–Wolf stepdown adjusted p-values, based on 9,999 wild cluster bootstrap replications, are reported in square brackets, treating all reported coefficients across the four outcomes within each panel as one family. ***(**)(*) indicates significance at the 1%(5%)(10%) level based on conventional p-values.

opment (Table 7). On soft soil, households in bombed villages earn about 23 percent less than their unbombed neighbors, hold less valuable durable goods, and are 7.6 percentage points more likely to report going hungry (Panel B). For context, about 88 percent of households in the soft-soil sample report being food secure. On hard soil,

income is about 32 percent higher and food security is also higher. The estimate for durable goods is positive but does not reach conventional significance (Panel C). These household results mirror the community-level findings, consistent with the *risk* and *reset* channels reaching individual families.

The field results show how persistent risk may constrain households' livelihoods (Table 8). Its first four columns report outcomes for each field a household cultivates. On soft soil, fields cultivated by households in bombed villages are about 30 percent smaller and generate about 28 percent less revenue than those of their unbombed neighbors. They also lose a larger share of the harvest after it is gathered, with the loss rate rising by 1.2 percentage points against a mean of 2.2 percent. Field investment is lower too, though the estimate is not significant. Together, these results describe farmers working less land, losing more of what they harvest, and earning less from it.

The field results could instead reflect households leaving farming, changing which fields we observe rather than how they are worked. The CSES includes non-farming households, so we can check whether bombing changes the probability that a household grows crops at all. The estimates are close to zero on both soils (Appendix Table G.1), supporting our interpretation of the field results as a contraction in activity among farming households rather than an effect driven by households leaving farming altogether.

Agricultural borrowing also points to constrained investment on soft soil. Among households that borrow, those in bombed villages borrow smaller amounts for agriculture and are less likely to use loans for this purpose, though the latter estimate is significant only at the ten percent level. This is consistent with persistent risk discouraging agricultural investment and the borrowing that finances it. On hard soil, none of the field or borrowing estimates are significant, consistent with the higher household incomes in Table 7 coming from activities outside farming.

Because both tables test several outcomes, we also report Romano–Wolf stepdown adjusted p-values in square brackets (Romano & Wolf 2005), correcting jointly within

Table 8: Field Productivity and Risk Taking Behavior

	Dependent variable is:					
	Field Productivity *Unit: Agricultural Fields*				Risk Taking *Unit: Households*	
	Ln(Area) (1)	Ln(Revenue) (2)	Loss Rate (3)	Investment (4)	Loan Taken (5)	Ln(Loan Amount) (6)
Panel A: All observations						
Bombing	-0.064	-0.106	0.003	-0.018	0.027	0.075
	(0.072)	(0.073)	(0.003)	(0.024)	(0.031)	(0.168)
	[0.818]	[0.679]	[0.818]	[0.818]	[0.510]	[0.691]
Bombing × Soft Soil	-0.204*	-0.116	0.008	-0.072	-0.070	-0.334
	(0.119)	(0.133)	(0.006)	(0.046)	(0.047)	(0.257)
	[0.580]	[0.818]	[0.695]	[0.663]	[0.337]	[0.407]
Mean	8.277	13.537	0.020	0.112	0.262	10.521
Observations	13175	13175	13175	13175	3707	3707
Clusters	602	602	602	602	648	648
Panel B: Soft soil						
Bombing	-0.362***	-0.324***	0.012**	-0.054	-0.073*	-0.446**
	(0.099)	(0.110)	(0.005)	(0.038)	(0.038)	(0.216)
	[0.008]	[0.042]	[0.117]	[0.209]	[0.084]	[0.080]
Mean	8.372	13.791	0.022	0.130	0.298	10.775
Observations	4635	4635	4635	4635	1530	1530
Clusters	236	236	236	236	266	266
Panel C: Hard soil						
Bombing	-0.091	-0.092	0.002	-0.013	0.041	0.129
	(0.071)	(0.071)	(0.002)	(0.023)	(0.030)	(0.162)
	[0.706]	[0.706]	[0.711]	[0.711]	[0.322]	[0.536]
Mean	8.225	13.399	0.018	0.103	0.237	10.343
Observations	8540	8540	8540	8540	2175	2175
Clusters	366	366	366	366	380	380

Note: All regressions include observations living within 2km of a treated-control interface (a bombing boundary). The unit of analysis for the first 4 columns is fields (agricultural land). *Area Cultivated* represents the size of the cultivated area (m^2). *Area Cultivated* is transformed ln(Area Cultivated +1). *Crop Revenue* indicates total revenue from the field: $Revenue = (QuantityHarvested - PostHarvestLoss) \times SalesPrice$. *Crop Revenue* is also log transformed ln(Cambodian riel +10000). *Loss Rate* is the post-harvest loss rate, defined as the ratio of quantity lost to total quantity. *Investment* refers to the number of investments on parcel. The last 2 columns cover CSES households with outstanding loans. *Loan Taken* indicates whether any of these loans were taken for agricultural activities. *Ln(Loan Amount)* is ln(agricultural loan amount in Cambodian riel + 10,000), with the agricultural amount set to zero for households whose loans were all for non-agricultural purposes. All regressions use a local linear polynomial in spatial coordinates with a triangular kernel weight and control for 50x50km grid fixed effects, province fixed effects, distance to the capital, distance to Vietnam borders, and other pre-bombing characteristics. Standard errors reported in parentheses are clustered at the village level. Romano–Wolf stepdown adjusted p-values, based on 9,999 wild cluster bootstrap replications, are reported in square brackets. Within each panel, columns (1)–(4) form the field-productivity outcome family and columns (5)–(6) form the household risk-taking outcome family; all reported coefficients within each family are corrected jointly. ***(**)(*) indicates significance at the 1%(5%)(10%) level based on conventional p-values.

each panel across outcomes that share a unit of observation.[20] The central pattern

[20]We use 9,999 wild cluster bootstrap replications at the village level (Cameron et al. 2008, Roodman et al. 2019), the standard choice when treatment varies only across clusters. This holds the spatial design, controls and fixed-effect structure fixed while generating bootstrap statistics under the null. Each outcome family is estimated on a common sample.

survives the correction. Household income remains lower on soft soil and higher on hard soil at the 10 percent level, while the soft-soil declines in assets, food security, cultivated area and crop revenue remain significant at 5 percent.

Allowing for correlation between nearby villages also preserves the central findings. With commune-clustered or Conley (1999) spatial standard errors, household income remains lower on soft soil and higher on hard soil, while the soft-soil declines in cultivated area and crop revenue remain significant (Appendix Tables G.2 and G.3).

6.6 Agricultural activity and livelihoods

The 2013 Agricultural Census lets us examine how these differences in income and production relate to households' use of land and labor. It does not record yields or revenue, but reports what households grow, the livestock they hold, and how they allocate their land and labor (Appendix Table G.4). Persistent risk should discourage activities that require working contaminated ground. On hard soil, rebuilding could instead create opportunities to produce for markets and work outside farming, consistent with the model's reset interpretation.

On soft soil, the Agricultural Census shows households reducing activities that require access to land. Households in bombed villages hold smaller herds, on the order of 8 percent fewer Tropical Livestock Units, and are 3.3 percentage points less likely to hold an above-median stock. Cattle and buffalo provide draft power for plowing, so smaller herds are consistent with the smaller cultivated fields in Table 8. Keeping them also requires access to the surrounding land. During the growing season, Cambodian cattle and buffalo are tethered near the home and fed grasses gathered from roadsides, paddy bunds and other uncultivated ground. After harvest, they graze on the paddies (Mong et al. 2013). Feeding livestock thus takes households beyond cultivated plots, where clearance has been less extensive. Forest use in bombed soft-soil areas is significantly lower, consistent with the same avoidance of uncleared ground.[21] Nor is there

[21] Cambodia's forests were heavily bombed during the war, and post-conflict clearance has concentrated on cultivated plots rather than forest interiors, leaving forest land among the most UXO-contaminated environments in the country.

evidence that households compensate by moving into non-farm work. Livestock labor falls, while the non-farm labor share shows no significant increase. The pattern is one of reduced land-based activity without an offsetting shift into other work. Allowing for correlation between neighboring villages by clustering standard errors at the commune rather than the village does not change this conclusion (Appendix Table G.5).

On hard soil, by contrast, households in bombed locations show a greater orientation toward markets and non-farm work compared to households in unbombed hard-soil locations. Those in bombed villages are more likely to report sale as the main purpose of production, significant at the ten percent level, and allocate a larger share of their labor to non-farm work, an increase of about 12 percent relative to a mean of 22.6 percent. Livestock holdings fall slightly, while the livestock labor share does not change significantly. This pattern suggests diversification beyond farming rather than merely a reallocation between agricultural activities. It helps explain how the income and food-security gains in Table 7 coexist with little evidence of improved field outcomes. Where bombing left little persistent hazard, the shift toward markets and off-farm livelihoods, rather than a return to pre-war subsistence farming, is consistent with the *reset* channel. Soft-soil households under persistent risk show no comparable shift. Post-war displacement and resettlement could also produce such changes in livelihoods, so we interpret this pattern as suggestive rather than conclusive.

As with the development gains in Section 6.3, movement toward safer soil alone does not explain why bombed hard-soil villages differ from their similarly safe unbombed neighbors. Our migration, population and NGO checks offer little evidence of a lasting concentration in bombed villages, although they cannot rule out earlier displacement and resettlement (Appendix Tables F.3, F.4 and F.2). Investment drawn to destroyed villages could also contribute to reset if it enabled more productive rebuilding. The shift toward markets and non-farm work therefore fits the model's interpretation of recovery and reorganization without identifying capital replacement itself.

6.7 Illustrating the productivity tax

The household and field results show how persistent hazard constrains productive activity. Our framework lets us illustrate how these constraints can accumulate into large long-run differences. The productivity tax τ reduces current output and the savings that finance investment, so its steady-state effect compounds the direct drag.

We calibrate this tax using the difference between the soft- and hard-soil bombing estimates, each measured against unbombed neighbors on the same soil. The exercise assumes a steady-state mapping with a capital share of 1/3 and a common hazard-free rebuilding benchmark. The tax captures the shortfall from that benchmark, including upgrading obstructed by hazards. It also requires the soil advantage to be additive in logs and unchanged by bombing, so differencing removes it. Appendix A.4 sets out the assumptions and outcome-specific calculations.

The illustrative tax ranges from 0.14–0.17 for crop revenue and cultivated area to 0.22–0.30 for durable goods and household income. Under the model's assumptions, these imply steady-state output gaps of about 21–42 percent. These gaps follow from the model rather than provide independent evidence, and the outcomes are different measures of activity and welfare, not four estimates of one output parameter. The exercise illustrates how a persistent drag on production and investment can compound into a much larger long-run shortfall.

7 Conclusion

Why do communities exposed to the same aerial bombing campaign experience diverging long-run outcomes? We study US bombing in Cambodia, using bombing boundaries and pre-bombing soil conditions that shifted the likelihood that ordnance detonated or remained buried. A simple vintage-capital framework predicts that persistent risk leaves bombed communities behind their unbombed neighbors in the long run, while destruction without lingering hazard can let them recover or even pull ahead.

The data confirm this divergence. In soft-soil areas, where bombs were less likely

to detonate, bombed communities face more UXO incidents, worse health, lower economic development, reduced household income, and constrained agricultural productivity, consistent with the *risk* channel. In hard-soil areas, where bombs were more likely to detonate, bombed communities instead show stronger health, higher economic development, and greater household welfare relative to their unbombed neighbors. Destruction without persistent hazard need not block recovery. Some bombed communities even ended up ahead of their unbombed neighbors, a pattern consistent with the model's *reset* interpretation. We trace these effects from community infrastructure through household welfare down to the productivity of individual agricultural fields.

These findings carry three implications. First, the long-run costs of UXO where it persists extend far beyond casualties. Persistent ordnance constrains productive activity and undermines health decades after conflict ends, so evaluations that focus on casualty reduction alone understate the burden. Our model illustrates how these costs can compound through reduced capital accumulation. Where ordnance was more likely to detonate and leave little hazard behind, bombed communities did not fall behind and, in some dimensions, surpassed their unbombed neighbors.[22] Our comparison uses differences in the likelihood of persistent contamination, not exposure to clearance, so for policy it speaks to the cost of a hazard that persists rather than the return to removing one. It does establish that destruction alone need not prevent long-run recovery, which places the persistent hazard left by unexploded ordnance at the center of the long-run cost.

Second, our household and field evidence shows that where UXO persists, farmers borrow less, cultivate less land, and earn less from their crops. Similar responses occur among farmers facing other risks, such as uninsured weather shocks and insecure property rights, in settings far from post-conflict Cambodia (Rosenzweig & Binswanger 1993, Karlan et al. 2014, Goldstein & Udry 2008). Persistent physical hazards thus represent a material risk holding back agricultural productivity.

Third, the distinction between destruction that can be rebuilt and hazards that con-

[22]For related evidence on recovery after bombing and economic gains from reconstruction following destruction, see Miguel & Roland (2011) and Hornbeck & Keniston (2017), respectively.

strain ongoing activity may help explain divergent post-conflict trajectories beyond Cambodia. Many conflicts leave both legacies, destruction in some places and in others, contaminated land or damaged institutions that impose ongoing costs. The balance between the two may shape whether recovery moves ahead or stalls. How fully communities gain from rebuilding can also depend on other conditions being in place. In Cambodia, for instance, the gains from clearing mined land reached local communities only where households held documented claims to that land (Borbely et al. 2026). Conflict itself can shape local accountability (Bühler & Madestam 2024), possibly as part of a reset of institutions. Two broader questions follow. For research, the question is whether the reset consistent with our hard-soil results also operates outside post-conflict settings, wherever vintage constraints may bind. For policy, it is whether similar gains from replacing outdated capital can be achieved without destruction.

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Data Availability

All data used in this study, together with the source code that reproduces every result, will be deposited in a public repository upon acceptance, except for datasets whose terms of use do not permit redistribution. These are the Cambodia Socio-Economic Surveys (2009 and 2014) and the 2013 National Census of Agriculture, which are available for purchase from the National Institute of Statistics, Ministry of Planning, Cambodia (https://nada.nis.gov.kh), and the Cambodia Demographic and Health Surveys (2000, 2005, 2010 and 2014), including their GPS data, which are freely available to registered users from the DHS Program (https://dhsprogram.com). The repository will include the code that processes these datasets, so that researchers who obtain them can reproduce all results. The sources of all data are cited in the manuscript.

Author Contributions

All authors contributed to the study conception and design. Data collection and analysis was performed by T.T.T. and J.N. The first draft of the manuscript was written by T.T.T., whereas all authors worked on later revisions of the manuscript. Writing of the final manuscript was led by J.N., while all authors contributed to the manuscript and the final revisions.

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A Conceptual framework

This appendix sets out the framework of Section 3 in full, followed by the details of the illustrative calibration in Section 6.7. The steady-state notation orders long-run tendencies; it does not describe a state any Cambodian community occupies.

A.1 Vintage capital and lock-in

A community produces $y = Ak^{\alpha}$, where k is capital per capita and A the productivity of the installed stock. A constant fraction s of output is saved and invested and capital depreciates at rate δ, so

$$\dot{k} = sAk^{\alpha} - \delta k. \tag{3}$$

Setting $\dot{k} = 0$ gives

$$k^* = \left(\frac{sA}{\delta}\right)^{\frac{1}{1-\alpha}}, \qquad y^* = A^{\frac{1}{1-\alpha}} \left(\frac{s}{\delta}\right)^{\frac{\alpha}{1-\alpha}}. \tag{4}$$

Steady-state output is proportional to $A^{1/(1-\alpha)}$, an exponent above one because a productivity gain raises output directly and again through the capital it finances.

Let A_o be the technology embodied in old-vintage capital and $A_n > A_o$ the current frontier. Communities on each sit at $k_o^* = (sA_o/\delta)^{1/(1-\alpha)}$ and $k_n^* = (sA_n/\delta)^{1/(1-\alpha)}$, with $k_o^* < k_n^*$. Upgrading is not automatic. Replacing infrastructure that still works means forgoing the output it produces, and where that infrastructure is collective no household captures enough of the gain to bear the cost alone. The community stays at k_o^*.

In the standard Solow model, with technology held fixed, destruction changes the initial capital stock but not the steady state to which the community converges (Davis & Weinstein 2002, Miguel & Roland 2011, Blattman & Miguel 2010). Embodied technology allows rebuilding to change productive capacity as well. Gilchrist & Williams (2004) analyze postwar catch-up in Germany and Japan through investment in capital embodying newer technology. Our reset interpretation adds the assumption that destruction relaxes the constraints keeping unbombed communities at the old technology.

A.2 The productivity tax

Suppose bombing destroys the installed stock. Frontier productivity is the benchmark for hazard-free rebuilding, and $\tau \in [0, 1)$ captures the shortfall due to persistent UXO:

$$y = (1-\tau)A_n k^{\alpha}, \qquad k_\tau^* = \left(\frac{s(1-\tau)A_n}{\delta}\right)^{\frac{1}{1-\alpha}}. \tag{5}$$

The tax is measured against hazard-free rebuilding. Here A_n is the productivity available without ordnance, so τ absorbs both what contaminated communities cannot do with the capital they hold and the upgrading the hazard prevents. The accumulation equation is then $\dot{k} = s(1-\tau)A_n k^{\alpha} - \delta k$.

Two cases follow. At $\tau = 0$ the community converges to $k_n^* > k_o^*$, the reset. At $\tau > 0$ it converges to k_τ^*, which falls below the pre-bombing k_o^* whenever

$$(1-\tau)A_n < A_o, \tag{6}$$

that is whenever the drag exceeds the potential vintage gain. Because the exponent in (4) is above one, a given tax costs more than itself in the long run. At $\alpha = 1/3$ a ten percent reduction in effective output leaves steady-state capital and output about fifteen percent lower, since $0.9^{3/2} = 0.854$.

A.3 Recovering the tax from the estimates

Compare two bombed communities that share A_n and differ only in the hazard. Equations (4) and (5) give

$$\frac{y^*_{\text{risk}}}{y^*_{\text{reset}}} = (1-\tau)^{\frac{1}{1-\alpha}}. \tag{7}$$

The empirical counterpart of the left-hand side in logs is $\hat{\beta}_1^{\text{soft}} - \hat{\beta}_1^{\text{hard}}$, the difference of the two within-soil bombing estimates. Each is measured against its own soil's unbombed communities. Under the assumption that the soil advantage is additive in logs and unchanged by bombing, differencing removes that advantage, as discussed in Appendix A.4. Inverting gives

$$\tau = 1 - \exp\left[(1-\alpha)\left(\hat{\beta}_1^{\text{soft}} - \hat{\beta}_1^{\text{hard}}\right)\right], \tag{8}$$

which is the expression used for the calibration in Section 6.7 at $\alpha = 1/3$.

We take $\alpha = 1/3$ from Gollin (2002), who shows that once self-employment income is treated correctly, factor shares are broadly stable across countries, including low-income ones. The choice matters in a known direction. A higher capital share makes the long-run loss from a given tax larger, through the exponent in (4), but by the same token implies a smaller tax behind any observed gap. At $\alpha = 0.4$ the gaps in Section 6.7 return values about a tenth below those at $\alpha = 1/3$.

A.4 Illustrative calibration

The calibration in Section 6.7 applies Equation (8) to the household and field estimates in Tables 7 and 8. We calibrate the tax against hazard-free rebuilding. The common A_n

is the productivity benchmark available without UXO, not a claim that contaminated communities complete the same upgrades. The tax captures their shortfall from that benchmark, including upgrading obstructed by hazards.

The mapping requires several assumptions. Both communities are evaluated at a steady state and, after allowing for persistent soil differences, share the hazard-free rebuilding benchmark A_n. We take the capital share to be 1/3 and attribute the remaining soil-adjusted gap to the tax. The last assumption carries the most weight, since soft soil is more fertile. Unbombed soft-soil locations out-produce unbombed hard-soil ones by around 0.3 log points on these outcomes. Each within-soil bombing estimate is measured against its own soil's unbombed counterpart. Differencing the estimates therefore removes this level advantage, provided it is additive in logs and does not itself change with bombing. The calibration thus uses an across-soil contrast, even though its inputs are the two within-soil estimates.

With $\alpha = 1/3$, Equation (8) becomes

$$\tau = 1 - \exp\left[\frac{2}{3}\left(\hat{\beta}_1^{\text{soft}} - \hat{\beta}_1^{\text{hard}}\right)\right].$$

Applied to the household and field estimates, it gives $\tau \approx 0.14$ from field-level crop revenue, 0.17 from cultivated area, 0.22 from the value of durable goods, and 0.30 from total household income.[1]

These are illustrative calibrations rather than point estimates. Through Equation (7), the range implies steady-state output gaps of about 21–42 percent at our assumed capital share. These gaps are a derivation from the model, not independent magnitudes. The four outcomes are also not comparable measures of output. Cultivated area is an activity margin measured as ln(area+1), durable goods a stock, and income and revenue enter as $\ln(y + 10{,}000)$. The range therefore indicates the different margins on which the UXO drag may operate rather than providing four estimates of one parameter.

The calibration works off the gap between the soils, but the soft-soil estimates add something on their own. They are negative on every outcome used here, so bombed soft-soil communities end up below their unbombed neighbors rather than merely short of the reset benchmark. That is the ordering in Equation 1.

[1] Both coefficients are taken as estimated, including the field-level hard-soil estimates that do not reach conventional significance; setting those to zero instead would raise the two agricultural figures to 0.19 and 0.21. Income, durables, and revenue enter as $\ln(y + 10{,}000)$ rather than raw log-output, and cultivated area enters as $\ln(\text{area} + 1)$, so the calibration also depends on these transformations.

B Data sources and variable construction

The data sources, geographic links and variable definitions are set out below, followed by descriptive statistics.

B.1 Bombing and the identification of bombing areas

The bombing dataset used in this study was compiled by the Yale University Cambodian Genocide Program and contains 115,273 records for Cambodia, of which 115,262 date from October 1965 to August 1973 and 11 from May 1975 (Cambodian Genocide Program 2005*b*). The data include strike dates and coordinates, aircraft numbers and types, bombing loads, and ordnance types.

We use the strike locations to define bombing areas that preserve their spatial clustering. The approach draws on spatial clustering methods used in geography, public health and ecology (Aldstadt 2010, Grubesic et al. 2014). A location is classified as bombed by whether its grid cell received air-delivered ordnance, not by whether it was heavily bombed.

Spatial clustering methods are widely used to study geographically concentrated events, including in epidemiology (Rushton & Elliott 2003, Elliott & Wartenberg 2004, Beale et al. 2008, Auchincloss et al. 2012). Grid-based methods, also called quadrat analysis, partition the study area into non-overlapping cells and count the events in each (Ankerst et al. 1999). Adjacent occupied cells can then be combined into larger areas (Boots & Getls 1988). The choice of cell size matters, since small cells can produce a noisy representation of exposure, while large cells can over-smooth it (Ankerst et al. 1999, Cheng et al. 2018).

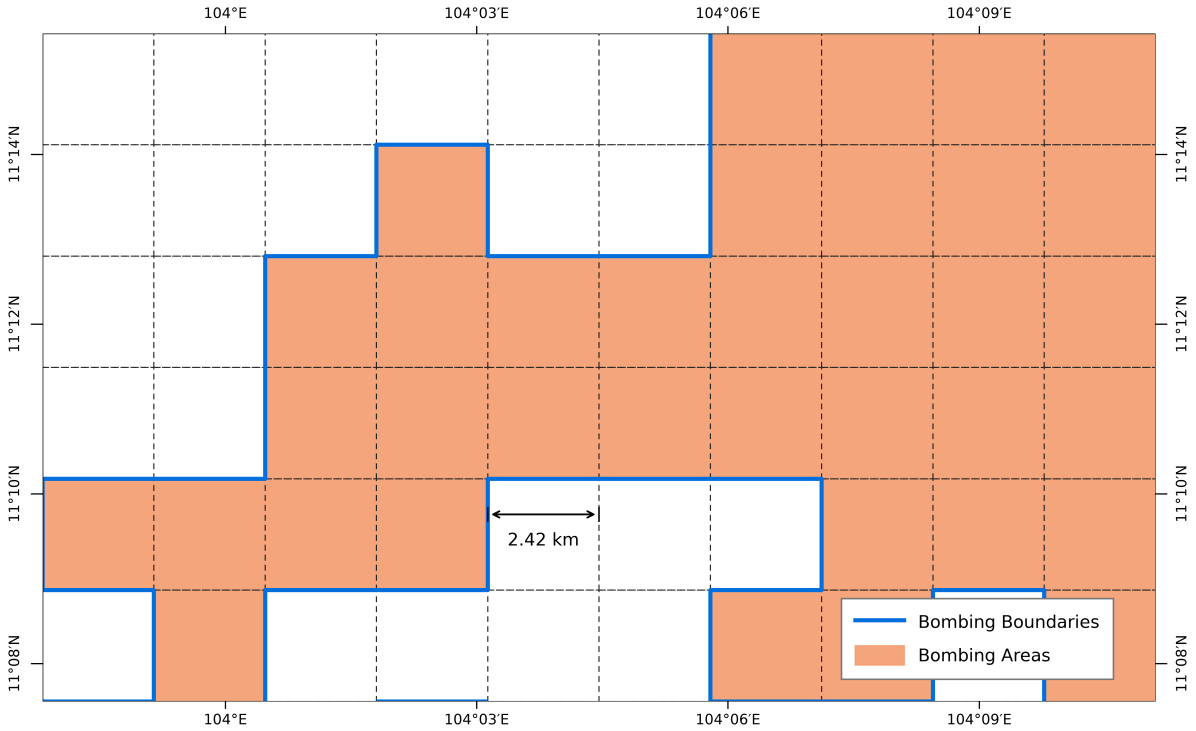
Following Boots & Getls (1988), we set the grid-cell area at twice the study area per recorded strike. The side length is therefore $I = \sqrt{2 \times A/n}$, where I denotes the calculated length of the side of a grid cell, A denotes the area of the focused region, and n denotes the number of features – airstrikes in our case – in the study area.

Each cell covers 5.856 km^2 (2.42 km on each side), using a rectangular analysis extent of 337,561 km^2 and the number of airstrikes being 115,273.[2] In total, the country is divided into 31,799 grid cells.[3] After identifying the bombing loads in each cell, only cells that have bombing loads greater than 0 are selected and defined as bombing areas. Figure B.1 demonstrates the way we construct bombing areas, zooming in on an actual portion of our data, and later how we construct bombing boundaries from the grid cells.

[2]This is similar to the cell size selected by Kohama et al. (2020) who examines how the economic characteristics of conflict zones influence the choice of military strategies.

[3]The full rectangular analysis extent contains approximately 57,600 grid cells. We restrict the grid to the study area by excluding cells outside the national boundary or overlapping permanent water bodies. After applying these exclusions, 31,799 grid cells remained and were used in the analysis.

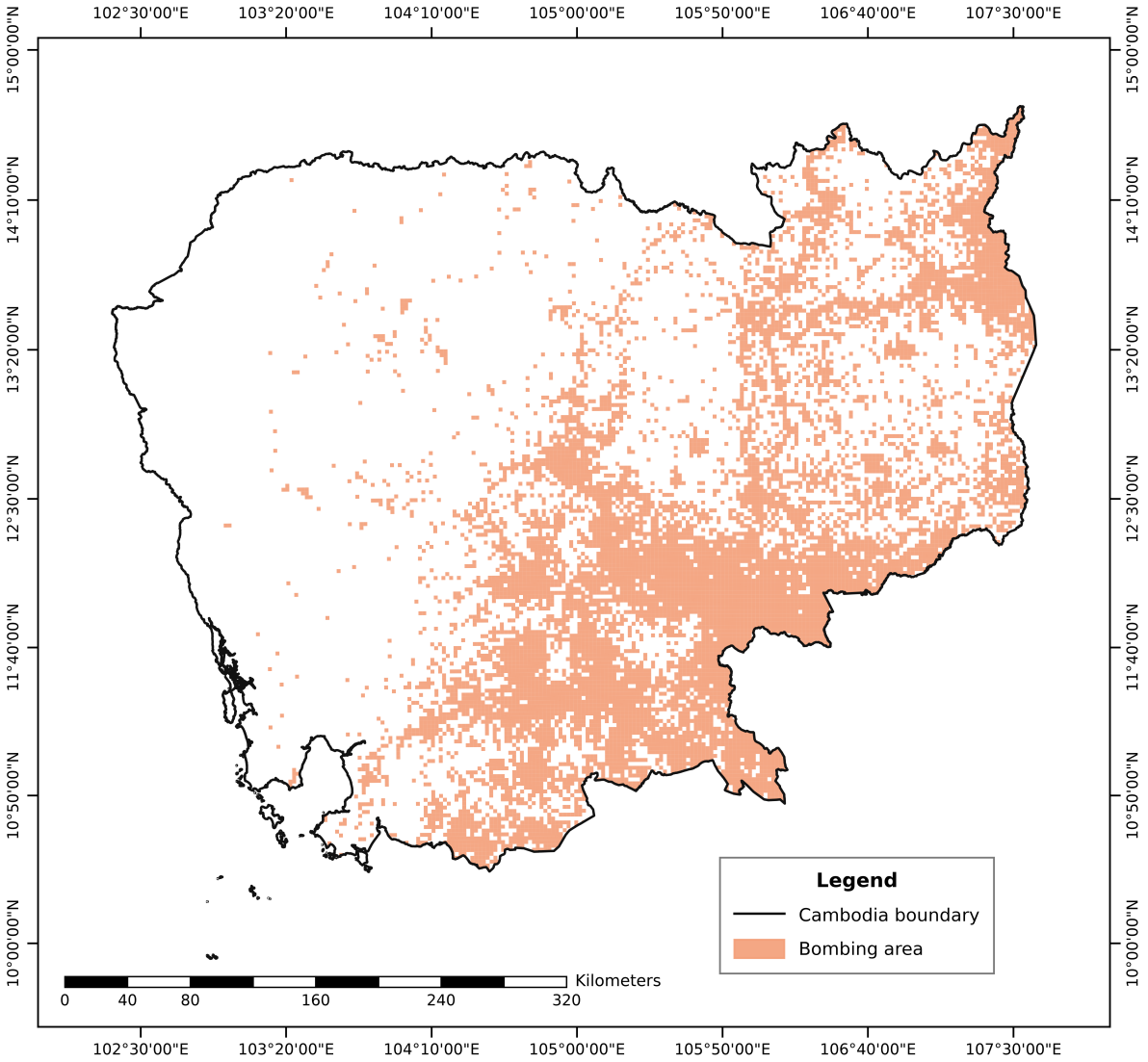
Figure B.1: Bombing boundaries and bombing areas



Notes: We compare nearby locations across bombing boundaries, which separate bombed and unbombed grid cells. A boundary is only the outline where bombed and unbombed cells meet; grid edges interior to either zone are not boundaries.

Figure B.2 then shows the spatial distribution of identified bombing areas in Cambodia, which captures the clustering of strike locations, with the remaining cells classified as unbombed.

Figure B.2: Areas of bombing in Cambodia (1965–1973)



Notes: The map depicts areas exposed to bombing during 1965–1973.

B.2 Outcome data, index construction and geographic linkage

B.2.1 Health Index: Cambodia Demographic and Health Surveys

Our analysis builds upon four waves of the Cambodia Demographic and Health Surveys (DHS 2000, 2005, 2010, and 2014) (National Institute of Statistics et al. 2001, National Institute of Public Health et al. 2006, National Institute of Statistics et al. 2011, 2015).[4] We rely on the DHS individual women's data, as it offers extensive health information that is not available for the male sample. DHS surveys provide a location for each sam-

[4]Demographic and Health Surveys (DHS) Program has conducted six surveys in Cambodia, including CDHS 1998, 2000, 2005, 2010, 2014, and 2021-2022. Data on the exact locations of DHS villages or GPS data is not available in Cambodia DHS 1998. Meanwhile, DHS 2021-2022 does not provide information on anemia level, one of the outcomes analyzed in this study.

pled group of 25–30 households. We refer to these sampling units as DHS villages. To keep respondents' confidentiality, GPS locations of DHS villages are displaced geospatially. Urban points are moved by up to 2 km and rural points by up to 5 km, with a random one percent of rural DHS villages moved up to 10 km. These are bounds rather than typical distances, and the 10 km case applies to one rural village in a hundred. At its bound the rural displacement is larger than the 2 km window we estimate in, so moving a DHS village can change its assigned bombing status, soil type, distance to the boundary or inclusion in the sample. Which of these moves, and in which direction, depends on how the bombed area sits around that village, so the error is not the classical kind and we do not claim it averages out.

The displacement concerns our DHS health outcomes. UXO incidents, the *Economic Development Index*, the CSES household and field outcomes and the 2013 Agricultural Census are all tied to 2008 Census village coordinates, which are exact, and together they show a similar pattern across soil types. The pattern therefore does not rest on displaced data. For the health result itself, the bandwidth checks in Appendix E.2 run from 1 to 5 km and the estimates hold throughout, which is reassuring against a displacement of this size.

We follow Anderson (2008) to generate a standardized weighted summary *health index* of women comprising the following components:

1. *Height-for-age Z-scores (HAZ).* We use the DHS-provided standardized height measure for women (`v440`, divided by 100), which expresses height relative to the DHS reference in standard-deviation units. Adult height reflects nutritional and health conditions during growth (Leroy & Frongillo 2019). The DHS label is height-for-age, although its reference does not vary with age for women aged 18 and above.[5]
2. *Weight-for-height Z-scores (WHZ).* We use the DHS-provided standardized weight-for-height measure for women (`v444a`, divided by 100). This DHS-specific measure expresses weight relative to its reference for height, with adjustments for age and pregnancy. Low values indicate low weight relative to that reference.
3. *Healthy Body Mass Index.* This variable is constructed based on women's Body Mass Index (BMI). BMI has been widely used in literature as an indicator of health status (Kountchou et al. 2019, Conti et al. 2024). Women are considered to have a healthy BMI if their BMI is between 18.5 and 25 kg/m^2 (Weir & Jan 2019).
4. *Healthy Corpulence Index.* Corpulence Index (CI), also known as Ponderal Index

[5]These are the adult women's measures supplied by DHS, not child growth scores calculated using WHO Anthro. See Thomas W. Pullum (2008), *An Assessment of the Quality of Data on Health and Nutrition in the DHS Surveys, 1993–2003*, DHS Methodological Reports No. 6, pp. 74–80.

(PI) or Rohrer's Index, is a measure of body corpulence calculated as weight divided by height cubed, unlike BMI which normalizes by height squared. This adjustment makes the index more appropriate for very short or very tall individuals (Roth 2018). We check the correlation between BMI and CI within our sample and find a moderate positive association (r = 0.367), indicating that while the two measures are related, they capture distinct aspects of body composition, which motivates us to use both indices. We then construct a variable called *Healthy CI*, which is defined as a CI between 11 and 15 (Mohajan & Mohajan 2023).

5. *No moderate or severe anemia*. Anemia is an indicator of inadequate nutrition and overall poor health, often associated with iron deficiency (World Health Organization 2008), that has been used as a health outcome in prior research (Aguilar & Vicarelli 2022, Rosales-Rueda 2018). Anemia status is classified by measuring hemoglobin levels, which are obtained through blood tests conducted by the DHS Program. Based on available DHS data, individuals are categorized into two groups: those with moderate or severe anemia and those with mild or no anemia. Our indicator equals one for the latter group.

By pooling information across multiple outcomes that capture the same latent dimension, a summary index reduces measurement error by averaging out idiosyncratic noise in individual outcome measures. We first recode each variable so that higher values indicate better health. We then standardize each variable and compute a weighted average. The weights are based on the inverse of the covariance matrix of the standardized variables, maximizing the information captured by the index (Anderson 2008).

B.2.2 Post-bombing economic development data

We utilize the following datasets to evaluate post-bombing economic growth and infrastructure development. We begin by using data from the 2008 Population Census and additional infrastructure datasets to construct an Economic Development Index that captures development at the census village level. We then draw on the Cambodia Socio-Economic Surveys from 2009 and 2014 to evaluate households' current levels of economic development.

Economic Development Index. Similar to the variable *Health Index*, we also use Anderson (2008)'s approach to construct a standardized weighted index of economic development based on the following components:

1. *School Locations in Cambodia (2012)*. We use this dataset to calculate the distance to the nearest school, which serves as a proxy for infrastructure development at the census village level. This dataset contains the geographic locations of schools

as well as detailed information on preschools, primary schools, high schools, and colleges.[6] We compute the distance from each census village to the nearest school. To ensure that higher values indicate better economic conditions, we reverse the sign of the distance variable so that larger values correspond to shorter distances to schools and, therefore, better economic development.

2. *Health facilities in Cambodia (2010)*. We use distance to hospitals as a proxy for health infrastructure development and healthcare accessibility. The dataset on health facilities provides comprehensive information on healthcare facilities in Cambodia, including national hospitals, referral hospitals, health centers, and health posts.[7] We then calculate the distances from each census village to the nearest hospital. We reverse the sign of the distance measure so that higher values correspond to shorter distances and thus better economic development.
3. *Radio Station in Cambodia (2012)*. This dataset provides information on radio stations across each province of Cambodia.[8] Based on this dataset, we construct a dummy variable equal to 1 if census villages are located within a big radio network.
4. *Data from 2008 Population Census*. Using data from the 2008 Population Census (National Institute of Statistics 2009*b*, 2011), we construct several indicators to measure development at the village level: (1) the percentage of households having access to improved water sources (including piped water, tube/pipe well, protected dug well and rain water); (2) the percentage of households having electricity as main source of light (including city power, generators, and both); (3) the percentage of households having toilet facility within premises; (4) dwelling tenure status (the proportion of households living in their own dwellings); (5) adult literacy rate (the percentage of literate individuals aged 15 and above among the total population aged 15 and above); (6) school attendance rate (the percentage of individuals (aged 6 and over) who are attending school); and (7) employment rate. Higher values of these indicators reflect higher levels of economic development.

[6] It was originally compiled by the Ministry of Education, Youth and Sports and later shared by the Office for the Coordination of Humanitarian Affairs (OCHA) through the Humanitarian Data Exchange (HDX). Open Development Cambodia (ODC) downloaded the data from HDX https://data.humdata.org in ESRI Shapefile format and converted it into other file formats (Open Development Cambodia n.d.*a*).

[7] Cambodia's Ministry of Health (MoH) originally compiled the data, which was subsequently contributed to the Humanitarian Data Exchange (HDX) by the Office for the Coordination of Humanitarian Affairs (OCHA). Open Development Cambodia's team gathered data from Google Maps and utilized references from Cambodia's Ministry of Health (Open Development Cambodia n.d.*d*).

[8] The data were collected from the fmscan.org website by Open Development Cambodia's mappers and were originally available in image format (Open Development Cambodia n.d.*f*).

Cambodia Socio-Economic Survey 2009 and 2014. To capture households' economic development at present, we use the Cambodia Socio-Economic Survey (CSES) 2009 and 2014 (National Institute of Statistics 2009*a*, 2014), which are nationally representative surveys covering 12,000 households across 720 villages in 2009 and 12,096 households across 1,008 villages in 2014. The surveys contain information about households' economic activities, agricultural and non-agricultural incomes, expenditures, vulnerability to food shortages, field productivity, and risk-taking behaviors. Although CSES 2009 & 2014 do not provide geo-locations of households, we geo-locate survey households by linking each sampled village to the 2008 Population Census using the official administrative village identifier, constructed from the province, district, commune, and village codes. The matched census record provides the geographic coordinates of each village. Because the linkage relies on unique administrative identifiers rather than village names, villages with identical names are distinguished correctly. We selected the 2009 and 2014 waves because they use a roughly equal sampling scheme and size, and they come after the 2008 census. Descriptive statistics for the household and field outcomes are in Table B.4.[9]

2013 Cambodia National Agricultural Census. We complement the CSES with the 2013 Cambodia National Agricultural Census (National Institute of Statistics 2013, National Institute of Statistics & Ministry of Agriculture, Forestry and Fisheries 2015), a full enumeration of agricultural households that we geolocate via the 2008 Population Census village identifiers. The Agricultural Census is structural rather than transactional, thus it does not record harvest yields, sale prices, or crop revenue, which we observe in the CSES. We use the Agricultural Census to construct household-level indicators of crop orientation (share of parcels in commercial crops, whether the main purpose of production is sale), livestock scale (log Tropical Livestock Units and an above-median indicator), forest land use, and the shares of household members whose primary activity is crop cultivation, livestock, or non-farm work.

B.2.3 Unexploded ordnance

Mine/ERW casualties (2005-2013). In addition to the above datasets, we use data on Mine/ERW casualties (2005-2013) to test whether UXO is more prevalent in regions with pre-bombing softer soil. This dataset provides detailed locations of landmine and remnants of war (ERW) incidents. The data also reports the number of victims and

[9]The 2009 CSES sampling frame is based on the 2008 Population Census. The 2014 CSES uses an updated national village sampling frame but retains the official administrative village identifiers.

casualties resulting from these incidents in Cambodia between 2005 and 2013.[10] In our analysis, we focus exclusively on incidents involving ERW likely originating from aerial bombing (CBU submunitions, fuses, and ammunition, other, and unknown) and exclude incidents related to landmines and other ground-delivered explosive weapons.[11] We merge this dataset with 2008 Population Census data. We count the number of incidents within 3km of census village locations and construct a binary outcome equal to 1 if there are any incidents within this 3-km radius.

B.2.4 Khmer Rouge burial sites

Cambodian Genocide Databases To test whether bombing exposure predicts the subsequent violence in Cambodia (i.e, the Khmer Rouge Genocide from April 1975 till January 1979), we use Yale University's Cambodian Genocide Databases, which provide geocoded information on 18,953 Khmer Rouge mass graves across 309 locations (Cambodian Genocide Program 2005*a*). We link these data to villages in the 2008 Population Census. Specifically, for each census village, we count the number of mass graves located within a 3 km radius and construct an indicator equal to one if at least one mass grave is located within 3 km of the village, and zero otherwise.

B.3 Pre-war soil classification

Soil types in 1962. We use the data on the distribution of soil types in Cambodia provided by Crocker (1962) to identify pre-bombing soil fertility.[12] Based on the discussion on soil type characteristics in White et al. (1997) and similar to the classification in Kohama et al. (2020), we create a dummy variable, *Soft Soil*, which equals 1 for villages on soil types classified as fertile and 0 otherwise. The empirical labels soft and hard soil refer to these two categories; mechanically soft and firm soils refer to the separate classification below.[13]

Crocker (1962) carried out an exploratory soil survey in Cambodia, followed by the publication of a general soil map at a 1:1,000,000 scale in 1963 (Kyuma & Kawaguchi 1966). The map classifies soil in Cambodia into 16 different soil types: red-yellow podzols, latosols, planosols, plinthite podzols, cultural hydromorphics, grey hydromorphics,

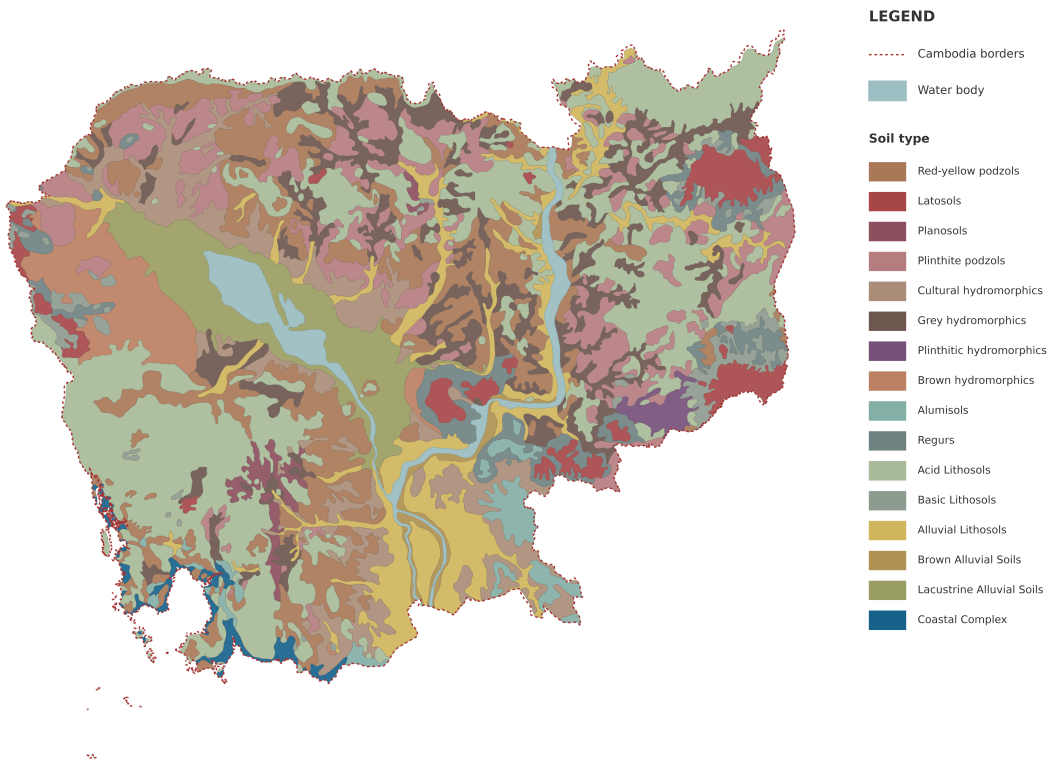
[10]The data was compiled by The Cambodia Mine/ERW Victim Information System (CMVIS) of the Cambodian Mine Action and Victim Assistance Authority (CMAA) and shared via the Office for the Coordination of Humanitarian Affairs (OCHA) on the Humanitarian Data Exchange (HDX) platform (Open Development Cambodia n.d.*e*).

[11]We note that UXO/ERW incidents can include those from separate causes than the US aerial bombing campaign (Roberts 2011, Martin et al. 2019).

[12]The data were provided to Open Development Cambodia in ESRI Shapefile format by Save Cambodia's Wildlife's 2013 Atlas Working Group (Open Development Cambodia n.d.*b*).

[13]Specifically, among sixteen different soil types given in Crocker (1962), the following six types of soils are classified as fertile: Latosols, Alluvial soils, Brown alluvial soils, Lacustrine alluvial soils, Regurs, and Brown hydromorphics (Kohama et al. 2020).

Figure B.3: Soil types in Cambodia (Crocker, 1962)



Notes: The map displays the distribution of soil types in Cambodia, as documented by Crocker (1962). The data, in vector format, were provided to Open Development Cambodia by Save Cambodia's Wildlife's Atlas Working Group (Open Development Cambodia n.d.*b*).

plinthitic hydromorphics, brown hydromorphics, alumisols, regurs, acid lithosols, basic lithosols, alluvial lithosols, brown alluvial soils, lacustrine alluvial soils, and coastal complex (Figure B.3).

White et al. (1997) discuss the characteristics of each soil group in Cambodia and its potential for rice production. For example, Prey Khmer soils correspond to Red-Yellow Podzols and occasionally to Planosols, as identified by Crocker (1962). These soils have low water-holding capacity and limited potential for high rice yields. On the contrary, Kompong Siem Soils which are Regurs in Crocker (1962) are considered fertile and well-suited for rice cultivation, with top soil having good water holding capacity.

Based on the characteristics of each soil type described in White et al. (1997) and similar to the classification in Kohama et al. (2020), we define these six soil types in Crocker (1962) as fertile (soft) soils: Latosols, Alluvial soils, Brown alluvial soils, Lacustrine alluvial soils, Regurs, and Brown hydromorphics. These soils often have good water-holding capacity and are suitable for rice production. Most importantly, since areas with soft or fertile soil are more likely to contain UXO (Moyes et al. 2002,

Lin 2022),[14] we use this classification of pre-bombing soil fertility in our analysis to disentangle the long-term effects of local area exposure to bombing in areas with fertile (soft) soil - a high likelihood of UXO versus areas with infertile (hard) soil - lower UXO occurrences.

B.3.1 Mechanical softness and fertility are different properties

Our soil indicator is built from fertility, while the channel it represents is mechanical. Soft ground reduces a falling bomb's deceleration on impact, making the fuze less likely to trigger and the bomb more likely to remain live (Lin 2022). Fertility is a matter of nutrients, organic matter and water-holding capacity. The two travel together in Cambodia, because both follow landscape position and drainage, but they are not the same thing and the distinction deserves to be made explicitly rather than assumed.

What governs whether a fuze functions is the deceleration the bomb experiences on impact, and that is set by the mechanical properties of the ground: bulk density, cohesion, compressibility, depth to a resistant layer, and above all moisture. The weapons literature summarizes these in a single penetrability index, the S-number of Young (1997), calibrated on several hundred instrumented penetration tests. Penetration depth is linear in S and mean deceleration varies as $1/S$, so S maps directly onto the quantity an inertial fuze responds to. It runs from about 2 for shallow soil over rock to above 30 for saturated clay.

Moisture can increase soil penetrability, although its effect depends on soil type and impact conditions. Penetration resistance depends on both soil compressibility and shear strength. In experiments with a 60-degree conical striker at impact velocities of 100–400 m/s, water-saturated sand exhibited lower penetration resistance than dry sand (Bragov et al. 2018, p. 206, Fig. 12). For clay, Young (1997, Table 4-2) assigns $S = 20$–30 to very soft, saturated clay with very low shear strength, compared with $S = 4$–6 for very stiff, dry clay, where higher S indicates greater penetrability. Although these categories do not constitute a controlled comparison of the same soil across seasons, they motivate the hypothesis that seasonal wetting increases penetrability in clay-rich ground. We examine the implications of this hypothesis in Appendix D.1.

Table B.1 maps Crocker's soil units to modern reference groups (Driessen et al. 2001) and classifies their mechanical properties. We fixed this classification before examining the estimates.

One unit needs comment. The shapefile labels a large unit "Alluvial Lithosols", a name that pulls in two directions: alluvium is deep unconsolidated sediment, while a lithosol is a shallow soil over rock. It is the former. Seng (2015), reporting Crocker's own numbering and areas, lists unit 13 as "Alluvials"; identifying the shapefile unit with

[14] See Section 6.1.

it reproduces his published area for the medium-fertility class to within 1.4 percent, and all three of his fertility-class areas to within 2.6 percent, while refusing the identification leaves that class 45.5 percent short with no other candidate available. The polygons themselves map as narrow dendritic strips following stream courses, together with the lower Mekong plain, which is how alluvium maps and is not how the Acid and Basic Lithosols on the same sheet map. The name is most plausibly a legacy of the 1938 USDA system, in which Lithosols, Regosols and Alluvial soils are the three great groups of the azonal order. This is also how the unit has been read in the fertility literature we follow, where it appears as "Alluvial soils".

Table B.1: Mechanical classification of Cambodian soil types

Soil type (Crocker 1962)	Modern equivalent	Mechanical basis	S	Class	Fertile
Cultural hydromorphics	Hydragric Anthrosols	Puddled paddy over plow pan; hard-sets dry	15–30	Soft	No
Alluvial soils[a]	Fluvisols (Gleyic)	Deep unconsolidated stratified sediment	10–25	Soft	Yes
Lacustrine alluvial soils	Gleyic Fluvisols	Fine lake sediment, inundated 3–5 months	20–40	Soft	Yes
Grey hydromorphics	Gleysols	Saturated, very low undrained shear strength	15–30	Soft	No
Brown hydromorphics	Stagnic Cambisols	Seasonally wet, fine-textured, aerated	10–20	Soft	Yes
Brown alluvial soils	Eutric Fluvisols	Deep unconsolidated, better-drained alluvium	8–20	Soft	Yes
Red-yellow podzols	Acrisols / Lixisols	Free-draining; hard-setting crust; traffic pan	4–9	Firm	No
Plinthite podzols	Plinthosols	Plinthite hardens irreversibly; gravel-rich	2–6	Firm	No
Acid lithosols	Leptosols	Hard rock within 25 cm; high stone content	2–6	Firm	No
Basic lithosols	Mollic Leptosols	Hard rock within 25 cm, stronger bedrock	2–5	Firm	No
Latosols	Ferralsols / Nitisols	Deep, free-draining, never saturated; stable	5–10	Firm	Yes
Regurs	Vertisols	Very hard dry, sticky and plastic wet	4–30	Ambiguous	Yes
Alumisols	Alisols	Hard dry, firm moist, sticky wet; plow pan	5–25	Ambiguous	No
Planosols	Planosols	Coarse horizon over dense clay subsoil	6–10	Ambiguous	No
Coastal complex	Arenosols + Gleysols	A complex: coastal sand and mangrove mud	4–60	Ambiguous	No
Plinthitic hydromorphics	Gleyic Plinthosols	Seasonal saturation over a dense plinthic layer	6–20	Ambiguous	No

[a] Labeled "Alluvial Lithosols" in the shapefile; Crocker's unit 13, discussed in the text.
Notes: S is the penetrability index of Young (1997); penetration depth is linear in S and deceleration varies as $1/S$, so higher S means softer ground and a greater chance the fuze fails. Ranges spanning dry and saturated states reflect seasonal variation. Modern equivalents follow Driessen et al. (2001). *Fertile* records membership of the six-type fertile set of White et al. (1997) and Kohama et al. (2020) used throughout the paper. The Tonle Sap water polygon is not a soil type and is omitted. Types classed *Ambiguous* are those where the evidence does not support a call: Regurs and Alumisols because they swing between states with moisture, the Coastal complex because it combines sand and mud, and Planosols and Plinthitic hydromorphics because contrasting surface and subsoil layers make a single mechanical classification uncertain.

The fertility classification is a useful, but imperfect, proxy for mechanical softness. In our estimation sample, 85 percent of villages classified as fertile lie on mechanically soft soils. However, some soils classified as infertile, particularly hydromorphic soils, are also mechanically soft. This overlap may reduce the estimated difference between the two groups, but it does not necessarily make our estimates conservative.

Fertility and mechanical softness cannot be cleanly separated by soil type because both depend on landscape position and drainage in Cambodia. Soil types where the two properties differ cover few villages: only 131 villages in our estimation sample have fertile but mechanically firm soils. Beyond this sample constraint, the two properties may reinforce each other. Softness affects how much ordnance remains unexploded, while fertility encourages cultivation, bringing households into contact with buried ordnance and increasing its economic cost. Fertility and softness therefore operate at different stages of the same mechanism, rather than necessarily offering competing explanations. This is why we approach the question through the timing of the bombing

instead.

Finally, the mechanical interpretation predicts a seasonal pattern that fertility alone does not readily explain. Wet ground is softer than the same ground when dry, and Cambodia's monsoon is sharp, so the risk channel predicts a stronger first stage where more bombing fell in the wet season. Appendix Table D.1 splits bombed communities at the median share of nearby strikes falling in those months. On soft soil, the higher-share group has an estimated UXO first stage nearly three times as large and a larger economic decline. Health estimates are negative in both groups and close in magnitude. On hard soil the estimates are similar across groups. We do not formally test differences between the seasonal coefficients, but their pattern supports the mechanical channel. Agricultural potential alone offers no clear reason why the consequences of bombing should depend on the season in which bombs fell.

B.4 Pre-war geography and settlement

We exploit several datasets from before the bombing or thereabouts for controls, heterogeneity dimensions, and balance testing.

Other geographic controls. We also control for elevation (Open Development Cambodia n.d.*c*) and distances to the capital and the Vietnamese border, measured at village locations.

1970 Indochina Atlas. We digitize three maps from the Indochina Atlas for location-specific measures of transportation links, agriculture, and population.[15] The Indochina Transportation map depicts the major roads and railways of the country in 1970 (Appendix, Figure B.4). We geo-reference the map and measure the distance from a village to 1970 roads and railways. The Indochina Agriculture map outlines regions where agricultural activities took place (Appendix, Figure B.5). We then match this information with village locations to determine whether agriculture or inland fishing was recorded around each village in 1970. Finally, the Indochina Population map displays population density (persons per square kilometer) across Indochina (Appendix, Figure B.6), enabling us to identify population density at the village level in 1970. These maps precede much of the bombing but do not predate the campaign (Figure 1). We therefore check that the health and economic results remain stable when the 1970 controls are removed (Appendix Table C.7).

[15]These maps were released in October 1970 by the Directorate of Intelligence, Office of Basic and Geographic Intelligence, U.S. Central Intelligence Agency (U.S. Central Intelligence Agency, Directorate of Intelligence 1970).

Global Agro-Ecological Zones (GAEZ). To assess the climate and potential crop productivity at the village level, we use the agro-ecological zones (AEZ) classification developed by the Food and Agriculture Organization of the United Nations (FAO) and the International Institute for Applied Systems Analysis (IIASA) (FAO & IIASA 2021, Fischer et al. 2021). This AEZ classification provides a comprehensive assessment of biophysical resources essential to agricultural production.[16] Based on the Dominant AEZ classification dataset, which is part of the GAEZ v4 Theme 1 Land and Water Resources, we match DHS villages and census villages with their corresponding AEZ zone (Appendix, Figure B.7). Subsequently, we create a binary variable indicating whether a village is located in a grid cell characterized as tropics and lowland.

Pre-war settlement. We use 618,927 building locations from the digitized data of Bühler & Madestam (2024), based on the US Army Map Service's L7016 1:50,000 topographic sheets hosted by Yale's Cambodian Genocide Program (United States Army Map Service 1963–1994). Map information on these sheets runs from 1966 to the early 1970s, so it records settlement at or near the bombing period rather than at a single dated moment. For each census village, we count all buildings within 3km and measure the largest single settlement within 2km, defining a settlement as buildings linked in steps of 150m or less. The 3km radius is the one we use elsewhere in the paper to count UXO incidents, landmines and Khmer Rouge grave sites. We also measure the number of nearby settlements, distance to the nearest building and building spacing. These measures enter the balance tests in Table C.5; Table F.6 splits villages at the median largest-settlement size and total building count within the estimation band. Figure B.8 illustrates the source maps.

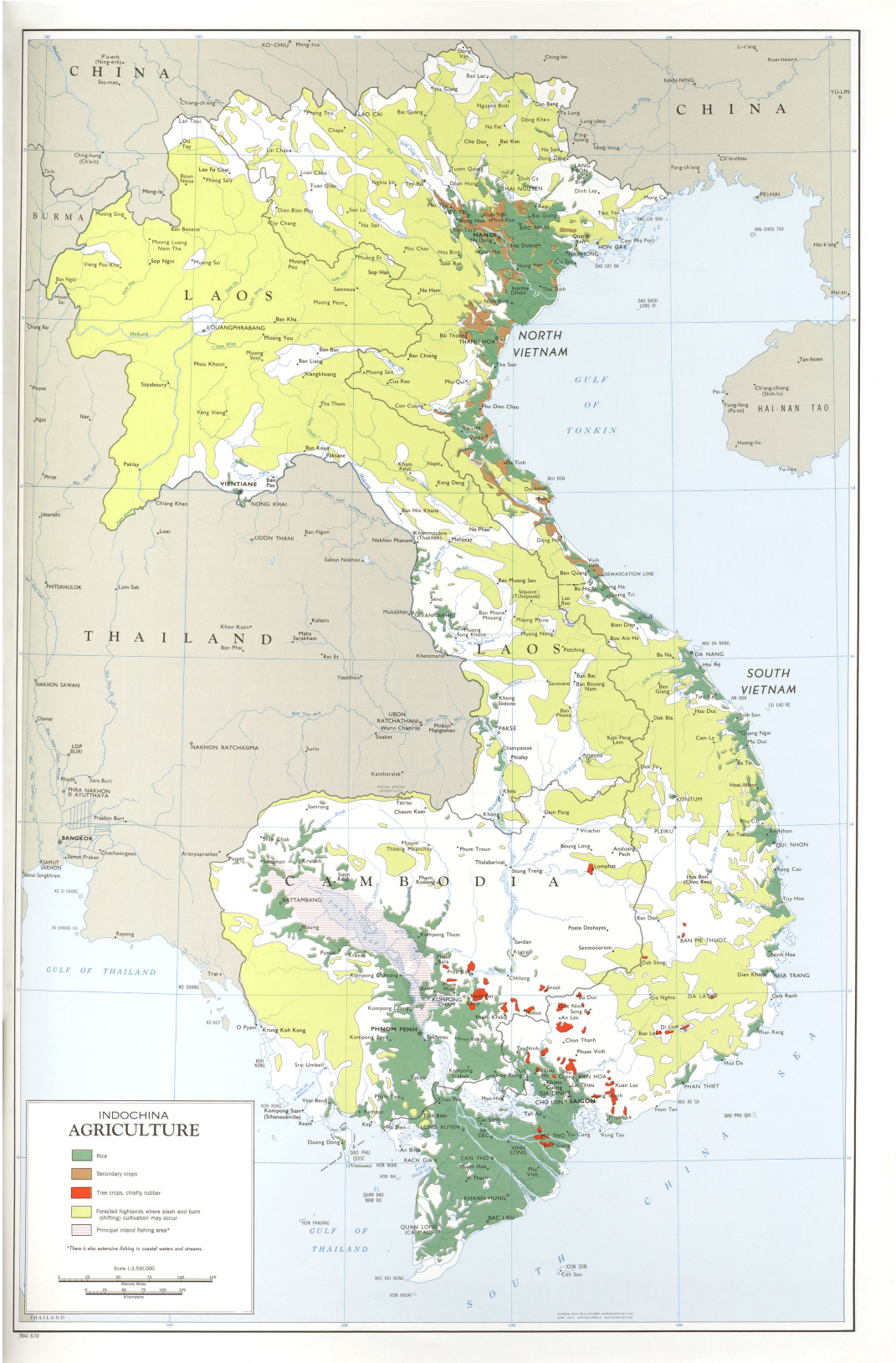
[16] The AEZ map incorporates thermal and moisture regimes, soil/terrain qualities, the presence of irrigated soils, and the identification of areas with significant bio-physical limitations such as extreme cold, arid deserts, steep terrains, and poor soil conditions. Geographical regions classified within the same AEZ category share similar climatic attributes, including rainfall and temperature patterns, which consequently contribute to comparable agricultural potentials. Several studies use GAEZ data to control for geographic characteristics or local agricultural suitability (e.g., Whatley & Gillezeau 2011, Cagé & Rueda 2016). Data can be accessed via https://gaez.fao.org.

Figure B.4: Indochina Transportation in 1970



Notes: Indochina Transportation map, which was in "Indochina Atlas", published in October 1970 by the Directorate of Intelligence, Office of Basic and Geographic Intelligence, U.S. Central Intelligence Agency (U.S. Central Intelligence Agency, Directorate of Intelligence 1970).

Figure B.5: Indochina Agriculture in 1970



Notes: Indochina Agriculture map, which was in "Indochina Atlas", published in October 1970 by the Directorate of Intelligence, Office of Basic and Geographic Intelligence, U.S. Central Intelligence Agency (U.S. Central Intelligence Agency, Directorate of Intelligence 1970).

Figure B.6: Indochina Population in 1970

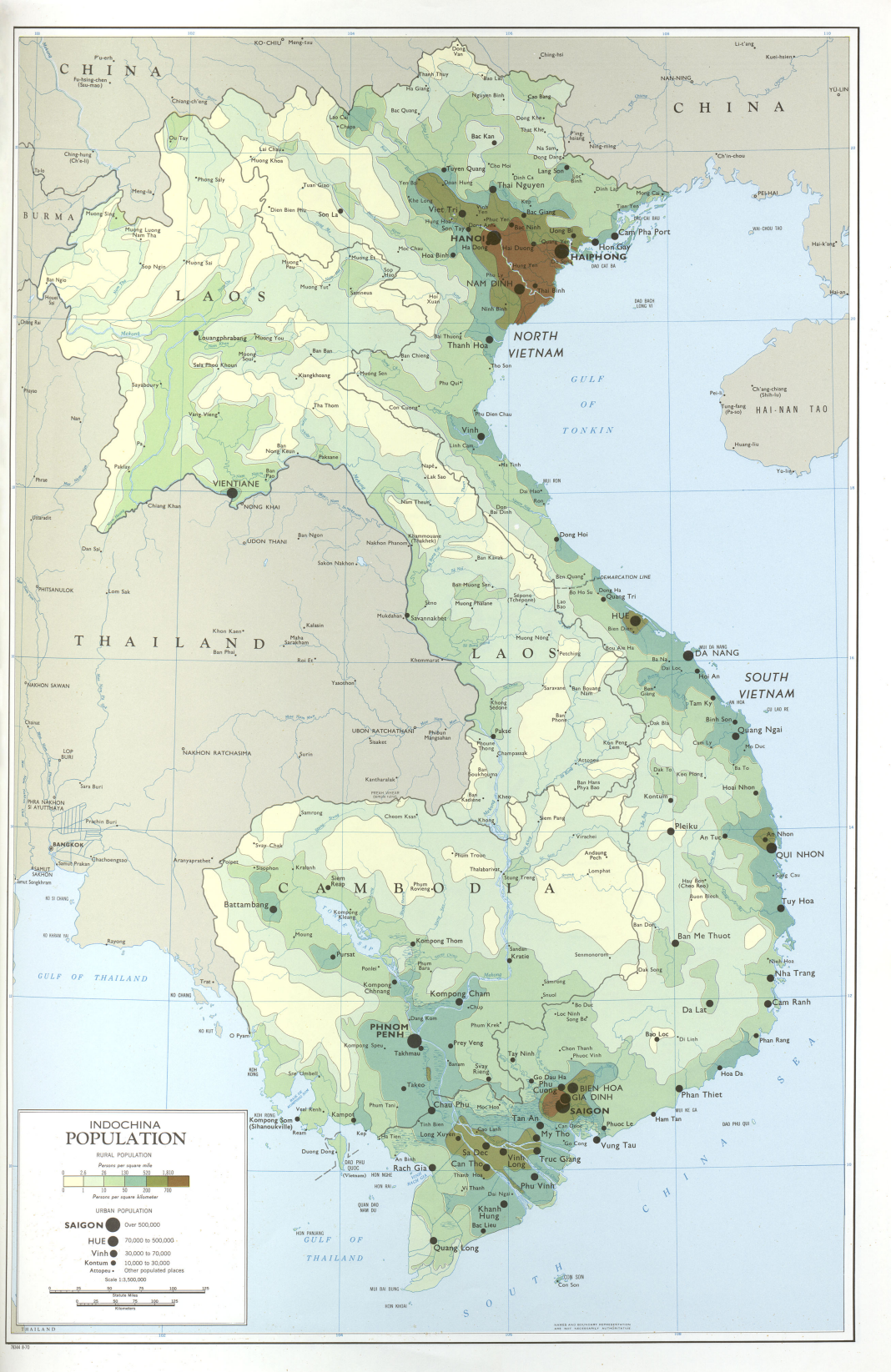
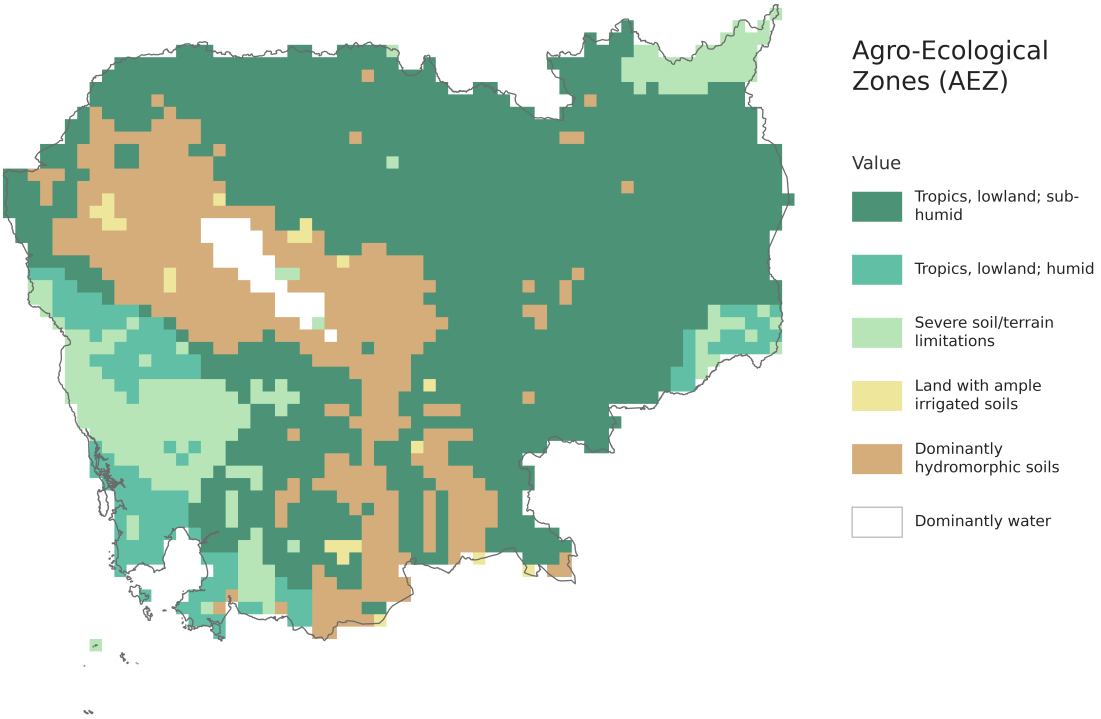
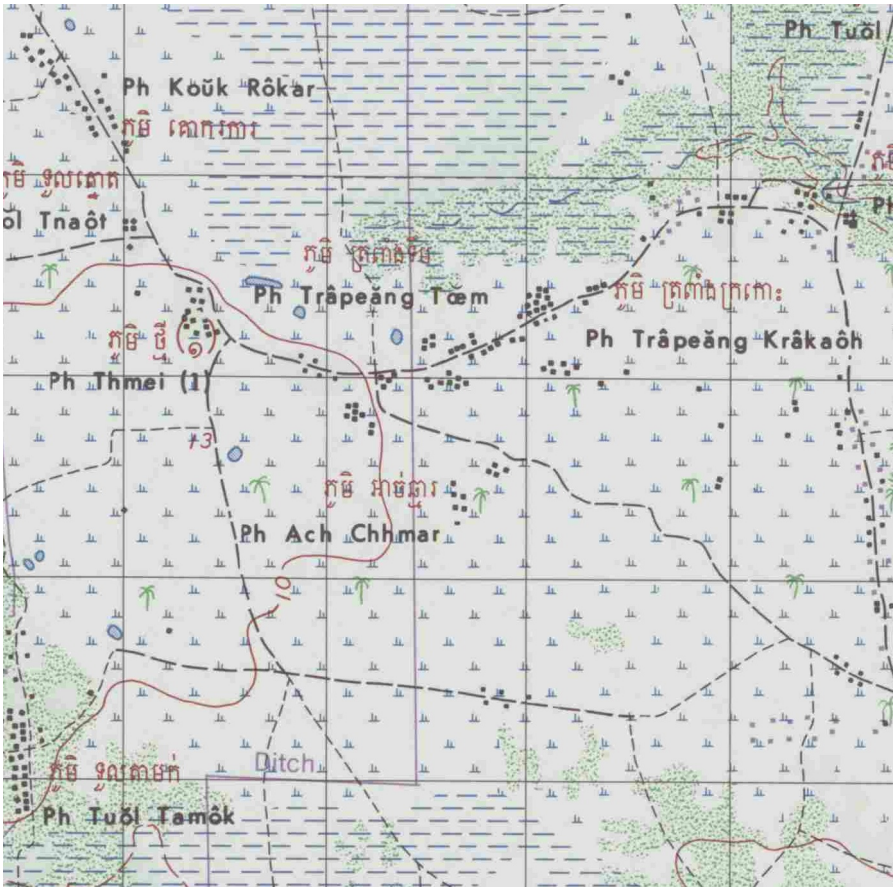

CHINA
CHINA
BURMA
LAOS
NORTH VIETNAM
GULF OF TONKIN
HAI-NAN TAO
THAILAND
LAOS
SOUTH VIETNAM
CAMBODIA
GULF OF THAILAND
SOUTH CHINA SEA
HANOI
HAIPHONG
Cam Pha Port
Hon Gay
NAM DINH
Thanh Hoa
Vinh
VIENTIANE
NONG KHAI
UDON THANI
HUE
DA NANG
QUI NHON
Quang Ngai
Tuy Hoa
Nha Trang
Cam Ranh
Da Lat
Ban Me Thuot
Pleiku
An Khe
Kontum
Hoai Nhon
An Nhon
Phan Thiet
Phan Rang
Vung Tau
BIEN HOA
GIA DINH
SAIGON
My Tho
Tan An
Vinh Long
Can Tho
Rach Gia
Phu Vinh
Khanh Hung
Quang Long
Truc Giang
Chau Phu
Takeo
PHNOM PENH
Kompong Cham
Kompong Thom
Battambang
Pursat
Kompong Chhnang
Kompong Speu
Prey Veng
Svay Rieng
Takhmau
Kompong Som (Shanoukville)
Tay Ninh
Loc Ninh
Song Be
Phuoc Le
Ham Tan
Bao Loc
Di Linh
Hoa Da
Tuyen Quang
Thai Nguyen
Bac Kan
Lang Son
Viet Tri
Hoa Binh
Phu Ly
Thai Binh
Hai Duong
Bac Giang
Bac Ninh
Dien Bien Phu
Son La
Luang Prabang
Sayaboury
Paksane
Thakhek
Savannakhet
Pakse
Attopeu
Saravane
Sekong
Khong
Stung Treng
Kratie
Sisophon
Battambang
Siem Reap
Pursat
Kompong Thom
Kompong Cham
Phnom Penh
Takeo
Kampot
Sihanoukville
Bangkok
Nakhon Ratchasima
Ubon Ratchathani
Surin
Khon Kaen
Udon Thani
Nakhon Phanom
Sakon Nakhon
Phitsanulok
Nakhon Sawan
Lom Sak
Chiang Mai
Chiang Rai
Lampang
Phrae
Nan
Tak
Mukdahan
Roi Et
Yasothon
Ubon Ratchathani
Samrong
Cheom Ksan
Khong
Sem Pang
Virachei
Andoung Pech
Lomphat
Stung Treng
Ban Hans
Phya Bao
Ban Souk Houang
Ban Kataine
Khong
Ban Phone
Pakse
Champassak
Attopeu
Kon Peng Lem
Dak To
Kon Plong
Dak Song
Ban Don
Senmonorom
Snuol
Loc Ninh
Song Be
Chon Thanh
Phuoc Vinh
Go Dau Ha
Phu Cuong
Go Cong
Cai Lay
Cai Be
Cao Lanh
Long Xuyen
Sa Dec
Tri Ton
Thanh Hoa
Vi Thanh
Dai Ngai
Bac Lieu
Ca Mau
Con Son
Hon Panjang
Hon Nghai
Mui Ba Lang
Mui Ke Ga
Dao Phu Quoc
Hon Nhong
Hon Kong
Koh Rong
Koh Kut
Koh Chang
Koh Si Chang
Rayong
Trat
Laem Singh
Ban Ngio
Houei Sai
Ban Pak
Ban Ban
Ban Cheng
Muong Soui
Muong Sai
Muong Sing
Muong Luong Nam Tha
Muong Khoa
Muong Et
Muong Peu
Muong Teu
Muong Yen
Muong Nong
Muong Phalane
Muong Son
Muong Soui
Muong Khoua

Figure B.7: Agro-ecological Zones classes



Notes: The map overlays Cambodia to the agro-ecological zones (AEZs) as classified by The Food and Agriculture Organization of the United Nations (FAO) and the International Institute for Applied Systems Analysis (IIASA). Geographic areas belonging to the same AEZ category exhibit analogous climatic characteristics, encompassing rainfall and temperature patterns, and thus possess equivalent agricultural capabilities (FAO & IIASA 2021).

Figure B.8: A 1:50,000 Sheet at Full Resolution



Notes: Series L7016, "Cambodia 1:50,000", prepared by the US Army Map Service and its successors, which covers the whole country in quarter-degree sheets of roughly 27km square. Shown is the south-west quadrant of 1:100,000 sheet 6132, covering 105.50–105.75°E and 11.50–11.75°N, whose marginalia credit compilation to the Army Map Service in 1967. The window is 4.4km square, reproduced at the scan resolution of 3.17m per pixel. Black squares are the series' *Village* symbol, one per building; blue tick marks are the ricefield symbol; green stipple and tufts are forest, plantation and palm; dashed black lines are cart tracks and footpaths; the straight black lines are the 1km grid; red lines are roads and red lettering is Khmer place names. Nothing is overlaid on the sheet.

B.5 Descriptive statistics

Tables B.2 and B.3 describe the DHS and census-village samples. In the unadjusted DHS comparisons, women inside bombing areas have a slightly lower mean health index. Across the full census sample, bombed villages have lower recorded UXO likelihood and a higher mean economic index, but neither difference is significant within the 2 km band. These pooled descriptive comparisons combine both soil types and do not condition on the geographic controls used in our regressions. Both samples place bombed locations closer to the Vietnamese border on average, consistent with the historical concentration of strikes.

Table B.2: Descriptive statistics *(Unit: DHS women)*

	All observations				Within 2km distance			
	All	Outside	Inside	Diff	All	Outside	Inside	Diff
Panel A: Health outcomes								
Health Index	0.00	0.01	-0.01	0.02***	-0.01	-0.01	-0.02	0.01*
	(0.42)	(0.42)	(0.42)	(4.73)	(0.43)	(0.42)	(0.43)	(2.23)
Height-for-age Z-score	-1.81	-1.79	-1.84	0.05***	-1.83	-1.82	-1.85	0.03*
	(0.89)	(0.88)	(0.89)	(4.94)	(0.89)	(0.88)	(0.90)	(2.05)
Weight-for-height Z-score	-0.96	-0.93	-1.01	0.08***	-1.03	-1.00	-1.06	0.06***
	(0.96)	(0.96)	(0.97)	(6.80)	(0.96)	(0.96)	(0.96)	(3.39)
Healthy BMI ($18.5 \leq BMI < 25$)	0.71	0.71	0.70	0.01	0.71	0.71	0.70	0.00
	(0.46)	(0.45)	(0.46)	(1.65)	(0.46)	(0.46)	(0.46)	(0.23)
Healthy Corpulence Index ($11 \leq CI < 15$)	0.69	0.69	0.70	-0.01	0.71	0.71	0.71	-0.01
	(0.46)	(0.46)	(0.46)	(-1.53)	(0.45)	(0.46)	(0.45)	(-0.84)
No moderate/severe anemia	0.91	0.90	0.91	-0.01*	0.91	0.90	0.91	-0.00
	(0.29)	(0.29)	(0.28)	(-2.27)	(0.29)	(0.29)	(0.29)	(-0.69)
Panel B: Pre-bombing characteristics								
Soft Soil	0.37	0.35	0.40	-0.05***	0.36	0.33	0.39	-0.06***
	(0.48)	(0.48)	(0.49)	(-12.14)	(0.48)	(0.47)	(0.49)	(-12.10)
Elevation/Altitude (meters)	42.22	37.83	47.73	-9.90***	40.76	36.61	44.98	-8.37***
	(76.74)	(59.67)	(93.53)	(-15.95)	(79.76)	(60.80)	(95.07)	(-9.42)
Tropics, lowland	0.55	0.49	0.62	-0.13***	0.60	0.55	0.65	-0.10***
	(0.50)	(0.50)	(0.48)	(-33.75)	(0.49)	(0.50)	(0.48)	(-18.15)
Agricultural activities (1970)	0.55	0.53	0.58	-0.05***	0.61	0.62	0.59	0.03***
	(0.50)	(0.50)	(0.49)	(-13.46)	(0.49)	(0.49)	(0.49)	(4.78)
Population density (1970)	0.75	0.68	0.83	-0.15***	0.80	0.80	0.81	-0.01
	(0.43)	(0.47)	(0.37)	(-46.80)	(0.40)	(0.40)	(0.40)	(-1.15)
Distance (km) to roads/railways (1970)	6.51	7.42	5.36	2.05***	6.32	6.26	6.38	-0.12
	(8.39)	(9.72)	(6.13)	(33.56)	(6.90)	(7.04)	(6.76)	(-1.55)
Panel C: Other characteristics								
Distance (km) to								
Vietnam borders	135.35	188.78	67.90	120.88***	89.97	105.33	74.25	31.08***
	(116.53)	(124.39)	(55.09)	(169.52)	(75.46)	(84.46)	(61.11)	(38.07)
Capital	135.45	160.72	103.55	57.17***	102.69	104.84	100.48	4.37***
	(101.41)	(103.31)	(89.28)	(77.37)	(89.26)	(88.73)	(89.74)	(4.41)
Thai borders	171.82	132.36	221.61	-89.25***	201.36	189.13	213.87	-24.73***
	(83.50)	(81.45)	(54.34)	(-170.85)	(60.40)	(61.61)	(56.48)	(-37.74)
Observations	68506	38423	30083	68506	32495	16435	16060	32495

Note: The table reports means with standard deviations in parentheses, except in the difference columns, where parentheses contain t-statistics. "All" means the whole sample, "Outside" means the sample includes observations located outside bombing areas, and "Inside" means the sample includes observations located inside bombing areas. "Within 2 km distance" means the sample is restricted to observations located within 2 km of bombing boundaries. "Diff" is the outside mean minus the inside mean. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table B.3: Descriptive statistics *(Unit: Census Villages)*

	All observations				Within 2km distance			
	All	Outside	Inside	Diff	All	Outside	Inside	Diff
Panel A: Unexploded Ordnance (UXO)								
UXO Likelihood	0.09	0.11	0.07	0.05***	0.07	0.07	0.06	0.01
	(0.29)	(0.32)	(0.25)	(9.25)	(0.25)	(0.25)	(0.24)	(0.89)
Number of incidents	0.11	0.15	0.08	0.08***	0.08	0.08	0.07	0.01
	(0.41)	(0.49)	(0.30)	(10.88)	(0.31)	(0.32)	(0.30)	(1.19)
Panel B: Post-bombing economic outcomes								
Economic Development Index	0.00	-0.03	0.03	-0.06***	0.02	0.02	0.02	-0.00
	(0.36)	(0.37)	(0.35)	(-9.17)	(0.32)	(0.33)	(0.31)	(-0.54)
Distance (km) to								
School (2012)	0.91	0.92	0.89	0.03*	0.92	0.91	0.92	-0.01
	(0.99)	(1.19)	(0.75)	(1.97)	(0.87)	(0.92)	(0.83)	(-0.62)
Hospital (2010)	14.14	15.73	12.66	3.07***	13.78	14.15	13.49	0.66*
	(11.87)	(13.68)	(9.64)	(15.31)	(11.80)	(12.92)	(10.83)	(2.27)
Radio Station (big network)	0.89	0.86	0.91	-0.06***	0.92	0.91	0.92	-0.01
	(0.32)	(0.35)	(0.28)	(-10.53)	(0.28)	(0.28)	(0.27)	(-1.00)
Percentage of households having								
Improved Water	49.67	46.68	52.46	-5.78***	52.63	52.68	52.59	0.09
	(38.14)	(38.46)	(37.62)	(-9.00)	(38.93)	(39.27)	(38.66)	(0.09)
Electricity	16.25	19.62	13.09	6.53***	14.07	17.94	11.03	6.91***
	(29.22)	(32.83)	(24.97)	(13.21)	(27.66)	(32.33)	(22.88)	(10.07)
Toilet within premises	25.57	27.62	23.65	3.96***	23.86	26.06	22.12	3.94***
	(28.42)	(31.42)	(25.13)	(8.23)	(27.88)	(31.02)	(25.00)	(5.74)
Own dwellings	94.87	94.02	95.67	-1.65***	95.53	94.81	96.09	-1.28***
	(8.98)	(8.67)	(9.20)	(-10.92)	(7.88)	(8.76)	(7.07)	(-6.62)
Literacy Rate	73.86	72.56	75.08	-2.52***	74.53	74.11	74.87	-0.76
	(17.43)	(18.29)	(16.50)	(-8.58)	(17.15)	(17.86)	(16.57)	(-1.82)
School Attendance Rate	28.98	28.62	29.33	-0.71***	29.61	29.36	29.81	-0.45**
	(7.05)	(7.24)	(6.84)	(-5.94)	(7.02)	(7.22)	(6.86)	(-2.64)
Employment Rate	98.79	98.49	99.07	-0.58***	98.92	98.65	99.14	-0.49***
	(3.09)	(3.53)	(2.58)	(-11.14)	(2.87)	(3.38)	(2.38)	(-6.81)
Panel C: Pre-bombing characteristics								
Soft Soil	0.39	0.39	0.39	0.00	0.36	0.36	0.36	-0.00
	(0.49)	(0.49)	(0.49)	(0.10)	(0.48)	(0.48)	(0.48)	(-0.39)
Elevation/Altitude (meters)	34.01	36.66	31.53	5.12***	31.18	30.37	31.83	-1.46
	(55.35)	(55.82)	(54.80)	(5.49)	(56.00)	(48.48)	(61.28)	(-1.11)
Tropics, lowland	0.52	0.46	0.57	-0.11***	0.53	0.49	0.55	-0.06***
	(0.50)	(0.50)	(0.50)	(-13.00)	(0.50)	(0.50)	(0.50)	(-4.63)
Agricultural activities (1970)	0.64	0.60	0.69	-0.09***	0.70	0.70	0.70	0.00
	(0.48)	(0.49)	(0.46)	(-11.38)	(0.46)	(0.46)	(0.46)	(0.31)
Population density (1970)	0.79	0.69	0.89	-0.21***	0.86	0.85	0.87	-0.02*
	(0.40)	(0.46)	(0.31)	(-31.04)	(0.35)	(0.36)	(0.34)	(-2.43)
Distance (km) to roads/railways (1970)	7.76	9.29	6.33	2.96***	7.46	7.62	7.33	0.29
	(8.90)	(10.63)	(6.60)	(19.69)	(7.37)	(7.68)	(7.12)	(1.65)
Panel D: Other characteristics								
Distance (km) to								
Vietnam borders	123.87	190.26	61.80	128.46***	80.57	95.65	68.68	26.97***
	(114.78)	(127.67)	(47.75)	(78.03)	(66.91)	(77.61)	(54.22)	(16.42)
Capital	113.19	153.75	75.26	78.49***	81.74	85.94	78.43	7.50***
	(95.44)	(104.87)	(66.07)	(52.71)	(71.33)	(74.78)	(68.31)	(4.33)
Thai borders	184.52	138.06	227.96	-89.90***	210.42	198.19	220.06	-21.87***
	(81.82)	(83.73)	(49.96)	(-76.69)	(56.90)	(59.07)	(53.19)	(-16.08)
Observations	14073	6800	7273	14073	7014	3091	3923	7014

Note: The table reports means with standard deviations in parentheses, except in the difference columns, where parentheses contain t-statistics. "All" means the whole sample, "Outside" means the sample includes observations located outside bombing areas, and "Inside" means the sample includes observations located inside bombing areas. "Within 2 km distance" means the sample is restricted to observations located within 2 km of bombing boundaries. "Diff" is the outside mean minus the inside mean. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table B.4: Descriptive statistics *(Unit: CSES households and agricultural fields)*

	All observations				Within 2km distance			
	All	Outside	Inside	Diff	All	Outside	Inside	Diff
Panel A: Households								
Ln(Income)	14.09	14.03	14.15	-0.12***	14.05	13.95	14.15	-0.20***
	(2.24)	(2.40)	(2.05)	(-4.20)	(2.30)	(2.47)	(2.11)	(-4.67)
Ln(Expenditure)	16.40	16.45	16.33	0.12***	16.41	16.48	16.34	0.14***
	(0.65)	(0.68)	(0.61)	(13.86)	(0.67)	(0.71)	(0.63)	(10.76)
Ln(Durable goods)	14.04	14.19	13.88	0.31***	14.10	14.26	13.95	0.30***
	(1.86)	(1.94)	(1.76)	(12.66)	(1.85)	(1.91)	(1.78)	(8.69)
Food secure	0.86	0.87	0.84	0.03***	0.86	0.87	0.85	0.02**
	(0.35)	(0.33)	(0.37)	(7.16)	(0.34)	(0.34)	(0.35)	(2.61)
Observations	22625	11790	10835	22625	11136	5520	5616	11136
Panel B: Agricultural borrowing								
Took an agricultural loan	0.27	0.29	0.25	0.04***	0.26	0.26	0.27	-0.01
	(0.44)	(0.45)	(0.43)	(4.03)	(0.44)	(0.44)	(0.44)	(-0.50)
Ln(Loan amount)	10.54	10.67	10.42	0.25***	10.52	10.53	10.52	0.01
	(2.30)	(2.40)	(2.20)	(4.70)	(2.29)	(2.33)	(2.26)	(0.19)
Observations	7747	3891	3856	7747	3707	1740	1967	3707
Panel C: Agricultural fields								
Ln(Cultivated area)	8.32	8.49	8.18	0.31***	8.27	8.37	8.19	0.17***
	(1.27)	(1.28)	(1.24)	(20.54)	(1.20)	(1.14)	(1.24)	(8.60)
Ln(Crop revenue)	13.28	13.46	13.13	0.32***	13.33	13.45	13.24	0.21***
	(1.63)	(1.57)	(1.67)	(16.73)	(1.49)	(1.42)	(1.55)	(8.41)
Post-harvest loss rate	0.02	0.02	0.02	-0.00*	0.02	0.02	0.02	-0.00***
	(0.06)	(0.05)	(0.06)	(-2.47)	(0.05)	(0.04)	(0.06)	(-4.15)
Field investment	0.10	0.10	0.10	-0.00	0.12	0.12	0.12	-0.00
	(0.33)	(0.31)	(0.34)	(-0.78)	(0.37)	(0.34)	(0.40)	(-0.10)
Observations	28106	12762	15344	28106	13856	6153	7703	13856

Note: The table reports means with standard deviations in parentheses, except in the difference columns, where parentheses contain t-statistics. Panel A describes CSES households in 2009 and 2014, Panel B agricultural borrowing among borrowers, and Panel C cultivated fields. These describe the data underlying Tables 7 and 8; those regressions additionally restrict each outcome family to a common estimation sample. Income, expenditure, durable goods, crop revenue and loan amounts enter as $\ln(y + 10{,}000)$. "All" means the whole sample, "Outside" means the sample includes observations located outside bombing areas, and "Inside" means the sample includes observations located inside bombing areas. "Within 2 km distance" means the sample is restricted to observations located within 2 km of bombing boundaries. "Diff" is the outside mean minus the inside mean. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

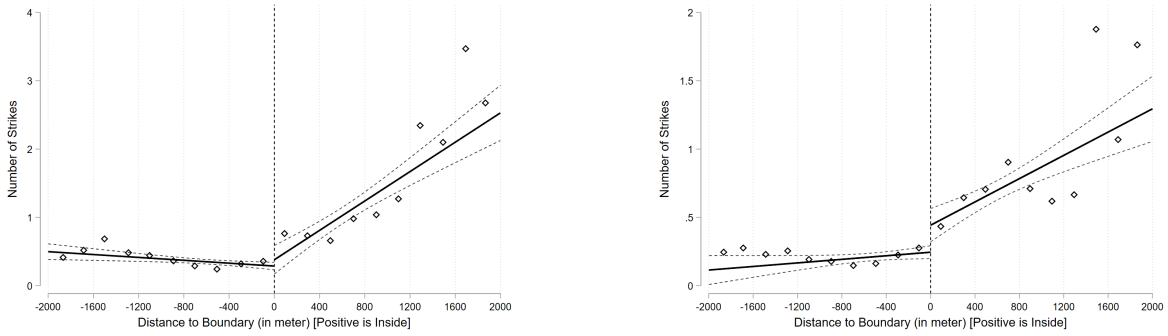
C Empirical design and validity

These checks concern the boundary comparison: bombing exposure, pre-war balance, specification choices and spillovers. Checks covering both health and economic development are kept together here, alongside spatial inference for the principal outcomes.

C.1 Bombing exposure at the boundary

Our treatment is defined from the grid, so we check that the boundary marks a real change in bombing using a measure the grid plays no part in: the number of recorded airstrikes within a fixed radius of each census village, counted from the strike coordinates. Table C.1 reports the discontinuity at five radii and Figure C.1 plots the 500m measure in each soil type. The discontinuity is significant at the 1 percent level everywhere, in both soil types. Its size relative to average exposure falls steadily as the disc widens, from 106 percent at 500 m to 52 percent at 3 km, as wider buffers increasingly count strikes on both sides of the boundary and dilute the exposure contrast.

Figure C.1: Bombing exposure at the boundary: strikes within 500m



(a) Soft soil (b) Hard soil

Note: Exposure is the number of recorded airstrikes within 500m of each census village, counted from strike coordinates rather than from the grid the boundary is drawn on. Points are binned residuals from a regression of the exposure measure on a linear polynomial in spatial coordinates, 50x50km grid fixed effects, province fixed effects and pre-bombing characteristics, weighted by a triangular kernel, with the sample mean added back so the axis reads in strikes. Solid lines are local linear regressions estimated separately on each side of the boundary, dashed lines 95% confidence intervals. Negative distances are locations outside the bombing area. Exposure is flat outside the boundary, steps up at it, and then climbs with depth into the bombed area. Because the fitted lines take a slope while the estimating equation has no term in distance, the discontinuity to read is the one in Table C.1. Table C.6 tests whether bombing near unbombed locations affects their outcomes.

Table C.1: Bombing exposure at the boundary, by buffer radius

	All areas (1)	Soft soil (2)	Hard soil (3)
500 m buffer			
Bombing	0.600***	0.770***	0.526***
	(0.040)	(0.081)	(0.048)
Mean exposure	0.57	0.73	0.47
Discontinuity / mean	106%	105%	111%
1 km buffer			
Bombing	2.213***	3.009***	1.829***
	(0.144)	(0.318)	(0.145)
Mean exposure	2.45	3.15	2.05
Discontinuity / mean	90%	95%	89%
1.5 km buffer			
Bombing	4.414***	5.878***	3.696***
	(0.306)	(0.733)	(0.279)
Mean exposure	5.50	6.94	4.69
Discontinuity / mean	80%	85%	79%
2 km buffer			
Bombing	6.828***	9.057***	5.713***
	(0.505)	(1.232)	(0.451)
Mean exposure	9.53	11.64	8.34
Discontinuity / mean	72%	78%	68%
3 km buffer			
Bombing	11.241***	15.107***	9.129***
	(1.063)	(2.626)	(0.982)
Mean exposure	21.61	25.20	19.60
Discontinuity / mean	52%	60%	47%

Note: The dependent variable is the number of recorded airstrikes within the stated radius of a census village, counted from strike coordinates rather than from the grid that defines the bombing boundary. Each panel is a separate set of regressions, reporting the coefficient on *Bombing* with its standard error beneath, then mean exposure among villages in the estimation band and the coefficient as a share of that mean. Column 1 pools both soils and includes a soil main effect; columns 2 and 3 split the sample. The sample is restricted to villages within 2km of a treated–control interface. All regressions use a local linear polynomial in spatial coordinates with a triangular kernel weight and control for 50x50km grid fixed effects, province fixed effects, distance to the capital, distance to the Vietnamese border and other pre-bombing characteristics. Standard errors reported in parentheses are clustered at the commune level. ***(**)(*) indicates significance at the 1%(5%)(10%) level.

C.2 Balance in pre-war characteristics and settlement

Table C.2: Balance check - DHS villages

	Dependent variable is:						
	(1) Elevation	(2) Tropics/lowland	(3) Econ. Activity	(4) Pop. Density	(5) Dist. to roads	(6) Dist. VN	(7) Dist. to capital
Panel A: All observations							
Bombing	-3.046	0.031	0.047	-0.010	0.083	-0.152	-0.342
	(2.208)	(0.039)	(0.036)	(0.015)	(0.442)	(0.489)	(0.965)
Bombing × Soft Soil	12.073**	0.049	-0.135**	0.007	-0.081	0.222	-0.466
	(5.250)	(0.065)	(0.064)	(0.028)	(0.776)	(0.825)	(1.533)
Mean	39.510	0.599	0.607	0.812	6.467	88.342	101.714
$\|\beta\|$/Mean	0.306	0.082	0.222	0.009	0.013	0.003	0.005
Observations	1051	1053	1053	1053	1053	1053	1053
Clusters	611	611	611	611	611	611	611
Panel B: Soft soil							
Bombing	9.951	0.080	-0.076	-0.024	-0.167	-0.140	-0.160
	(6.120)	(0.051)	(0.051)	(0.023)	(0.638)	(0.683)	(1.143)
Mean	46.150	0.559	0.656	0.814	6.020	74.872	88.164
$\|\beta\|$/Mean	0.216	0.144	0.116	0.029	0.028	0.002	0.002
Observations	381	381	381	381	381	381	381
Clusters	252	252	252	252	252	252	252
Panel C: Hard soil							
Bombing	-2.277	0.025	0.056	-0.009	0.089	-0.092	-0.131
	(2.030)	(0.039)	(0.036)	(0.015)	(0.444)	(0.460)	(0.982)
Mean	35.752	0.622	0.579	0.810	6.719	95.946	109.362
$\|\beta\|$/Mean	0.064	0.040	0.097	0.011	0.013	0.001	0.001
Observations	670	672	672	672	672	672	672
Clusters	418	418	418	418	418	418	418

Note: The unit of analysis is DHS villages. The sample is restricted to villages within 2km of a treated-control interface (a bombing boundary). All regressions use a local linear polynomial in spatial coordinates with a triangular kernel weight. We control for 50x50km grid fixed effects and province fixed effects in all regressions. *Tropics/lowland* is a dummy variable indicating whether this location falls within tropical lowland areas classified as humid or sub-humid in the agro-ecological zones classification. Columns (3) (4) (5) use data from the Indochina Atlas, published in October 1970. *Econ. Activity* indicates whether there was any agricultural activity or inland fishing in these areas in 1970. *Pop. Density* is a binary variable reflecting if the population density in 1970 was at least 10 inhabitants per square kilometer. *Dist. to roads* refers to distance (in km) to 1970 main roads/railways. *Dist. VN* indicates distance to Vietnam border. *Dist. to capital* refers to distance to the capital of Cambodia. Standard errors reported in parentheses are clustered at the commune level. ***(**)(*) indicates significance at the 1%(5%)(10%) level.

Table C.3: Balance check - Population Census Villages

	Dependent variable is:						
	(1) Elevation	(2) Tropics/lowland	(3) Econ. Activity	(4) Pop. Density	(5) Dist. to roads	(6) Dist. VN	(7) Dist. to capital
Panel A: All observations							
Bombing	-0.924	-0.020	-0.001	0.004	-0.006	0.231	-0.616
	(0.642)	(0.016)	(0.015)	(0.008)	(0.215)	(0.191)	(0.419)
Bombing × Soft Soil	2.200*	0.074***	-0.026	-0.023*	-0.390	-0.241	1.254*
	(1.130)	(0.029)	(0.025)	(0.012)	(0.364)	(0.340)	(0.691)
Mean	31.183	0.525	0.700	0.861	7.456	80.567	81.738
$\|\beta\|$/Mean	0.071	0.142	0.037	0.027	0.052	0.003	0.015
Observations	7013	7014	7014	7014	7014	7014	7014
Clusters	1105	1105	1105	1105	1105	1105	1105
Panel B: Soft soil							
Bombing	0.023	0.056**	-0.026	-0.017**	-0.316	-0.159	0.601
	(0.785)	(0.025)	(0.021)	(0.007)	(0.286)	(0.247)	(0.498)
Mean	31.023	0.440	0.710	0.859	7.393	76.766	76.621
$\|\beta\|$/Mean	0.001	0.127	0.037	0.020	0.043	0.002	0.008
Observations	2523	2523	2523	2523	2523	2523	2523
Clusters	560	560	560	560	560	560	560
Panel C: Hard soil							
Bombing	-0.253	-0.010	-0.008	0.007	-0.029	0.031	-0.679*
	(0.608)	(0.016)	(0.015)	(0.008)	(0.202)	(0.177)	(0.405)
Mean	31.273	0.573	0.695	0.863	7.492	82.703	84.612
$\|\beta\|$/Mean	0.008	0.017	0.011	0.009	0.004	0.000	0.008
Observations	4490	4491	4491	4491	4491	4491	4491
Clusters	828	828	828	828	828	828	828

Note: The unit of analysis is 2008 Population Census Villages. The sample is restricted to villages within 2km of a treated-control interface (a bombing boundary). All regressions use a local linear polynomial in spatial coordinates with a triangular kernel weight. We control for 50x50km grid fixed effects and province fixed effects in all regressions. *Tropics/lowland* is a dummy variable indicating whether this location falls within tropical lowland areas classified as humid or sub-humid in the agro-ecological zones classification. Columns (3) (4) (5) use data from the Indochina Atlas, published in October 1970. *Econ. Activity* indicates whether there was any agricultural activity or inland fishing in these areas in 1970. *Pop. Density* is a binary variable reflecting if the population density in 1970 was at least 10 inhabitants per square kilometer. *Dist. to roads* refers to distance (in km) to 1970 main roads/railways. *Dist. VN* indicates distance to Vietnam border. *Dist. to capital* refers to distance to the capital of Cambodia. Standard errors reported in parentheses are clustered at the commune level. ***(**)(*) indicates significance at the 1%(5%)(10%) level.

Table C.4: Balance check at a 1 km bandwidth — Population Census Villages

	Dependent variable is:						
	(1) Elevation	(2) Tropics/lowland	(3) Econ. Activity	(4) Pop. Density	(5) Dist. to roads	(6) Dist. VN	(7) Dist. to capital
Panel A: All observations							
Bombing	-0.783	-0.023	-0.003	0.006	0.250	0.315	-0.737
	(0.854)	(0.017)	(0.017)	(0.010)	(0.243)	(0.227)	(0.463)
Bombing × Soft Soil	2.439*	0.062**	-0.023	-0.023*	-0.283	-0.375	1.496*
	(1.454)	(0.031)	(0.028)	(0.014)	(0.401)	(0.397)	(0.771)
Mean	31.624	0.529	0.690	0.859	7.648	78.913	81.684
$\|\beta\|$/Mean	0.077	0.117	0.034	0.027	0.037	0.005	0.018
Observations	4396	4397	4397	4397	4397	4397	4397
Clusters	935	936	936	936	936	936	936
Panel B: Soft soil							
Bombing	0.552	0.042*	-0.031	-0.014*	0.070	-0.072	0.826
	(0.933)	(0.024)	(0.022)	(0.007)	(0.293)	(0.273)	(0.529)
Mean	31.338	0.446	0.696	0.863	7.910	73.839	77.749
$\|\beta\|$/Mean	0.018	0.095	0.045	0.016	0.009	0.001	0.011
Observations	1511	1511	1511	1511	1511	1511	1511
Clusters	433	433	433	433	433	433	433
Panel C: Hard soil							
Bombing	-0.129	-0.015	-0.009	0.007	0.242	0.131	-0.721
	(0.813)	(0.017)	(0.017)	(0.010)	(0.229)	(0.209)	(0.451)
Mean	31.774	0.573	0.687	0.856	7.510	81.570	83.744
$\|\beta\|$/Mean	0.004	0.027	0.013	0.008	0.032	0.002	0.009
Observations	2885	2886	2886	2886	2886	2886	2886
Clusters	693	694	694	694	694	694	694

Note: This repeats Table C.3 within 1 km of a bombing boundary rather than 2 km. The unit of analysis is 2008 Population Census villages. All regressions use a local linear polynomial in spatial coordinates with a triangular kernel weight and control for 50x50km grid fixed effects and province fixed effects. Variable definitions are as in Table C.3. Standard errors reported in parentheses are clustered at the commune level. ***(**)(*) indicates significance at the 1%(5%)(10%) level.

Table C.5: Extended balance tests on pre-war local build-up — Population Census Villages

	Dependent variable is:				
	(1) Buildings within 3km	(2) Largest settlement within 2km	(3) Distinct settlements within 2km	(4) Distance to nearest building	(5) Spacing between buildings
Panel A: All observations					
Bombing	8.978	2.338	0.618	-9.785	-0.762
	(6.367)	(1.769)	(0.629)	(10.423)	(2.060)
Bombing × Soft Soil	-27.782**	-9.363**	-0.234	6.553	3.298
	(12.430)	(4.159)	(1.041)	(15.101)	(3.110)
Mean	337.892	46.445	30.018	172.909	94.764
$\|\beta\|$/Mean	0.082	0.202	0.008	0.038	0.035
Observations	7014	7014	7014	7011	6958
Clusters	1105	1105	1105	1104	1101
Panel B: Soft soil					
Bombing	-17.690*	-5.677	0.080	0.607	2.066
	(10.301)	(3.751)	(0.778)	(10.890)	(2.321)
Mean	345.218	56.830	26.812	166.512	91.803
$\|\beta\|$/Mean	0.051	0.100	0.003	0.004	0.023
Observations	2523	2523	2523	2522	2508
Clusters	560	560	560	559	555
Panel C: Hard soil					
Bombing	6.430	2.263	0.439	-9.784	-0.533
	(6.005)	(1.648)	(0.601)	(10.523)	(2.097)
Mean	333.776	40.611	31.819	176.504	96.433
$\|\beta\|$/Mean	0.019	0.056	0.014	0.055	0.006
Observations	4491	4491	4491	4489	4450
Clusters	828	828	828	827	825

Note: The unit of analysis is 2008 Population Census villages within 2km of a bombing boundary. Each column puts a measure of pre-war settlement on the left-hand side of the boundary regression used in Table C.3, with a local linear polynomial in spatial coordinates, a triangular kernel weight, 50x50km grid fixed effects and province fixed effects. The measures are built from the pre-war building locations described in Section 4. A settlement is a set of buildings linked in steps of 150m or less. Columns (1) to (3) count, respectively, all buildings within 3km, the buildings in the largest settlement within 2km, and the number of distinct settlements within 2km. Columns (4) and (5) are in meters: the distance to the nearest building, censored at a 4km search radius that binds for 3 villages, and the mean distance between neighboring buildings within 2km, undefined for the 56 villages holding fewer than two buildings within 2km. Standard errors reported in parentheses are clustered at the commune level. ***(**)(*) indicates significance at the 1%(5%)(10%) level.

C.3 Cross-boundary spillovers

Exposure changes sharply at the boundary but does not stop there, so the communities we treat as controls are not wholly unexposed. We test whether that matters. Among locations classified as unbombed, we ask whether the amount of bombing nearby predicts their outcomes, using the same 3km strike count and splitting it into terciles with those having no strikes within 3km as the omitted category. The buffer crosses the boundary, which is what makes the test possible: 2,738 of the 3,091 unbombed census villages within 2km of a boundary record positive strike counts. We use the 3km count because at shorter radii the control side is almost clean, with only 3.0% having a strike within 500m. Distance to the boundary enters directly, since nearby strike counts partly track proximity to it.

Table C.6 reports the result. Nearby bombing does not predict outcomes among the unbombed, in either sample or within either soil type, and no joint test comes close to rejecting ($p = 0.764$ for census villages and $p = 0.324$ for DHS women). The point estimates nonetheless fall the way the two mechanisms imply. Every soft-soil coefficient in both panels is negative, which is what residual hazard reaching a control location would do. Every hard-soil series rises with nearby bombing, from 0.014 to 0.051 among census villages and from -0.007 to 0.026 among DHS women, which is what destruction without a lasting hazard would do. We read this as no statistically significant evidence of a spillover rather than as ruling one out, since the confidence intervals do not exclude effects of the size we estimate at the boundary. The directional pattern supports our attenuation reasoning: lower outcomes among soft-soil controls would narrow the negative bombing effect, while higher outcomes among hard-soil controls would narrow the positive effect. These imprecise estimates do not, however, establish formal bounds on either effect.

Table C.6: Do cross-boundary spillovers contaminate the comparison group?

Panel A: Census villages, economic development index

	All unbombed	Unbombed, soft	Unbombed, hard
	(1)	(2)	(3)
Low strikes within 3km	0.011	-0.051	0.014
	(0.028)	(0.071)	(0.030)
Medium strikes within 3km	-0.005	-0.088	0.032
	(0.032)	(0.074)	(0.032)
High strikes within 3km	0.015	-0.087	0.051
	(0.036)	(0.079)	(0.041)
Joint test p-value	0.764	0.505	0.472
Mean	0.020	0.063	-0.002
Observations	3091	1101	1986
Communes	825	360	603

Panel B: DHS women, health index

	All unbombed	Unbombed, soft	Unbombed, hard
	(1)	(2)	(3)
Low strikes within 3km	-0.021	-0.025	-0.007
	(0.018)	(0.039)	(0.021)
Medium strikes within 3km	0.001	-0.007	0.009
	(0.021)	(0.043)	(0.026)
High strikes within 3km	-0.016	-0.038	0.026
	(0.026)	(0.054)	(0.034)
Joint test p-value	0.324	0.699	0.650
Mean	-0.006	0.017	-0.017
Observations	8665	2891	5774
Clusters	521	173	348

Note: The sample is restricted to locations that were *not* bombed and that lie within 2km of a treated–control interface (a bombing boundary). Column (1) pools both soil types; columns (2) and (3) split them. The regressor is the number of recorded airstrikes within 3km of the location, split into terciles among locations with a positive count inside the band, with those having no strikes within 3km as the omitted category. Distance to the boundary enters the regression directly. "Joint test" reports the p-value from a test that all three exposure coefficients are zero. Panel A is census villages with standard errors clustered at the commune; Panel B is DHS women, adding survey-year fixed effects and clustering at the DHS village. Both include 50x50km grid fixed effects, province fixed effects, a local linear polynomial in spatial coordinates with a triangular kernel weight, distance to the capital, distance to the Vietnamese border and other pre-bombing characteristics. ***(**)(*) indicates significance at the 1%(5%)(10%) level.

C.4 Controls, local fixed effects and boundary geometry

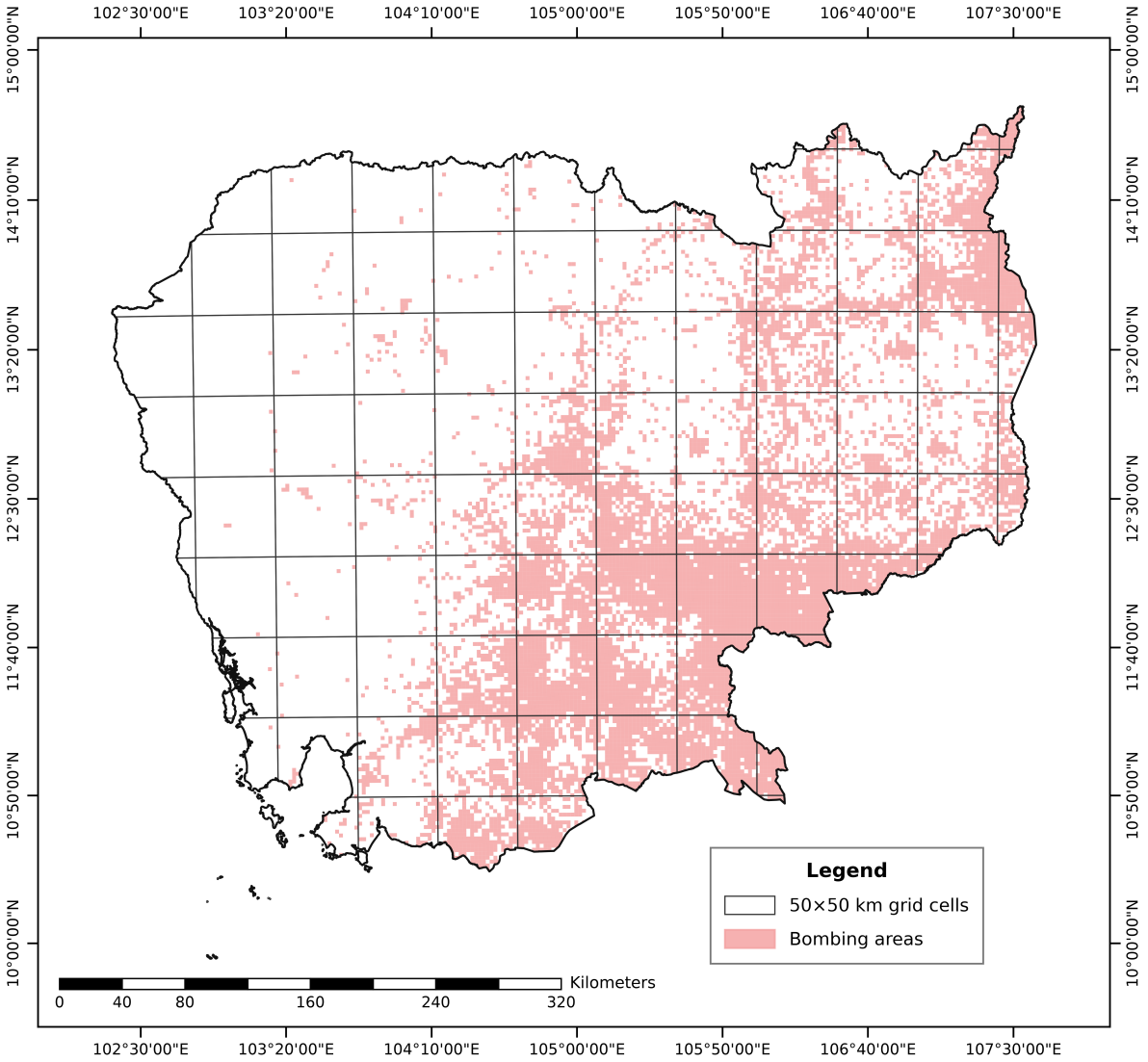
Removing the controls. Table C.7 strips out the covariates in stages and reports the headline health and economic estimates at each stage; Sections 6.2 and 6.3 discuss them alongside the baseline. Every column keeps the local linear polynomial in spatial coordinates, the triangular kernel weights, the 50x50km grid fixed effects and the province fixed effects, because these define the local comparison rather than adjust it, so the exercise varies the covariate set while retaining the design. Dropping controls can also admit a small number of observations with missing covariates.

Table C.7: Headline estimates with the controls removed in stages

	Health index (DHS women)			Economic index (Census villages)		
	All areas	Soft soil	Hard soil	All areas	Soft soil	Hard soil
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Full controls (specification used in the paper)						
Bombing	0.024**	-0.032**	0.027***	0.014	-0.040**	0.018*
	(0.010)	(0.014)	(0.010)	(0.010)	(0.017)	(0.009)
Bombing × Soft Soil	-0.052***			-0.046**		
	(0.017)			(0.021)		
Mean	-0.0131	-0.00982	-0.0149	0.0230	0.0279	0.0205
Observations	17268	6242	11026	7012	2521	4489
Clusters	1051	381	670	1104	559	827
Panel B: No 1970 (pre-bombing) controls						
Bombing	0.025**	-0.032**	0.027***	0.014	-0.039**	0.018*
	(0.010)	(0.014)	(0.010)	(0.010)	(0.017)	(0.010)
Bombing × Soft Soil	-0.054***			-0.044**		
	(0.017)			(0.022)		
Mean	-0.0131	-0.00982	-0.0149	0.0230	0.0279	0.0205
Observations	17268	6242	11026	7012	2521	4489
Clusters	1051	381	670	1104	559	827
Panel C: No controls						
Bombing	0.024**	-0.027*	0.027***	0.016	-0.038**	0.019**
	(0.010)	(0.015)	(0.010)	(0.010)	(0.017)	(0.010)
Bombing × Soft Soil	-0.054***			-0.051**		
	(0.017)			(0.022)		
Mean	-0.0132	-0.00982	-0.0151	0.0230	0.0279	0.0204
Observations	17311	6242	11069	7013	2521	4490
Clusters	1053	381	672	1104	559	827

Note: Columns (1)–(3) report the health index on DHS women, with survey-year fixed effects and standard errors clustered at the DHS village. Columns (4)–(6) report the economic development index on 2008 Population Census villages, with standard errors clustered at the commune. Both restrict to units within 2 km of a bombing boundary. *Every* column retains the local linear polynomial in spatial coordinates, the triangular kernel weights, the 50x50km grid fixed effects and the province fixed effects. The covariate set varies, with small changes in sample size where dropping controls admits observations with missing covariates. Panel A is the specification used in the paper. Panel B drops the 1970 controls, retaining elevation, tropics/lowland, distance to Vietnam and distance to the capital. Panel C drops all covariates. ***(**)(*) indicates significance at the 1%(5%)(10%) level.

Figure C.2: Generating 50x50km grid cells



Notes: The map illustrates how the country was divided into 50x50km grid-cells.

Border-segment fixed effects. Each edge of a 2.42km bombing-grid cell along a treated–control interface defines a border segment. We assign each village to its nearest segment, so segment fixed effects restrict comparisons to villages sharing the same local stretch of boundary. Given the large number of fixed effects, we apply this demanding check only to the pooled interaction specification described in Section 5.1. Table C.8 reports the results; Sections 6.2 and 6.3 discuss them alongside the baseline estimates.

Moving the boundary. A cell is bombed if a bomb fell in it, and the boundary follows from that, running wherever a bombed cell meets an unbombed one. It is therefore a staircase along the edges of 2.42 km cells, and the staircase belongs to the grid rather than to the bombing. Redrawing the grid would move the boundary and reassign treatment at once, so to isolate the geometry we smooth the staircase while holding

Table C.8: Control for border segment fixed effects

	UXO Likelihood		Health Index		Economic Index	
	(1)	(2)	(3)	(4)	(5)	(6)
Bombing	-0.006	-0.006	0.043**	0.047**	0.012	0.018
	(0.006)	(0.006)	(0.018)	(0.019)	(0.012)	(0.012)
Bombing × Soft Soil	0.075***	0.076***	-0.049*	-0.053*	-0.027	-0.033
	(0.028)	(0.029)	(0.028)	(0.028)	(0.025)	(0.025)
Mean	0.065	0.059	-0.013	-0.013	0.023	0.029
Observations	5656	5374	17311	15690	5656	5374
Clusters	965	900	1054	957	965	900
Exclude K5 Belt	No	Yes	No	Yes	No	Yes

Note: The unit of analysis for columns (1) (2) (5) (6) is census villages. The unit of analysis for columns (3) (4) is DHS individuals. The sample is restricted to observations living within 2km of a treated-control interface (a bombing boundary). All regressions use a local linear polynomial in spatial coordinates with a triangular kernel weight and control for distance to the capital, distance to Vietnam borders, border segment fixed effects and other pre-bombing characteristics. Regressions (2) (4) (6) exclude provinces along the K-5 mine belt. Standard errors reported in parentheses are clustered at the commune level for regressions (1) (2) (5) (6) and at the DHS village level for regressions (3) (4). *UXO Likelihood* is a dummy variable equal to 1 if any UXO incidents occur within a 3 km buffer, and 0 otherwise. The health and economic development indices are defined in Tables 4 and 5. ***(**)(*) indicates significance at the 1%(5%)(10%) level.

every cell's treatment at its assigned value, so the only thing that changes is each unit's distance to the line, which sets the estimation band and the kernel weights. No unit ends up on the far side of the boundary from its own assignment, and between 29 and 38 percent of those in the band move by more than 100 m. Smoothing also shortens the line, so a few units sitting near 2km fall outside the band and the sample declines slightly from row to row; those units are the farthest from the boundary and carry the least kernel weight. Table C.9 reports the health and economic estimates at the cell-edge boundary and at three degrees of smoothing, the corners cut once, twice and three times, which shifts the line without displacing it systematically inward or outward. The estimates change little across these boundary definitions.

Table C.9: Same treatment, smoothed boundary

	Health index (DHS women)			Economic index (census villages)		
Boundary	Soft	Hard	Difference	Soft	Hard	Difference
Cell edges (baseline)	-0.032**	0.027***	-0.058	-0.040**	0.018*	-0.057
	(0.014)	(0.010)		(0.017)	(0.009)	
Rounded, one pass	-0.031**	0.028***	-0.059	-0.038**	0.018*	-0.056
	(0.015)	(0.010)		(0.017)	(0.010)	
Rounded, two passes	-0.031**	0.029***	-0.060	-0.038**	0.017*	-0.055
	(0.015)	(0.010)		(0.017)	(0.010)	
Rounded, three passes	-0.031**	0.029***	-0.060	-0.037**	0.017*	-0.055
	(0.015)	(0.010)		(0.017)	(0.010)	
Observations	6242	11026		2521	4489	

Note: Treatment is the baseline assignment in every row and no cell changes status between rows; only the boundary moves. The first row measures distance to the boundary as drawn, along the edges of the 2.42km cells. The remaining rows smooth that staircase by corner-cutting, once, twice and three times. The reported observation and cluster counts are those of the first row. Health estimates are DHS women with survey-year fixed effects, clustered at the DHS village; economic estimates are census villages clustered at the commune. All regressions use a local linear polynomial in spatial coordinates with a triangular kernel weight and control for 50x50km grid fixed effects, province fixed effects, distance to the capital, distance to the Vietnamese border and other pre-bombing characteristics. ***(**)(*) indicates significance at the 1%(5%)(10%) level.

C.5 Subsequent conflict: Khmer Rouge violence

Table C.10: Khmer Rouge Violence*(Unit: census villages)*

	All areas		Soft soil		Hard soil	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Any Mass Grave within 3 km Buffer (Dummy)						
Bombing	0.008	0.007	-0.010	-0.016	0.013	0.010
	(0.015)	(0.015)	(0.023)	(0.023)	(0.014)	(0.014)
Bombing × Soft Soil	-0.046	-0.052*				
	(0.028)	(0.029)				
Mean	0.159	0.155	0.231	0.232	0.118	0.111
Observations	7012	6613	2521	2396	4489	4215
Clusters	1104	1013	559	522	827	752
Exclude K5 Belt	No	Yes	No	Yes	No	Yes
Panel B. ln(Number of Mass Graves + 1)						
Bombing	0.069	0.053	0.020	-0.023	0.081	0.061
	(0.062)	(0.063)	(0.088)	(0.081)	(0.059)	(0.059)
Bombing × Soft Soil	-0.138	-0.163				
	(0.105)	(0.106)				
Mean	0.524	0.511	0.698	0.702	0.426	0.402
Observations	7012	6613	2521	2396	4489	4215
Clusters	1104	1013	559	522	827	752
Exclude K5 Belt	No	Yes	No	Yes	No	Yes

Note: The unit is 2008 Population Census villages within 2km of a treated–control interface (a bombing boundary). All regressions use a local linear polynomial in spatial coordinates with a triangular kernel weight and control for 50x50km grid fixed effects, province fixed effects, distance to the capital, distance to the Vietnamese border and other pre-bombing characteristics; standard errors are clustered at the commune. Columns (2), (4) and (6) exclude provinces along the K-5 mine belt. Panel A reports whether there are any mass graves within 3km of the village and Panel B the log of one plus the number of mass graves. ***(**)(*) indicates significance at the 1%(5%)(10%) level.

Table C.11: Control for Khmer Rouge Violence

	Health Index *Unit: DHS women*			Economic Index *Unit: Census villages*		
	All areas (1)	Soft soil (2)	Hard soil (3)	All areas (4)	Soft soil (5)	Hard soil (6)
Bombing	0.024**	-0.032**	0.027***	0.013	-0.040**	0.018*
	(0.010)	(0.015)	(0.010)	(0.010)	(0.016)	(0.009)
Bombing × Soft Soil	-0.052***			-0.044**		
	(0.017)			(0.021)		
Mean	-0.013	-0.010	-0.015	0.023	0.027	0.020
Observations	17268	6242	11026	7012	2521	4489
Clusters	1051	381	670	1104	559	827

Note: Columns (1)–(3) report the health index on DHS women, with survey-year fixed effects and standard errors clustered at the DHS village. Columns (4)–(6) report the economic index on census villages (2008 Population Census), with standard errors clustered at the commune. In both cases the sample is restricted to units within 2km of a treated–control interface (a bombing boundary). All regressions use a local linear polynomial in spatial coordinates with a triangular kernel weight and control for 50x50km grid fixed effects, province fixed effects, distance to the capital, distance to the Vietnamese border and other pre-bombing characteristics. We also control for the logged number of nearby Khmer Rouge mass graves in all regressions. In the "All areas" columns the reported interaction is bombing × soft soil; the soil-split columns estimate the bombing effect separately within each soil type. ***(**)(*) indicates significance at the 1%(5%)(10%) level.

C.6 Spatial inference for the principal outcomes

Table C.12: Conley Standard Errors

	Dependent variable is:		
	UXO Likelihood	Health Index	Economic Index
	(1)	(2)	(3)
Bombing	-0.016**	0.025***	0.015*
	(0.007)	(0.003)	(0.008)
Bombing × Soft Soil	0.050**	-0.052***	-0.046***
	(0.024)	(0.019)	(0.016)
Mean	0.065	-0.013	0.023
Observations	7013	17268	7013

Note: All regressions use a triangular kernel weight and control for 50x50km grid fixed effects, province fixed effects, distance to the capital, distance to Vietnam borders and other pre-bombing characteristics. Standard errors are computed using Conley (1999)'s spatial correction. ***(**)(*) indicates significance at the 1%(5%)(10%) level.

D UXO: additional evidence and sensitivity

The seasonal comparison links soil conditions at the time of bombing to later UXO, health and economic outcomes. We then vary the radius used to count UXO incidents around each village.

D.1 Soil conditions and the season of bombing

The soil classification is described in Appendix B.3. The concern for interpretation is not simply that fertile land supports better outcomes. Our within-soil design compares bombed and unbombed communities in the same soil category. The remaining question is whether bombing affects fertile land differently through destruction and reconstruction, rather than through detonation failure. Three features bear on this. Soil type is balanced with respect to bombing treatment and intensity (Table 2), arguing against different bombing loads as the source of the soil pattern. The first stage links the same soil split to residual UXO (Section 6.1). Under a reconstruction account in which fertile land offers higher returns to upgrading, one would also expect larger gains on soft soil, rather than the decline we observe. These arguments motivate the seasonal comparison, which uses differences in ground moisture at the time of bombing to examine the mechanical explanation more directly. We split bombed units in two by the share of their strikes that fell between May and November, computing the cut once among bombed units inside the estimation band and holding it fixed across outcomes. We cut at the median rather than at a fixed half share, because the campaign itself was wet-weighted, with 64 percent of all strikes falling in those months.

Table D.1 reports this comparison for UXO and both indices. On soft soil, the health estimate is −0.034 in the higher wet-season-share group and −0.027 in the lower-share group, with only the former significant at 5 percent. Both hard-soil health estimates are 0.026 and significant at 5 percent. The economic decline on soft soil is also larger in the higher-share group, while hard-soil estimates are similar across groups. We do not formally test wet–dry coefficient differences, and the health estimates in particular are close. Their direction alongside the larger UXO first stage in the higher-share group supports the mechanical interpretation.

Table D.1: Bombing Effects by Wet-Season Share

	UXO likelihood (1)	Health index (2)	Economic index (3)
Panel A. Seasonal bombing exposure: soft soil			
Lower wet-season share	0.020	-0.027	-0.027
	(0.017)	(0.022)	(0.018)
Higher wet-season share	0.055**	-0.034**	-0.051**
	(0.024)	(0.016)	(0.022)
Panel B. Seasonal bombing exposure: hard soil			
Lower wet-season share	-0.017	0.026**	0.020*
	(0.012)	(0.013)	(0.010)
Higher wet-season share	-0.019*	0.026**	0.016
	(0.012)	(0.012)	(0.012)

Note: Column (1) is an indicator for any UXO incident within 3 km of the village; column (2) is the health index on DHS women; column (3) is the economic development index on census villages. Bombed units are split in two by the share of strikes within 2.42 km that fell in the wet season, May to November, with unbombed units as the omitted category, so each row reports the bombing effect for locations with a lower or higher wet-season share. The split is at the median wet share among bombed units inside the estimation band, computed once and held fixed across panels. The pooled model includes indicators for both bombing groups and their interactions with soil type. Soft and hard soil are the 1962 classification used throughout. All regressions restrict to units within 2 km of a bombing boundary and include a local linear polynomial in spatial coordinates with a triangular kernel weight, 50x50km grid fixed effects, province fixed effects and the pre-bombing controls; census regressions cluster at the commune and DHS regressions cluster at the DHS village. ***(**)(*) indicates significance at the 1%(5%)(10%) level.

D.2 The radius used to count UXO incidents

Our first stage counts UXO incidents within three kilometers of a village. Figures D.1 and D.2 re-estimate the pooled and within-soil specifications at radii from one to five kilometers in quarter-kilometer steps. The outcomes are an indicator for any incident and the inverse hyperbolic sine of the number of victims. The pooled estimates follow a similar profile, changing little between roughly 2.75 and 3.75 km before declining at wider radii. The baseline radius therefore lies within a broad plateau.

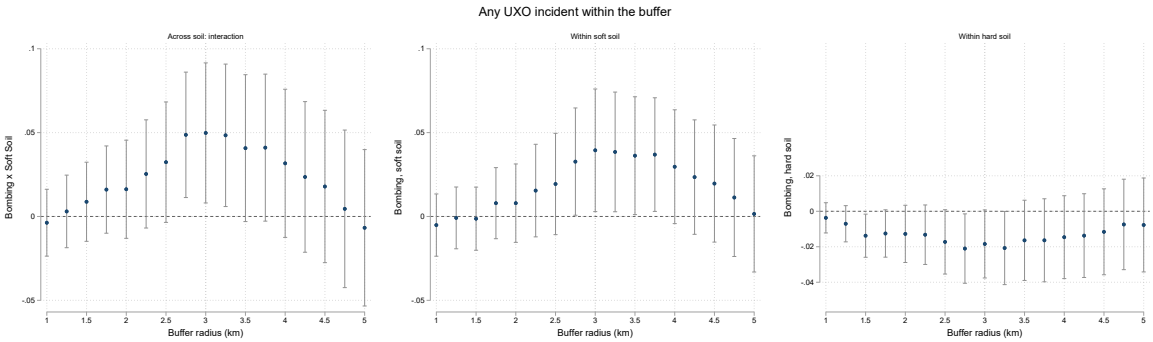
At narrow radii, recorded incidents are sparse: at one kilometer, 98.8 percent of villages in the estimation sample have no recorded casualty within the buffer. Such a narrow window measures encounters over a much smaller area and leaves little outcome variation for comparing villages. This limits what we can learn from the weaker estimate at that radius about the first stage around three kilometers.

Wider buffers increasingly pool exposure across the bombing boundary. All sampled villages lie within two kilometers of a boundary, so at five kilometers their buffers

necessarily extend to the other side. The share of identifying incidents counted by villages on both sides rises from 15 percent at one kilometer to 45 percent at three kilometers and 63 percent at five. Over the same range, the treated-minus-control difference in the share of buffer area inside the bombed zone falls from 0.72 to 0.21. Wider buffers thus compress the exposure contrast between treated and control villages. The counting radius must balance capturing encounters with preserving this contrast, and the three-kilometer specification falls within the range where the estimated first stage changes little.

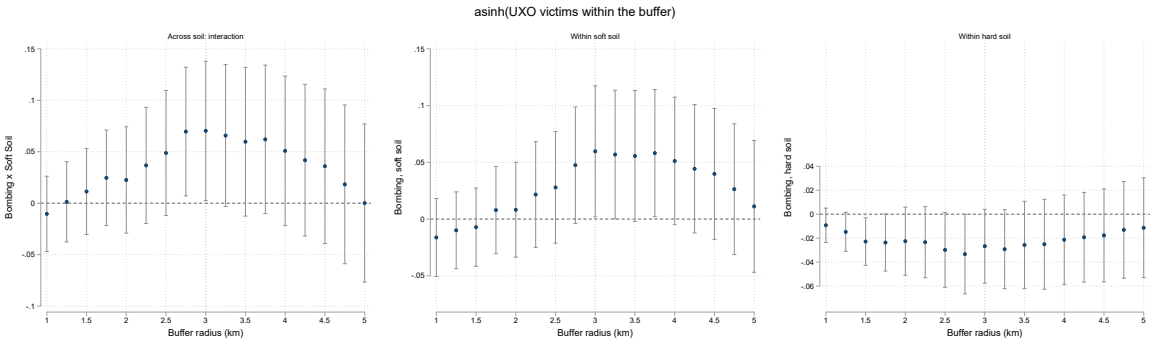
Weighting incidents by the number of victims gives a similar profile (Figure D.2). We transform the victim count using $\text{asinh}(v) = \ln(v + \sqrt{v^2 + 1})$, which behaves like a logarithm at large values but is defined at zero. Most villages have no casualty within the buffer. With so many zeros, these coefficients do not carry the percentage interpretation of log coefficients (Chen & Roth 2024), so we focus on the profile across radii rather than the magnitude at any one radius.

Figure D.1: UXO first stage across counting radii: any incident



Note: Each point is the bombing × soft soil interaction (left), the bombing coefficient estimated within soft soil (center), or within hard soil (right), from the first-stage specification of Section 6.1 re-run with the outcome defined at the radius on the horizontal axis. The outcome is an indicator equal to one if any qualifying ERW casualty incident falls within that radius of the village. Spikes are 95% confidence intervals; standard errors are clustered at the commune. The dotted line marks the three-kilometer radius used in the paper. The unit is 2008 Census villages within 2km of a bombing boundary.

Figure D.2: UXO first stage across counting radii: victims



Note: As Figure D.1, but the outcome is the inverse hyperbolic sine of the number of victims of the incidents falling within the radius, $\text{asinh}(v) = \ln(v + \sqrt{v^2 + 1})$, which is defined at zero where a logarithm is not. Between 85 and 99 percent of villages have no casualty within the buffer depending on the radius, so the height of these coefficients does not carry a percentage interpretation (Chen & Roth 2024); the shape across radii is what the figure is for.

E Health: additional results and robustness

The health components, displacement checks, alternative specifications and cohort estimates are collected here. Evidence bearing on both health and economic development also appears in Appendices C, D and F.

E.1 Health index components and child mortality

Table E.1 reports the five components of the *Health Index*. On soft soil, weight-for-height declines significantly, with negative but imprecise estimates for height-for-age and the indicator for no moderate or severe anemia. Healthy BMI and corpulence estimates are close to zero. On hard soil, height-for-age and the probability of having no moderate or severe anemia improve significantly, while weight-for-height and the two body-composition estimates are close to zero. The components therefore broadly support the index pattern without each yielding a precisely estimated effect. Precision is limited in these separate regressions, reducing our power to detect changes in individual health measures. Combining information across them is the reason we use the index as our primary health outcome.

Table E.1: Effects on health: Component Outcomes

	Dependent variable is:				
	(1)	(2)	(3)	(4)	(5)
	Height-for-Age	Weight-for-Height	Healthy BMI	Healthy CI	No moderate/ severe anemia
Panel A: All areas					
Bombing	0.082***	-0.003	0.012	0.002	0.017**
	(0.025)	(0.029)	(0.010)	(0.011)	(0.008)
Bombing × Soft Soil	-0.113***	-0.087*	-0.014	0.003	-0.021*
	(0.042)	(0.049)	(0.017)	(0.019)	(0.012)
Mean	-1.833	-1.027	0.705	0.708	0.905
Relative magnitude	0.092	0.003	0.018	0.003	0.018
Observations	17238	13514	17237	17237	15179
Clusters	1051	941	1051	1051	1051
Panel B: Soft soil					
Bombing	-0.043	-0.092**	0.004	0.006	-0.009
	(0.035)	(0.041)	(0.013)	(0.016)	(0.008)
Mean	-1.830	-1.001	0.697	0.691	0.920
Relative magnitude	0.049	0.093	0.005	0.008	0.010
Observations	6231	4882	6231	6231	5458
Clusters	381	344	381	381	381
Panel C: Hard soil					
Bombing	0.073***	-0.006	0.015	0.006	0.017**
	(0.026)	(0.030)	(0.011)	(0.011)	(0.008)
Mean	-1.834	-1.042	0.710	0.718	0.898
Relative magnitude	0.082	0.007	0.021	0.008	0.019
Observations	11007	8632	11006	11006	9721
Clusters	670	597	670	670	670

Note: The unit of analysis is DHS survey respondents. The sample is restricted to those living within 2km of a treated-control interface (a bombing boundary). All regressions use a local linear polynomial in spatial coordinates with a triangular kernel weight and control for 50x50km grid fixed effects, province fixed effects, distance to the capital, distance to Vietnam borders and other pre-bombing characteristics. *No moderate/severe anemia* equals one for mild or no anemia. *Healthy BMI* is defined as a BMI value between 18.5 and 25. *Healthy Corpulence Index (CI)* or Rohrer's Index is defined as a CI between 11 and 15. *Relative magnitude* is the coefficient divided by the outcome mean for the three binary components, and by the outcome standard deviation for height-for-age and weight-for-height, which are Z-scores whose mean is not a scale. Standard errors reported in parentheses are clustered at the DHS village level. ***(**)(*) indicates significance at the 1%(5%)(10%) level.

Table E.2: Child Mortality

	All areas		Soft soil		Hard soil	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Child Mortality (Dummy)						
Bombing	-0.025***	-0.019***	0.010	0.016	-0.022***	-0.018***
	(0.007)	(0.007)	(0.010)	(0.010)	(0.007)	(0.007)
Bombing × Soft Soil	0.040***	0.038***				
	(0.012)	(0.012)				
Mean	0.179	0.177	0.174	0.176	0.182	0.177
Observations	32332	29373	11652	10889	20680	18484
Clusters	1051	954	381	355	670	599
Exclude K5 Belt	No	Yes	No	Yes	No	Yes
Panel B: Number of deaths						
Bombing	-0.044***	-0.033**	0.008	0.013	-0.040***	-0.030**
	(0.014)	(0.014)	(0.020)	(0.020)	(0.014)	(0.014)
Bombing × Soft Soil	0.060**	0.053**				
	(0.024)	(0.024)				
Mean	0.288	0.284	0.287	0.292	0.289	0.279
Observations	32332	29373	11652	10889	20680	18484
Clusters	1051	954	381	355	670	599
Exclude K5 Belt	No	Yes	No	Yes	No	Yes

Note: The unit of analysis is DHS individuals (DHS women). The sample is restricted to those living within 2km of a treated-control interface (a bombing boundary). All regressions use a local linear polynomial in spatial coordinates with a triangular kernel weight and control for 50x50km grid fixed effects, province fixed effects, distance to the capital, distance to Vietnam borders and other pre-bombing characteristics. Regressions (2) (4) (6) exclude provinces along the K-5 mine belt. Standard errors reported in parentheses are clustered at the DHS village level. Child mortality is a binary indicator equal to 1 if a woman has experienced the death of at least one child, and 0 otherwise. The number of deaths measures the total number of a woman's children who have died. ***(**)(*) indicates significance at the 1%(5%)(10%) level.

E.2 DHS displacement and bandwidth sensitivity

Urban DHS villages. DHS village coordinates are displaced by at most 2 km in urban areas, compared with 5 km for most rural villages. Restricting the sample to urban villages therefore examines the health pattern where the displacement bound is smaller, although assignment error remains possible. Table E.3 reports the results. The bombing × soft soil interaction is −0.048, and −0.053 excluding the K-5 belt, compared with −0.052 at baseline. Neither urban interaction is statistically significant. The hard-soil estimate is positive and significant at 0.059, compared with 0.027 at baseline. The sample falls from 17,268 women in 1,051 DHS villages to 3,967 women in 228 villages, and the standard error of the interaction rises from 0.017 to 0.035. The smaller sample therefore gives a less precise estimate of the across-soil contrast. We would not expect the soft-soil penalty at full strength among urban households in any case, since the channel runs through working contaminated land and the mechanism evidence in Section 6 is cultivated area, crop revenue, livestock and forest use.

Table E.3: Health Index: urban DHS villages only

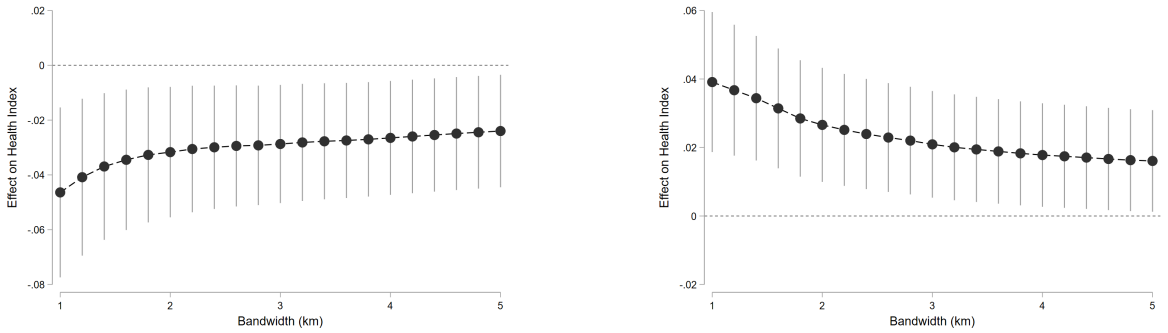
	All areas		Soft soil		Hard soil	
	(1)	(2)	(3)	(4)	(5)	(6)
Bombing	0.055**	0.062***	-0.020	-0.015	0.059**	0.067***
	(0.022)	(0.023)	(0.029)	(0.032)	(0.025)	(0.025)
Bombing × Soft Soil	-0.048	-0.053				
	(0.035)	(0.037)				
Mean	0.066	0.069	0.079	0.077	0.054	0.061
Observations	3967	3630	1944	1734	2023	1896
Clusters	228	207	113	101	115	106
Exclude K5 Belt	No	Yes	No	Yes	No	Yes

Note: The unit of analysis is DHS women and the sample is restricted to DHS villages classified as urban, where the confidentiality displacement is at most 2 km rather than 5 km. Columns 1–2 pool both soils and include a soil main effect; columns 3–4 and 5–6 split the sample. Regressions (2) (4) (6) exclude provinces along the K-5 mine belt. All regressions use a local linear polynomial in spatial coordinates with a triangular kernel weight and control for 50x50km grid fixed effects, province fixed effects, survey-year fixed effects, distance to the capital, distance to the Vietnamese border and other pre-bombing characteristics. The sample is restricted to units within 2km of a treated–control interface (a bombing boundary). Standard errors reported in parentheses are clustered at the DHS village level. ***(**)(*) indicates significance at the 1%(5%)(10%) level.

Bandwidth sensitivity. We conduct sensitivity checks to different choices of bandwidth ranging from 1 to 5 km with 0.2km intervals. Figure E.1 shows that our results are robust to any choice of bandwidths with negative health effects in soft soil regions and positive health effects in hard soil regions. The range also bears on the DHS con-

fidentiality displacement described in Section 4. Displacement misassigns bombing status only when it carries a DHS village across a bombing boundary, which is likeliest for villages recorded close to one, so the share of the estimation sample exposed to that error is largest at a 1 km band and falls as the band widens toward the 5 km displacement bound. That both estimates keep their sign and significance as this exposure to misassignment changes supports the health pattern.

Figure E.1: Sensitivity of Results to Bandwidth Choice



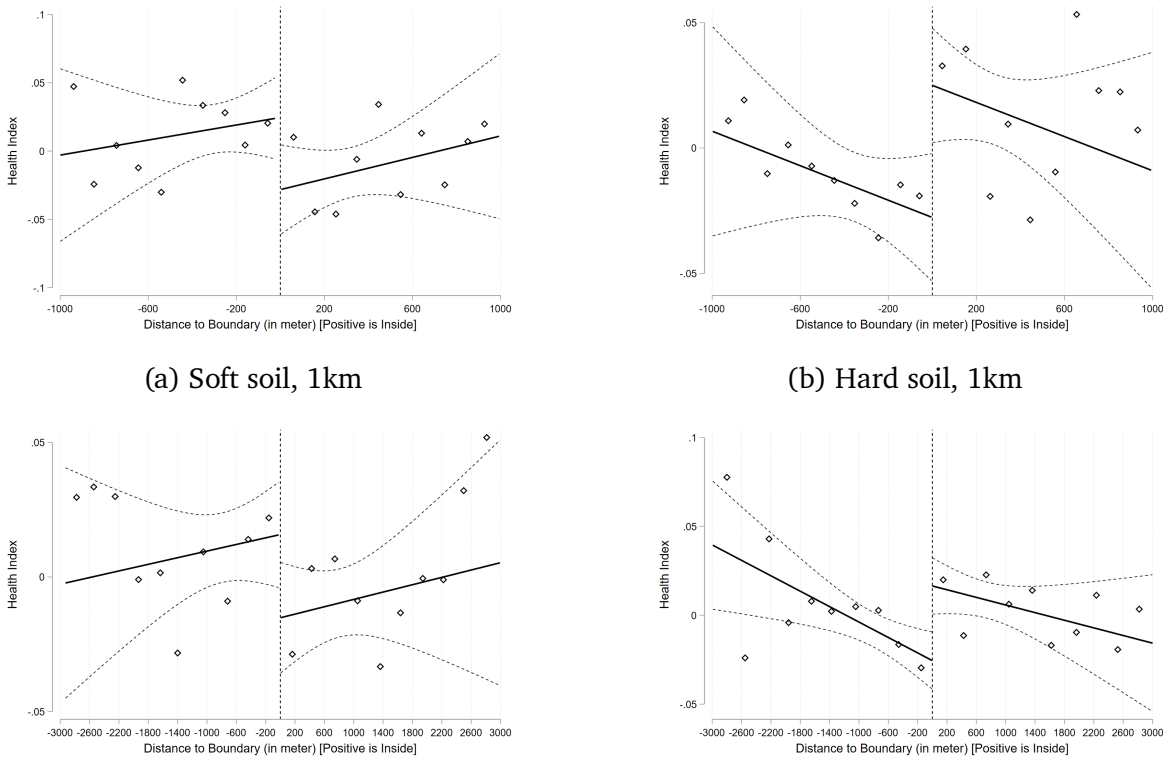
(a) Soft soil

(b) Hard soil

Note: Dependent variable is Health Index. Each sub-graph reports coefficient estimates and confidence intervals for different bandwidth levels ranging from 1 to 5 kilometers (horizontal axis) with 0.2km intervals. Each dot indicates the estimate using the specified bandwidth. Range spikes represent 90% confidence intervals of the estimates.

The window used to draw the boundary plots. Figure 3 plots binned residuals over a 2 km window either side of the boundary. Figure E.2 redraws it at 1 km and 3 km. Narrowing the window leaves fewer observations in each bin and widening it admits units further from the line, but the picture does not turn on the choice: on soft soil the outcome drops at the boundary and on hard soil it jumps at every window. The corresponding economic-index plots are in Appendix F.1.

Figure E.2: Health Index: binned residuals at 1km and 3km windows



(a) Soft soil, 1km

(b) Hard soil, 1km

(c) Soft soil, 3km

(d) Hard soil, 3km

Note: As Figure 3, redrawn at 1km and 3km. Points are binned residuals from a regression of the *Health Index* on a linear polynomial in spatial coordinates and the control variables. Solid lines are local linear regressions estimated separately either side of the bombing boundary, dashed lines 95% confidence intervals. Negative distances are locations outside the bombing areas.

E.3 Placebo boundaries and specification sensitivity

Shifted-boundary placebo. We test whether the health effects are specific to the bombing boundaries by shifting the boundary set 5 km north, east, west and south in turn. After each shift, we reassign treatment and re-estimate the pooled and within-soil specifications. Table E.4 shows no significant placebo-boundary effect on the *Health Index* within either soil type or in the pooled sample. This supports locating the health pattern at the bombing boundaries rather than at arbitrary lines through the same landscape.

Table E.4: Health Index: shifted-boundary placebo

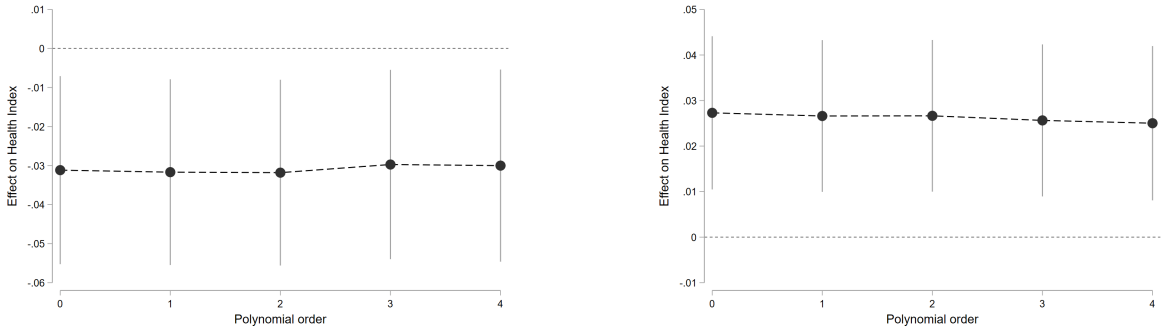
	Shift east (1)	Shift west (2)	Shift north (3)	Shift south (4)
Panel A: All observations				
Bombing	0.002	-0.010	-0.012	0.008
	(0.010)	(0.010)	(0.010)	(0.010)
Bombing × Soft Soil	0.013	0.002	0.018	-0.012
	(0.017)	(0.016)	(0.016)	(0.016)
Mean	0.402	0.390	0.407	0.399
Observations	18323	18327	18857	18339
Clusters	1113	1107	1144	1111
Panel B: Soft soil				
Bombing	0.011	-0.013	0.008	-0.005
	(0.014)	(0.014)	(0.013)	(0.014)
Mean	0.440	0.410	0.416	0.382
Observations	7418	7004	7214	6548
Clusters	448	415	433	395
Panel C: Hard soil				
Bombing	-0.005	-0.012	-0.016	0.008
	(0.010)	(0.010)	(0.010)	(0.010)
Mean	0.380	0.379	0.402	0.408
Observations	10905	11323	11643	11791
Clusters	665	692	711	716

Note: The table shows the results of placebo tests which shift bombing borders by 5 kilometers to four different directions: east-west-north-south. The unit of analysis is survey respondents. All regressions control for 50x50km grid fixed effects, province fixed effects, distance to the capital, distance to Vietnam borders and other pre-bombing characteristics. Standard errors reported in parentheses are clustered at the DHS village level. ***(**)(*) indicates significance at the 1%(5%)(10%) level.

Choice of polynomial orders. Figure E.3 plots the main coefficients for different orders of the functional form in spatial coordinates. The results are consistent when higher orders of polynomials are used in the main regression with a substantial decline in the health index for soft-soil areas and consistently positive effects for hard-soil areas.

Grid-cell fixed effects. To guarantee a comparison of observations in very close geographic proximity, we push even further than our main specification. Within our context, because there are numerous bombing borders widespread across the country, we employ

Figure E.3: Sensitivity of Results to Different Orders of Polynomial

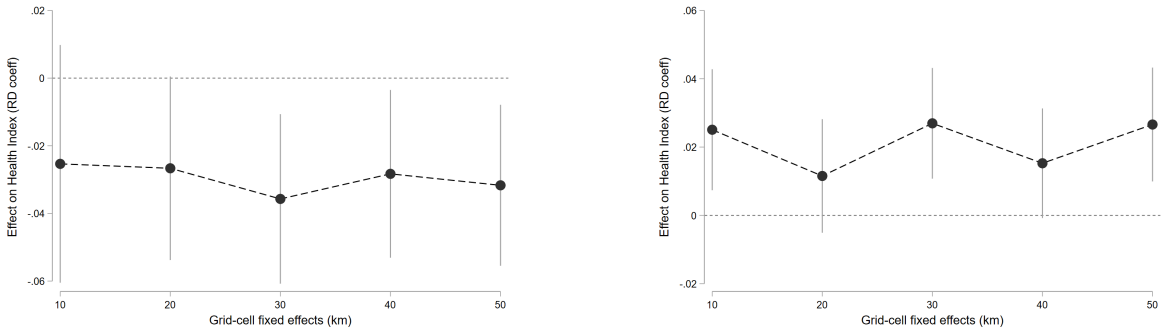


(a) Soft soil (b) Hard soil

Note: Dependent variable is Health Index. Each dot represents the estimate using the specified order of polynomial in spatial coordinates. Range spikes represent 90% confidence intervals of the estimates.

50x50km grid-fixed effects along with province fixed effects to ensure that we compare individuals within a confined area. However, we are also concerned that 50x50km grids may be too broad to account for between-area heterogeneity. Therefore, instead of using 50x50km grids to control for spatial variation, we divide the country into smaller grid-cells, ranging from 10x10km, 20x20km up to 50x50km, and control for grid-cell fixed effects in our main regressions. Figure E.4 presents our results. When smaller-grid fixed effects are employed, the results remain similar: living in bombing areas is associated with better health in hard-soil areas and worse health in soft-soil areas.

Figure E.4: Sensitivity of Results to Different Grid Fixed Effects



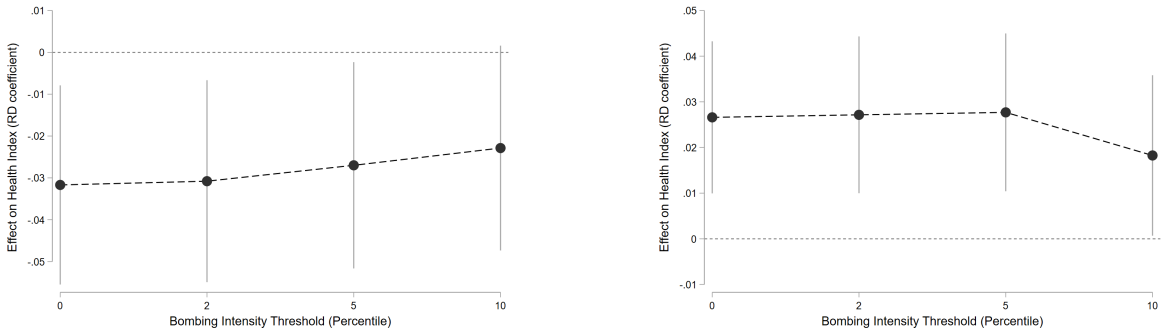
(a) Soft soil (b) Hard soil

Note: Dependent variable is Health Index. Each dot represents the estimate controlling for a specified grid-fixed-effects. Range spikes represent 90% confidence intervals of the estimates.

Alternative Bombing Intensity Thresholds To assess whether the results are sensitive to how bombing exposure is defined, we re-define treatment using thresholds based

on the distribution of positive bombing intensity rather than a binary indicator of any exposure (tonnage > 0). Specifically, we classify a grid as treated (bombed) only if its total bombing load exceeds cutoffs corresponding to the 2nd, 5th, and 10th percentiles of positive bombing distribution. This approach excludes areas with minimal exposure and focuses on locations substantially affected by bombing. Cells at or below each cutoff are reclassified as controls. Boundaries are reconstructed from the retained treated cells, and distances, sample inclusion within the 2 km band and kernel weights are recalculated for each threshold. Figure E.5 presents our results. The estimated effects remain stable across all thresholds, suggesting that the health pattern is not specific to the baseline definition of bombing exposure.

Figure E.5: Sensitivity of Results to Alternative Bombing Intensity Thresholds



(a) Soft soil (b) Hard soil

Note: Dependent variable is Health Index. Each dot represents the estimate using a specified bombing intensity threshold (0, 2%, 5%, or 10%). Range spikes represent 90% confidence intervals of the estimates.

Uni-dimensional specification We use a uni-dimensional specification to cross-verify our results. The regressions take the same form as our main specification. However, in this setting, the functional form $f(Geo_c)$ uses a uni-dimensional measure, namely distance to bombing boundaries as a running variable. Drawing from the spatial RD literature, we use a local linear functional form, $f(Geo_c) = \eta dist_c + \theta Bombing_c \times dist_c$, with the forcing variable $dist_c$ denoting the Euclidean distance between a household location and the closest point on bombing boundaries. Higher-order polynomials will take the following form: $f(Geo_c) = \sum_{k=1}^{a} \left(\eta_k dist_c^k + \theta_k Bombing_c \times dist_c^k\right)$. The interaction term of the treatment variable with the distance to the bombing areas is of great importance because it allows for different slopes of the functions on two sides of bombing boundaries. Table E.5 shows that the uni-dimensional specification yields similar results, with coefficients even larger in magnitude.

Table E.5: Uni-dimensional specification

	All areas		Soft soil		Hard soil	
	Linear (1)	Quad. (2)	Linear (3)	Quad. (4)	Linear (5)	Quad. (6)
Bombing	0.057***	0.076***	-0.043*	-0.069*	0.059***	0.079***
	(0.017)	(0.026)	(0.024)	(0.036)	(0.017)	(0.026)
Bombing × Soft Soil	-0.089***	-0.143***				
	(0.030)	(0.044)				
Mean	-0.013	-0.013	-0.010	-0.010	-0.015	-0.015
Observations	17311	17311	6242	6242	11026	11026
Clusters	1053	1053	381	381	670	670

Note: The unit of analysis is DHS survey respondents. The sample is restricted to those living within 2km of a treated-control interface (a bombing boundary). All regressions control for 50x50km grid fixed effects, province fixed effects, distance to the capital, distance to Vietnam borders and other pre-bombing characteristics. Regressions (1) (3) (5) control for a linear polynomial in distance to the bombing boundaries. Regressions (2) (4) (6) control for a quadratic polynomial in distance to the bombing boundaries. Standard errors reported in parentheses are clustered at the DHS village level. ***(**)(*) indicates significance at the 1%(5%)(10%) level.

Alternative specifications. In Table E.6, we investigate robustness to a wide range of specifications. The first column controls for 50x50km grid fixed effects and district fixed effects instead of province fixed effects.[17] Column (2) shows the results when observations are clustered at bombing grids instead of DHS villages. Column (3) includes the distance to Thai border among the control variables. The following column presents outcomes without a triangular kernel weight.[18] Columns (5) (6) exclude the distance to the capital and the distance to Vietnam's borders in the main regressions. The last column runs a *donut* exercise by removing all observations close to bombing boundaries (within 0.1km) and keeping the remaining data to fit the spatial matching model. The objective of the *donut* exercise is to address the potential issue of systematic disparities between border populations and populations further away from bombing boundaries. Table E.6 shows that our estimates are robust in terms of magnitude and significance level across different specification choices. Even in the more restrictive *donut* model, the effects remain strong and significant.

[17] In our dataset, households are located in 176 districts.

[18] A triangular kernel involves decreasing the weight assigned to each observation as the distance from the boundaries increases.

Table E.6: Robustness checks: different specifications with x- and y- coordinates as spatial controls

	Dependent variable is Health Index						
	Grid-&District-FE (1)	Cluster(Bombing Grid) (2)	With Dist. Thai (3)	No weight (4)	No Dist. Capital (5)	No Dist. Vietnam (6)	Donut 0.1km (7)
Panel A: All observations							
Bombing	0.027***	0.024**	0.023**	0.016	0.024**	0.025**	0.018*
	(0.010)	(0.010)	(0.010)	(0.010)	(0.010)	(0.010)	(0.011)
Bombing × Soft Soil	-0.049***	-0.052***	-0.051***	-0.043***	-0.052***	-0.052***	-0.051***
	(0.018)	(0.018)	(0.017)	(0.015)	(0.017)	(0.017)	(0.018)
Mean	0.0945	0.0945	0.0945	0.0945	0.0945	0.0945	0.0948
Observations	17311	17268	17268	17268	17268	17268	16067
Clusters	1054	842	1051	1051	1051	1051	975
Panel B: Soft soil							
Bombing	-0.028*	-0.032**	-0.032**	-0.028**	-0.032**	-0.032**	-0.035**
	(0.015)	(0.015)	(0.014)	(0.013)	(0.014)	(0.014)	(0.015)
Mean	0.0802	0.0802	0.0802	0.0802	0.0802	0.0802	0.0801
Observations	6242	6242	6242	6242	6242	6242	5824
Clusters	381	290	381	381	381	381	357
Panel C: Hard soil							
Bombing	0.025**	0.027***	0.025**	0.016*	0.027***	0.027***	0.019*
	(0.010)	(0.010)	(0.010)	(0.010)	(0.010)	(0.010)	(0.010)
Mean	0.102	0.102	0.102	0.102	0.102	0.102	0.103
Observations	11069	11026	11026	11026	11026	11026	10243
Clusters	673	567	670	670	670	670	618

Note: The unit of analysis is DHS survey respondents. Regressions (1) control for grid-fixed effects and district-fixed effects (instead of province fixed effects). Regressions (2) show the estimates when observations are clustered at bombing grids instead of DHS villages. Regressions (3) additionally control for distance to Thai borders. Regressions (4) exclude the triangular kernel weight. Regressions (5) (6) exclude the distance to the capital and the distance to Vietnam's borders, correspondingly. Regressions (7) conduct a donut exercise that excludes observations within 0.1 km of bombing boundaries. Standard errors reported in parentheses are clustered at the DHS village level except in column (2), which clusters at the bombing-grid cell. ***(**)(*) indicates significance at the 1%(5%)(10%) level.

E.4 Birth cohorts and direct exposure to conflict

The cohort comparison asks whether the health results reflect direct exposure to war during childhood or persistent differences in community conditions. Women born before 1980 include those alive during the bombing or Khmer Rouge period. We add a second cutoff at 1990 because conflict continued after the Khmer Rouge, with Vietnamese forces present until 1989 and civil war continuing to the 1991 Paris accords. The later cohort was born well after the bombing and around the transition out of prolonged conflict. It is not a claim that everyone born from 1990 was unexposed to subsequent fighting.

Table E.7 shows negative soft-soil and positive hard-soil estimates for both the oldest and youngest cohorts. The pooled interaction is −0.066 among women born before 1980 and −0.082 among those born from 1990, significant at 1 and 5 percent respectively. The youngest within-soil estimates are imprecise, particularly with only 1,081 women on soft soil. The middle cohort has small, insignificant estimates on both soils and a pooled interaction of −0.007. Its early childhood coincided with continued conflict and deprivation across Cambodia, although the estimates do not establish why it differs. The important finding is that the health pattern remains among women born well after the bombing, rather than appearing only among those who could have experienced it directly.

Table E.7: Health Index: birth cohort heterogeneity

	All areas		Soft soil		Hard soil	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Born before 1980						
Bombing	0.030**	0.029**	-0.032*	-0.028	0.033**	0.033**
	(0.014)	(0.015)	(0.019)	(0.020)	(0.014)	(0.015)
Bombing × Soft Soil	-0.066***	-0.062***				
	(0.023)	(0.024)				
Mean	-0.0822	-0.0796	-0.0737	-0.0775	-0.0871	-0.0808
Observations	8633	7881	3156	2980	5477	4901
Clusters	1050	953	380	354	670	599
Panel B: Born 1980–1989						
Bombing	0.001	0.007	-0.009	-0.009	0.004	0.006
	(0.015)	(0.015)	(0.021)	(0.022)	(0.015)	(0.016)
Bombing × Soft Soil	-0.007	-0.013				
	(0.025)	(0.026)				
Mean	0.0377	0.0381	0.0407	0.0375	0.0360	0.0384
Observations	5615	5092	2003	1864	3612	3228
Clusters	1037	941	374	349	663	592
Panel C: Born 1990 or later						
Bombing	0.046*	0.052*	-0.034	-0.032	0.040	0.051*
	(0.025)	(0.027)	(0.034)	(0.036)	(0.026)	(0.028)
Bombing × Soft Soil	-0.082**	-0.092**				
	(0.040)	(0.042)				
Mean	0.0908	0.0889	0.0827	0.0772	0.0954	0.0954
Observations	3018	2672	1081	955	1935	1715
Clusters	688	623	245	225	441	396
Exclude K5 Belt	No	Yes	No	Yes	No	Yes

Note: The unit of analysis is DHS women, split by year of birth. Panel A is women born before 1980, Panel B those born from 1980 to 1989, and Panel C those born in 1990 or later. The additional cut at 1990 distinguishes women born during the 1980s conflict from those born near the transition out of prolonged war and well after the bombing. Columns 1–2 pool both soils and include a soil main effect; columns 3–4 and 5–6 split the sample. Regressions (2) (4) (6) exclude provinces along the K-5 mine belt. The sample is restricted to women within 2km of a treated–control interface (a bombing boundary). All regressions use a local linear polynomial in spatial coordinates with a triangular kernel weight and control for 50x50km grid fixed effects, province fixed effects, survey-year fixed effects, distance to the capital, distance to the Vietnamese border and other pre-bombing characteristics. Standard errors reported in parentheses are clustered at the DHS village level. ***(**)(*) indicates significance at the 1%(5%)(10%) level.

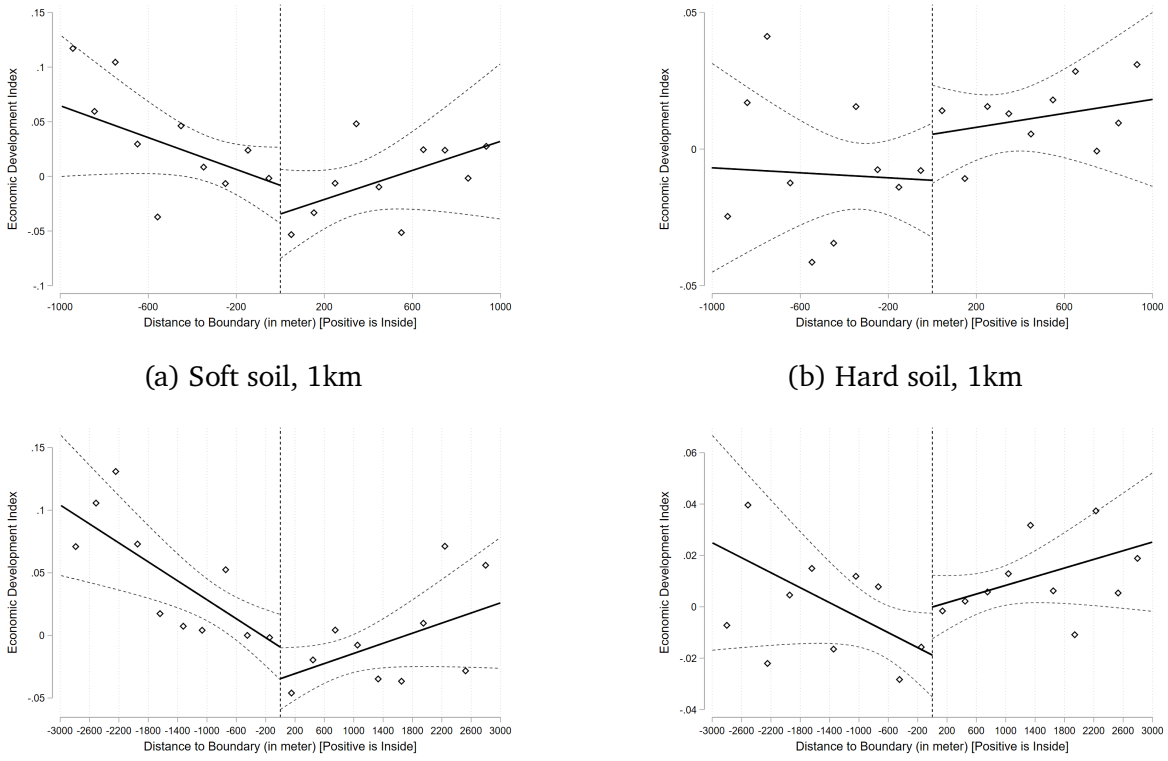
F Additional results and mechanisms for health and economic development

We report additional boundary plots and the economic index components, then turn to aid, migration and population as possible explanations for the health and economic patterns. The final subsection relates the economic results to pre-war settlement.

F.1 Economic development: bandwidth sensitivity

Figure F.1 redraws the economic-index boundary plots at 1 km and 3 km, alongside the 2 km window in Figure 4. As with health (Appendix E.2), the drop on soft soil and the jump on hard soil remain visible at both windows.

Figure F.1: Economic Development Index: binned residuals at 1km and 3km windows



(a) Soft soil, 1km

(b) Hard soil, 1km

(c) Soft soil, 3km

(d) Hard soil, 3km

Note: As Figure 4, redrawn at 1km and 3km. The unit is 2008 Census villages. Points are binned residuals from a regression of the *Economic Development Index* on a linear polynomial in spatial coordinates and the control variables. Solid lines are local linear regressions estimated separately either side of the bombing boundary, dashed lines 95% confidence intervals. Negative distances are locations outside the bombing areas.

F.2 Economic development index components

The components in Table F.1 broadly support the index results. On soft soil, bombed villages are significantly farther from health facilities. Access to water, electricity and sanitation, and dwelling ownership also point toward weaker development, though these estimates are not individually significant. On hard soil, bombed villages have higher employment, school attendance and literacy rates, significant at the 5, 5 and 10 percent levels respectively. Most component estimates are imprecise, which limits what we can establish separately for each outcome and motivates combining them in the index.

Table F.1: Economic Index (2008 Population Census Villages): Individual Outcomes

	Dependent variable is:									
	Dist. School (1)	Dist. Hospital (2)	Radio Station (3)	Improved Water (4)	Electricity (5)	Toilet (6)	Own Dwellings (7)	Literacy (8)	School Attendance (9)	Employment (10)
Panel A: All observations										
Bombing	-0.009	-0.264	0.001	-0.874	-0.909	-0.206	0.311	0.924**	0.501**	0.196**
	(0.026)	(0.226)	(0.005)	(1.120)	(0.751)	(0.783)	(0.226)	(0.469)	(0.207)	(0.087)
Bombing × Soft Soil	0.027	0.737*	-0.010	-1.410	-1.168	-0.526	-0.172	-0.194	-0.291	-0.364*
	(0.044)	(0.379)	(0.007)	(1.922)	(1.592)	(1.456)	(0.539)	(0.771)	(0.384)	(0.199)
Mean	0.915	13.780	0.916	52.628	14.074	23.857	95.530	74.531	29.613	98.924
Observations	7012	7012	7012	7012	7012	7012	7012	7012	7012	7012
Clusters	1104	1104	1104	1104	1104	1104	1104	1104	1104	1104
Panel B: Soft soil										
Bombing	0.011	0.711***	-0.007	-2.192	-1.952	-1.017	-0.215	0.782	0.170	-0.151
	(0.031)	(0.270)	(0.004)	(1.563)	(1.369)	(1.197)	(0.422)	(0.612)	(0.301)	(0.172)
Mean	0.793	11.723	0.922	57.570	20.158	29.237	93.667	76.218	29.418	98.484
Observations	2521	2521	2521	2521	2521	2521	2521	2521	2521	2521
Clusters	559	559	559	559	559	559	559	559	559	559
Panel C: Hard soil										
Bombing	-0.009	-0.304	0.001	-0.387	-0.677	-0.013	0.324	0.888*	0.462**	0.198**
	(0.026)	(0.226)	(0.005)	(1.102)	(0.731)	(0.769)	(0.207)	(0.466)	(0.208)	(0.087)
Mean	0.984	14.936	0.912	49.852	10.656	20.835	96.576	73.583	29.723	99.171
Observations	4489	4489	4489	4489	4489	4489	4489	4489	4489	4489
Clusters	827	827	827	827	827	827	827	827	827	827

Note: The unit of analysis is census villages. All regressions use a local linear polynomial in spatial coordinates with a triangular kernel weight and control for 50x50km grid fixed effects, province fixed effects, distance to the capital, distance to Vietnam borders and other pre-bombing characteristics. *Economic Development Index* is the standardized weighted summary index variable created from multiple outcomes in this table. *Dist. School* indicates the distance from each village to the nearest school (in km). *Dist. Hospital* indicates the distance from each village to the nearest hospital (in km). *Radio Station* indicates location within big radio networks. The subsequent columns report the percentage of households in the village having access to improved water (column 4), electricity (column 5), and toilets (column 6); the percentage living in owner-occupied dwellings (column 7); literacy rate (column 8); school attendance rate (column 9); and employment rate (column 10). Standard errors reported in parentheses are clustered at the commune level. ***(**)(*) indicates significance at the 1%(5%)(10%) level.

F.3 Aid, migration, and population

We examine whether the hard-soil gains coincide with aid or population concentrating in bombed villages. Table F.2 uses the 2009 and 2014 CSES with year fixed effects. Within hard soil, neither project presence nor project counts differs significantly across bombing boundaries, and the count estimate is negative (−0.079, Panel B, column 5). Both estimates are positive within soft soil, significant at 10% but not when excluding K-5 provinces. The results provide little evidence of NGO concentration in bombed hard-soil villages. These later project observations do not measure government reconstruction spending or establish how aid was allocated earlier in the recovery.

Table F.3 reports migration status among the people observed in the DHS and CSES. Within hard soil, the bombing coefficients are −0.036 and 0.028, respectively, neither significant. The pooled interaction is small and insignificant in the DHS, but negative and significant in the CSES, where residents of bombed soft-soil locations are less likely to report migration. These residential histories do not measure out-migration from the villages and therefore cannot establish whether people previously fled contaminated locations.

To assess whether the hard-soil development gains reflect population concentrating in bombed villages, Table F.4 examines village population in both 1998 and 2008. The censuses cover somewhat different sets of villages, so we retain each year's coverage and estimate within-soil differences separately rather than population change for a fixed panel. Neither census shows a significant population difference within either soil type, and the hard-soil estimates are negative in both years. These censuses cannot rule out earlier displacement or changes in population composition. Together with the migration evidence, however, they weigh against the development gains being driven by a lasting concentration of people in bombed hard-soil villages.

Table F.2: NGO Projects (2009 and 2014 CSES Village Data)

	All areas		Soft soil		Hard soil	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: NGO Project (Dummy)						
Bombing	0.014	-0.016	0.101*	0.087	0.018	-0.008
	(0.044)	(0.043)	(0.060)	(0.060)	(0.044)	(0.043)
Bombing × Soft Soil	0.067	0.084				
	(0.066)	(0.066)				
Mean	0.243	0.240	0.207	0.206	0.271	0.265
Observations	775	741	338	330	424	404
Clusters	530	500	222	216	317	299
Exclude K5 Belt	No	Yes	No	Yes	No	Yes
Panel B: Number of NGO Projects						
Bombing	-0.073	-0.124	0.148*	0.130	-0.079	-0.125
	(0.081)	(0.081)	(0.085)	(0.087)	(0.081)	(0.080)
Bombing × Soft Soil	0.206*	0.243**				
	(0.118)	(0.118)				
Mean	0.381	0.378	0.311	0.309	0.434	0.426
Observations	775	741	338	330	424	404
Clusters	530	500	222	216	317	299
Exclude K5 Belt	No	Yes	No	Yes	No	Yes

Note: Observations are CSES village-years from 2009 and 2014, restricted to villages within 2km of a treated-control interface (a bombing boundary). Panel A measures whether an NGO development project is currently functioning in the village, and Panel B counts these projects. All regressions use a local linear polynomial in spatial coordinates with triangular kernel weights, 50x50km grid, province and year fixed effects, distance to the capital, distance to Vietnam borders and other pre-bombing characteristics. Regressions (2), (4) and (6) exclude provinces along the K-5 mine belt. Means refer to each regression sample. Standard errors in parentheses are clustered at the commune level. ***(**)(*) indicates significance at the 1%(5%)(10%) level.

Table F.3: Migration Status

	All areas		Soft soil		Hard soil	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: DHS Individuals						
Bombing	-0.036*	-0.022	-0.019	-0.017	-0.036	-0.021
	(0.022)	(0.021)	(0.029)	(0.030)	(0.023)	(0.022)
Bombing × Soft Soil	0.002	-0.012				
	(0.034)	(0.033)				
Mean	0.340	0.331	0.343	0.338	0.339	0.326
Observations	15545	14184	5597	5336	9948	8848
Clusters	496	450	178	168	318	282
Exclude K5 Belt	No	Yes	No	Yes	No	Yes
Panel B: CSES Individuals						
Bombing	0.035	0.039	-0.056*	-0.039	0.028	0.028
	(0.023)	(0.024)	(0.030)	(0.032)	(0.023)	(0.024)
Bombing × Soft Soil	-0.095***	-0.093***				
	(0.034)	(0.035)				
Mean	0.354	0.357	0.401	0.407	0.325	0.326
Observations	23487	22354	9081	8620	14406	13734
Clusters	325	312	137	132	188	180
Exclude K5 Belt	No	Yes	No	Yes	No	Yes

Note: The unit of analysis is individuals. The sample is restricted to those living within 2km of a treated-control interface (a bombing boundary). All regressions use a local linear polynomial in spatial coordinates with a triangular kernel weight and control for 50x50km grid fixed effects, province fixed effects, distance to the capital, distance to Vietnam borders and other pre-bombing characteristics. Regressions (2) (4) (6) exclude provinces along the K-5 mine belt. Standard errors reported in parentheses are clustered at the village level. The dependent variable, *Migration*, equals one if the individual reports having moved, and zero otherwise, based on self-reported residential history. ***(**)(*) indicates significance at the 1%(5%)(10%) level.

Table F.4: Village Population, 1998 and 2008 Censuses

	1998 Census			2008 Census		
	All	Soft soil	Hard soil	All	Soft soil	Hard Soil
	(1)	(2)	(3)	(4)	(5)	(6)
Bombing	-0.025	-0.034	-0.013	-0.032	-0.009	-0.024
	(0.022)	(0.031)	(0.022)	(0.022)	(0.035)	(0.022)
Bombing × Soft Soil	-0.009			-0.000		
	(0.038)			(0.041)		
Mean	6.427	6.524	6.375	6.570	6.694	6.501
Observations	6490	2236	4250	7012	2521	4489
Clusters	1034	508	782	1104	559	827

Note: The dependent variable is ln(Total Population). Regressions are estimated separately for 1998 and 2008 Population Census villages within 2km of a treated–control interface (a bombing boundary), using all provinces. All regressions use a local linear polynomial in spatial coordinates with triangular kernel weights and control for 50x50km grid fixed effects, province fixed effects, distance to the capital, distance to the Vietnamese border and other pre-bombing characteristics. Standard errors in parentheses are clustered at the commune level. ***(**)(*) indicates significance at the 1%(5%)(10%) level.

F.4 Development in 1998 and 2008

The 1998 Population Census lets us examine development a decade earlier, after conflict that continued into the 1990s. We construct indices from the same six dimensions in each census, namely adult literacy, current school attendance, dwelling ownership, water access, electricity and toilet facilities, using the closest available census categories.[19] Components are standardized and index weights calculated separately within each census, following the procedure used for the main results. Restricting to villages matched across censuses would discard a substantial share of each year's coverage, so we retain each census's villages and estimate the bombing effects separately. With census-specific categories, standardization and weights, pooling the indices would mix changes in development with changes in measurement. We instead apply our within-soil design in each census to ask whether the bombing pattern was already present in 1998, reporting the results side by side in Table F.5.

The soft-soil penalty is already evident in 1998, at −0.040, and persists in 2008 at −0.034. Excluding the K-5 belt provinces strengthens it in both years, to −0.050 and −0.041. The latter is close to −0.043 for the full 2008 index with the same exclusion (Table 5). Finding the penalty a decade earlier and again in 2008 provides further support for the risk channel, under which buried ordnance continues to constrain activity long after the bombing.

Bombed hard-soil villages also lag behind in 1998, with an estimate of −0.029, but by 2008 the estimate is positive at 0.015. This is close in magnitude to the main estimate of 0.018 in Table 5, despite restricting the index to six dimensions rather than ten. Excluding the K-5 belt provinces gives 0.018 in 2008, again close to the corresponding full-index estimate of 0.022. The restriction leaves out some of the development captured by the full index, and the all-province estimate is less precise, with a standard error of 0.012 rather than 0.009. It is not statistically significant, but its direction and magnitude remain consistent with the main results. These separate estimates are not a formal test of change between years or direct evidence of capital replacement. The earlier disadvantage and later positive estimate nevertheless fit recovery that takes time after destruction and are consistent with the model's reset interpretation.

[19] The categories are not identical. In particular, the 1998 water measure includes wells without distinguishing protection, whereas the 2008 measure distinguishes protected from unprotected wells and also includes rainwater and purchased water.

Table F.5: Village Development, 1998 and 2008 Censuses

	1998 Census			2008 Census		
	(1) All	(2) Soft soil	(3) Hard soil	(4) All	(5) Soft soil	(6) Hard soil
Panel A: Village development index, all provinces						
Bombing	-0.032***	-0.040*	-0.029***	0.010	-0.034*	0.015
	(0.011)	(0.022)	(0.011)	(0.013)	(0.020)	(0.012)
Bombing × Soft Soil	-0.012			-0.029		
	(0.026)			(0.026)		
Mean	0.008	-0.019	0.023	0.024	0.032	0.019
Observations	6490	2236	4250	7012	2521	4489
Clusters	1034	508	782	1104	559	827
Panel B: Village development index, excluding K5 belt provinces						
Bombing	-0.030***	-0.050**	-0.027**	0.012	-0.041**	0.018
	(0.011)	(0.023)	(0.011)	(0.014)	(0.020)	(0.013)
Bombing × Soft Soil	-0.021			-0.036		
	(0.027)			(0.027)		
Mean	0.006	-0.022	0.021	0.027	0.032	0.023
Observations	6102	2126	3971	6613	2396	4215
Clusters	948	475	709	1013	522	752

Note: Panels A and B report development indices combining six common dimensions: water access, electricity, toilet facilities, dwelling ownership, adult literacy and current school attendance. Components are standardized and inverse-covariance weights calculated separately within each census following Anderson (2008). The unit is 1998/2008 Population Census villages within 2km of a treated–control interface (a bombing boundary). Panel A uses all provinces; Panel B drops the seven provinces along the Thai border where the K-5 Plan laid a mine barrier in the 1980s. All regressions use a local linear polynomial in spatial coordinates with a triangular kernel weight and control for 50x50km grid fixed effects, province fixed effects, distance to the capital, distance to the Vietnamese border and other pre-bombing characteristics; standard errors are clustered at the commune. ***(**)(*) indicates significance at the 1%(5%)(10%) level.

F.5 Pre-war settlement and the destruction interpretation

Table F.6 splits the within-soil estimates of Table 5 by historical settlement size, using the largest pre-war settlement within 2 km in Panel A and all pre-war buildings within 3 km in Panel B. On soft soil we do not expect a gradient. Bombs there were less likely to detonate, and the penalty reflects unexploded ordnance that constrains the use of land long after the bombing, a constraint that does not depend on the amount of built capital before the war. The estimates are consistent with this prediction, although they do not establish that the effects are equal. In both panels, the penalty is similar in areas with small and large pre-war settlements and is close to the soft-soil estimate of -0.040 in Table 5; p-values for equality range from 0.72 to 0.87.

On hard soil we do expect a gradient. Bombs there were more likely to detonate, so larger settlements had more old capital to destroy and more room to rebuild with newer,

more productive capital. The estimates follow this pattern. Where settlements were small, the hard-soil estimates are close to zero and insignificant. Where they were large, the estimates are 0.027 and 0.026 in Panel A and B respectively, about one and a half times the main estimate of 0.018, or roughly 8% rather than 6% of a standard deviation of the index. Excluding the K-5 provinces sharpens the pattern, as it does for the main estimate. The large-settlement estimates increase to 0.033 (Panel A) and 0.031 (Panel B), about 10% of a standard deviation, and both are significant at the 5% level. The small-settlement estimates remain small and insignificant. The differences between small and large settlements are not statistically significant, with p-values between 0.34 and 0.44, since splitting each soil sample in two leaves less power. Their direction nonetheless fits rebuilding where bombs were more likely to detonate and more stood to be destroyed.

Table F.6: Bombing and Pre-War Settlement: Economic Development Index

	Economic Development Index			
	Soft Soil		Hard Soil	
	(1)	(2)	(3)	(4)
Exclude K-5 provinces	No	Yes	No	Yes
Panel A. Largest pre-war settlement within 2 km				
Small pre-war settlement	-0.043**	-0.047**	0.013	0.016
	(0.018)	(0.018)	(0.011)	(0.012)
Observations	1,034	981	2,580	2,418
Large pre-war settlement	-0.039*	-0.036	0.027*	0.033**
	(0.023)	(0.023)	(0.016)	(0.015)
Observations	1,482	1,412	1,906	1,794
p-value, small = large	0.869	0.723	0.439	0.366
Panel B. Buildings within 3 km				
Small pre-war settlement	-0.039*	-0.044*	0.010	0.012
	(0.022)	(0.023)	(0.012)	(0.013)
Observations	1,243	1,186	2,268	2,076
Large pre-war settlement	-0.034	-0.033	0.026*	0.031**
	(0.023)	(0.024)	(0.015)	(0.015)
Observations	1,277	1,208	2,219	2,137
p-value, small = large	0.874	0.747	0.395	0.342

Note: Each estimate is a bombing effect on the Economic Development Index within a soil and settlement group, using 2008 Census villages within 2km of a bombing boundary. Panel A measures the largest settlement within 2km, linking buildings in steps of at most 150m. Panel B counts all buildings within 3km. Small/large groups are at or below/above the full-sample medians of 30 and 319 buildings, respectively; these cut-offs remain fixed when K-5 provinces are excluded in columns (2) and (4). Building data are from Bühler & Madestam (2024), digitized from US Army maps hosted by Yale with information dating from 1966 to the early 1970s. Regressions use linear spatial controls, triangular kernel weights, 50x50km grid and province fixed effects, and the baseline pre-bombing controls. Standard errors in parentheses are clustered at the commune. The p-value tests equality of the small- and large-settlement bombing effects, allowing all controls and fixed effects to vary by settlement group. ***(**)(*) indicates significance at the 1%(5%)(10%) level.

G Household welfare and agricultural mechanisms

These tables supplement the household and field results with evidence on selection into farming and the allocation of agricultural resources. They also report the checks on clustering and spatial correlation.

G.1 Selection into crop farming

Table G.1: The probability that a household engages in crop farming

	Dependent variable is: Farming Household *Unit of Analysis: Households*		
	All areas (1)	Soft soil (2)	Hard soil (3)
Bombing	0.003	-0.002	-0.000
	(0.025)	(0.037)	(0.023)
Bombing × Soft Soil	-0.003		
	(0.043)		
Mean	0.560	0.395	0.680
Observations	11136	4675	6461
Clusters	746	323	423

Note: The unit of analysis is CSES households, and the sample covers all of them rather than only those who farm. The dependent variable is an indicator equal to one if the household produces crops, set to zero for households not administered the crop-production module. Column 1 pools both soils and includes a soil main effect; columns 2 and 3 split the sample. All regressions use a local linear polynomial in spatial coordinates with a triangular kernel weight and control for 50x50km grid fixed effects, province fixed effects, survey-year fixed effects, distance to the capital, distance to the Vietnamese border and other pre-bombing characteristics. The sample is restricted to households within 2km of a treated–control interface (a bombing boundary). Standard errors reported in parentheses are clustered at the village level. ***(**)(*) indicates significance at the 1%(5%)(10%) level.

G.2 Inference for CSES household and field outcomes

The level of clustering, and spatial correlation. Bombing is assigned at the 2.42 km grid cell, which is coarser than the village at which our household, field and Agricultural Census estimates cluster. We therefore re-estimate them at the commune, which averages roughly 115 km^2 and so spans about twenty of the cells where treatment is assigned, allowing the errors to correlate across them. For the two CSES samples we add Conley (1999) standard errors, which drop clusters altogether and let the covariance decay with distance instead of stopping at an administrative edge. We take the Agricultural Census at the commune alone, since Conley builds its covariance from pairwise distances and 969,674 households puts that computation out of practical reach.

Tables G.2, G.3 and G.5 report the results. The alternative standard error methods preserve the central conclusions, although some individual estimates change significance. Both within-soil income effects remain significant under commune clustering and Conley inference, as do the soft-soil declines in cultivated area and crop revenue. The soft-soil pattern of reduced land-based activity in the Agricultural Census also remains under commune clustering. For comparison, the bombing $\times$ soft soil interaction on household income has a standard error of 0.160 when clustered at the village and 0.151 at the commune. Conley raises it to 0.228, and the interaction still holds at the five percent level. The central findings therefore do not depend on the village-level clustering used in the main tables.

Table G.2: Inference: CSES Household Development

	Dependent variable is: *Unit of Analysis: Households*			
	Ln(Income) (1)	Ln(Expenditure) (2)	Ln(Durable) (3)	Food Secure (4)
Panel A: All observations				
Bombing	0.297***	0.020	0.126*	0.041*
	(0.100)	(0.033)	(0.069)	(0.022)
	[0.130]**	[0.042]	[0.076]*	[0.025]*
Bombing × Soft Soil	-0.541***	-0.054	-0.337***	-0.106***
	(0.151)	(0.049)	(0.112)	(0.035)
	[0.228]**	[0.060]	[0.115]***	[0.035]***
Observations	11135	11135	11135	11135
Communes	548	548	548	548
Panel B: Soft soil				
Bombing	-0.264**	-0.049	-0.255***	-0.076***
	(0.113)	(0.038)	(0.091)	(0.026)
	[0.104]**	[0.035]	[0.082]***	[0.026]***
Observations	4674	4674	4674	4674
Communes	234	234	234	234
Panel C: Hard soil				
Bombing	0.275***	0.020	0.114*	0.042*
	(0.100)	(0.033)	(0.068)	(0.022)
	[0.124]**	[0.039]	[0.068]*	[0.025]
Observations	6461	6461	6461	6461
Communes	336	336	336	336

Note: Commune-clustered standard errors are in parentheses and Conley (1999) spatial standard errors in brackets. Conley standard errors use a 10km distance cutoff. Stars beside the coefficient refer to the commune-clustered standard error and those beside the bracketed figure to the Conley one. Panel A reports the interacted model and Panels B and C each soil type separately. All regressions use a local linear polynomial in spatial coordinates with a triangular kernel weight and control for 50x50km grid fixed effects, province fixed effects, survey-year fixed effects, distance to the capital, distance to the Vietnamese border and other pre-bombing characteristics. The sample is restricted to units within 2km of a treated–control interface (a bombing boundary). ***(**)(*) indicates significance at the 1%(5%)(10%) level.

Table G.3: Inference: CSES Field Productivity and Risk Taking

	Dependent variable is:					
	Field Productivity *Unit: Agricultural Fields*				Risk Taking *Unit: Households*	
	Ln(Area) (1)	Ln(Revenue) (2)	Loss Rate (3)	Investment (4)	Loan Taken (5)	Ln(Loan Amount) (6)
Panel A: All observations						
Bombing	-0.064	-0.106	0.003	-0.018	0.027	0.075
	(0.070)	(0.070)	(0.002)	(0.024)	(0.030)	(0.165)
	[0.063]	[0.068]	[0.002]	[0.025]	[0.031]	[0.162]
Bombing × Soft Soil	-0.204*	-0.116	0.008	-0.072	-0.070	-0.334
	(0.118)	(0.124)	(0.006)	(0.044)	(0.048)	(0.261)
	[0.124]	[0.126]	[0.006]	[0.053]	[0.047]	[0.251]
Observations	13175	13175	13175	13175	3707	3707
Communes	496	496	496	496	508	508
Panel B: Soft soil						
Bombing	-0.362***	-0.324***	0.012**	-0.054	-0.073*	-0.446**
	(0.100)	(0.103)	(0.006)	(0.036)	(0.039)	(0.219)
	[0.093]***	[0.097]***	[0.006]**	[0.039]	[0.037]**	[0.216]**
Observations	4635	4635	4635	4635	1530	1530
Communes	200	200	200	200	216	216
Panel C: Hard soil						
Bombing	-0.091	-0.092	0.002	-0.013	0.041	0.129
	(0.067)	(0.067)	(0.002)	(0.023)	(0.030)	(0.160)
	[0.062]	[0.066]	[0.002]	[0.024]	[0.031]	[0.162]
Observations	8540	8540	8540	8540	2175	2175
Communes	308	308	308	308	310	310

Note: Commune-clustered standard errors are in parentheses and Conley (1999) spatial standard errors in brackets. Conley standard errors use a 10km distance cutoff. Stars beside the coefficient refer to the commune-clustered standard error and those beside the bracketed figure to the Conley one. Panel A reports the interacted model and Panels B and C each soil type separately. All regressions use a local linear polynomial in spatial coordinates with a triangular kernel weight and control for 50x50km grid fixed effects, province fixed effects, survey-year fixed effects, distance to the capital, distance to the Vietnamese border and other pre-bombing characteristics. The sample is restricted to units within 2km of a treated–control interface (a bombing boundary). ***(**)(*) indicates significance at the 1%(5%)(10%) level.

G.3 Agricultural production and labor allocation

Table G.4: Agricultural Census (2013): crop orientation, livestock, forest, and labor allocation

	Dependent variable is:							
	Comm. Share (1)	Main Sale (2)	Ln Livestock (3)	Stock > Median (4)	Uses Forest (5)	Labor: Crop (6)	Labor: Livestock (7)	Labor: Non-farm (8)
Panel A: All observations								
Bombing	0.008*	0.015	-0.014	-0.007	-0.010	-0.000	0.008	0.024**
	(0.004)	(0.010)	(0.009)	(0.007)	(0.008)	(0.006)	(0.007)	(0.011)
Bombing × Soft Soil	-0.002	-0.006	-0.025	-0.023*	-0.021*	-0.008	-0.037***	-0.006
	(0.009)	(0.019)	(0.017)	(0.012)	(0.012)	(0.011)	(0.014)	(0.019)
Mean	0.085	0.211	0.783	0.564	0.085	0.664	0.569	0.228
Observations	937079	969674	969674	969674	966308	969674	969674	969674
Clusters	6569	6583	6583	6583	6583	6583	6583	6583
Panel B: Soft soil								
Bombing	-0.000	0.002	-0.039***	-0.033***	-0.024***	-0.009	-0.029**	0.009
	(0.008)	(0.016)	(0.014)	(0.010)	(0.007)	(0.009)	(0.011)	(0.014)
Mean	0.106	0.290	0.690	0.490	0.056	0.651	0.512	0.231
Observations	323929	339539	339539	339539	337999	339539	339539	339539
Clusters	2262	2266	2266	2266	2266	2266	2266	2266
Panel C: Hard soil								
Bombing	0.006	0.017*	-0.017*	-0.009	-0.010	0.000	0.006	0.028**
	(0.004)	(0.010)	(0.009)	(0.006)	(0.008)	(0.006)	(0.007)	(0.011)
Mean	0.074	0.168	0.834	0.604	0.100	0.671	0.600	0.226
Observations	613150	630135	630135	630135	628309	630135	630135	630135
Clusters	4307	4317	4317	4317	4317	4317	4317	4317

Note: The unit of analysis is agricultural households in the 2013 Cambodia National Agricultural Census. The sample is restricted to households in villages within 2km of a treated-control interface (a bombing boundary). *Comm. Share* is the share of a household's temporary-crop parcels planted with commercial (cash) crops. *Main Sale* is an indicator for whether the household's main purpose of production is sale rather than home consumption. *Ln Livestock* is ln(TLU + 1), where TLU is total livestock in Tropical Livestock Units (FAO weights). *Stock > Median* is an indicator equal to 1 if the household's TLU exceeds the sample median. *Uses Forest* is an indicator for whether the household uses forest land. *Labor: Crop*, *Labor: Livestock*, and *Labor: Non-farm* are the shares of enumerated household members whose primary activity is crop agriculture, livestock, or non-farm work (handicrafts, trade, other industry and services), respectively. All regressions use a local linear polynomial in spatial coordinates with a triangular kernel weight and control for 50x50km grid fixed effects, province fixed effects, distance to the capital, distance to Vietnam borders, and other pre-bombing characteristics. Standard errors reported in parentheses are clustered at the census village. ***(**)(*) indicates significance at the 1%(5%)(10%) level.

G.4 Inference for the Agricultural Census

Table G.5: Inference: Agricultural Census

	Dependent variable is:							
	Comm. Share	Main Sale	Ln Livestock	Stock > Median	Uses Forest	Labor: Crop	Labor: Livestock	Labor: Non-farm
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: All observations								
Bombing	0.008*	0.015	-0.014	-0.007	-0.010	-0.000	0.008	0.024*
	(0.005)	(0.011)	(0.010)	(0.007)	(0.007)	(0.007)	(0.008)	(0.012)
Bombing × Soft Soil	-0.002	-0.006	-0.025	-0.023*	-0.021*	-0.008	-0.037**	-0.006
	(0.010)	(0.020)	(0.018)	(0.013)	(0.012)	(0.012)	(0.015)	(0.019)
Observations	937079	969674	969674	969674	966308	969674	969674	969674
Communes	1050	1050	1050	1050	1050	1050	1050	1050
Panel B: Soft soil								
Bombing	-0.000	0.002	-0.039***	-0.033***	-0.024***	-0.009	-0.029**	0.009
	(0.008)	(0.017)	(0.015)	(0.010)	(0.009)	(0.010)	(0.013)	(0.015)
Observations	323929	339539	339539	339539	337999	339539	339539	339539
Communes	516	516	516	516	516	516	516	516
Panel C: Hard soil								
Bombing	0.006	0.017	-0.017*	-0.009	-0.010	0.000	0.006	0.028**
	(0.004)	(0.010)	(0.010)	(0.007)	(0.008)	(0.007)	(0.008)	(0.012)
Observations	613150	630135	630135	630135	628309	630135	630135	630135
Communes	803	804	804	804	804	804	804	804

Note: Standard errors are clustered at the commune. Panel A reports the interacted model and Panels B and C each soil type separately. All regressions use a local linear polynomial in spatial coordinates with a triangular kernel weight and control for 50x50km grid fixed effects, province fixed effects, distance to the capital, distance to the Vietnamese border and other pre-bombing characteristics. The sample is restricted to units within 2km of a treated–control interface (a bombing boundary). ***(**)(*) indicates significance at the 1%(5%)(10%) level.